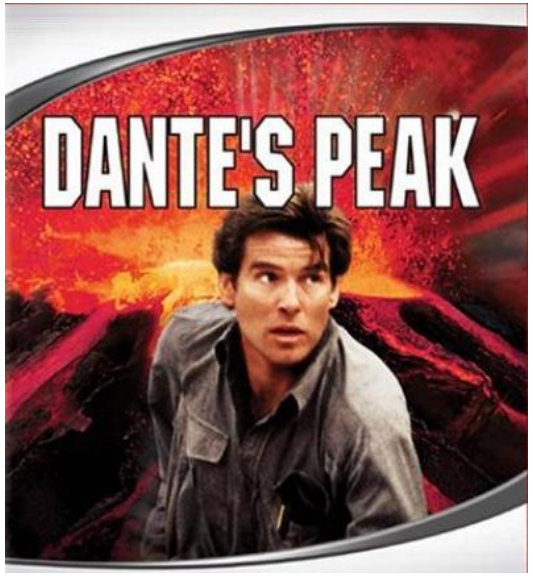
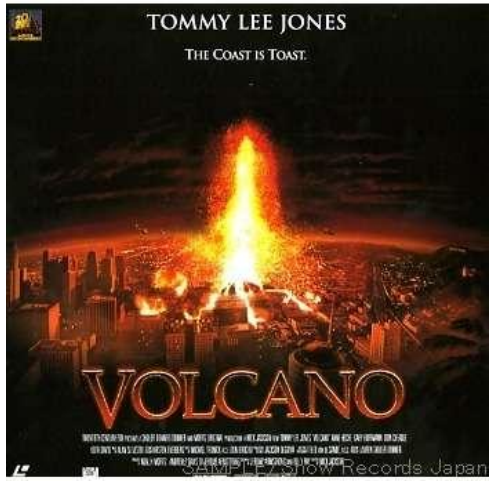
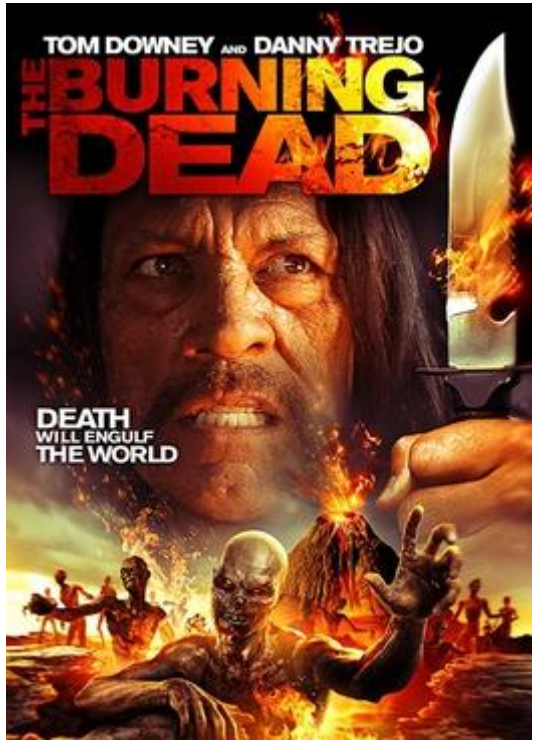
Your *Mystery Volcano* Project
Generic Instructions















Gather information

- From your homework and exercises you should know/have:
  - What volcano you're working on
  - The eruptive history of the volcano
  - The history of lahars, PDCs, and ballistics (how big? how often? what triggers them?)
  - Any existing geological maps
  - Any existing hazard maps
  - If there is a reason to update the hazard map (different scenario? different audience?)
  - What scenario or scenarios you will focus on (Worst-case? Most-likely? Range of sizes? what type of lahars, PDCs? what volume of lahars and PDCs? what size eruption, if any?)
  - A DEM of your volcano
  - Some Google Earth images of your volcano
  - A population map if available
  - A topographic or other map that shows roads, rivers, lakes, villages, cities, etc.
  - Who is your target audience (General public? Mining company or transportation agency? Tourists? Emergency managers? Land-use planners?)

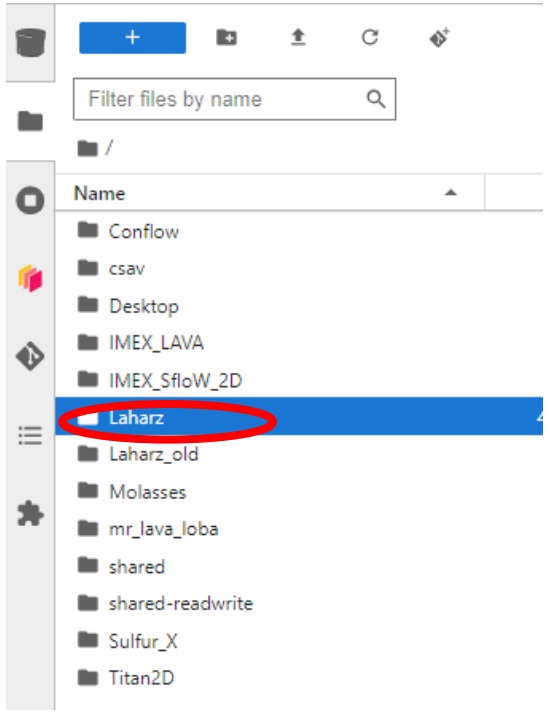
What you need for modeling

- Lahar (Presentation 8) and PDC volumes (Presentation 9)
- Stream threshold and H/L value for LAHARZ (remember the advice in Presentation 7)
- Basal friction value for the PDC volume(s) (remember the regression equation and basal friction table in Presentation 9)
- Big DEM for LaharZ, smaller DEM for Titan2D
- Summit coordinates (also think about how pointy the summit is or if there is a crater or multiple peaks)

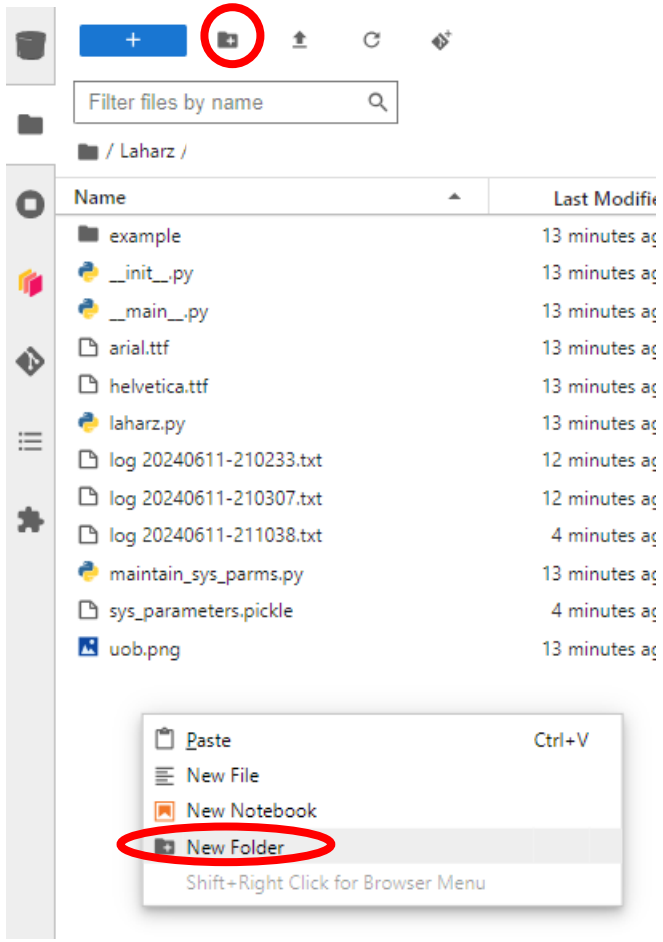
Gather more information and write it down (you can also use a sheet like this to keep track of different model runs, just copy the slide)

- What are the coordinates of the summit?
  - Latitude GCS (degrees):
  - Longitude GCS (degrees):
  - Latitude UTM (m):
  - Longitude UTM (m):
- For LAHARZ
  - What Lahar volumes will you model:
  - What stream network thresholds will you try:
  - What H/L ratio(s) will you try:
  - If your volcano has a complex summit, what elevation will you add:
- For PDC modeling:
  - Will you use energy cones and/or TITAN2D:
  - What type(s) of PDCs will you model:
  - What PDC volume(s) will you model:
  - For energy cones, which H/L values will you use (related to volume):
  - For TITAN2D, which basal friction angles (convert H/L to degrees):
  - For TITAN2D, what is your pile geometry (height, max radius, min radius):
- For ballistic modeling:

Let’s make a folder to work in!

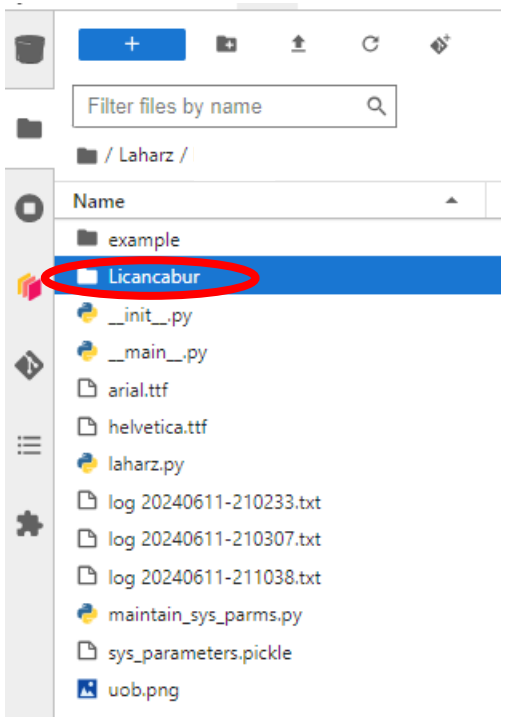


Double click on “Laharz”



Right click in a blank space and choose “New Folder”

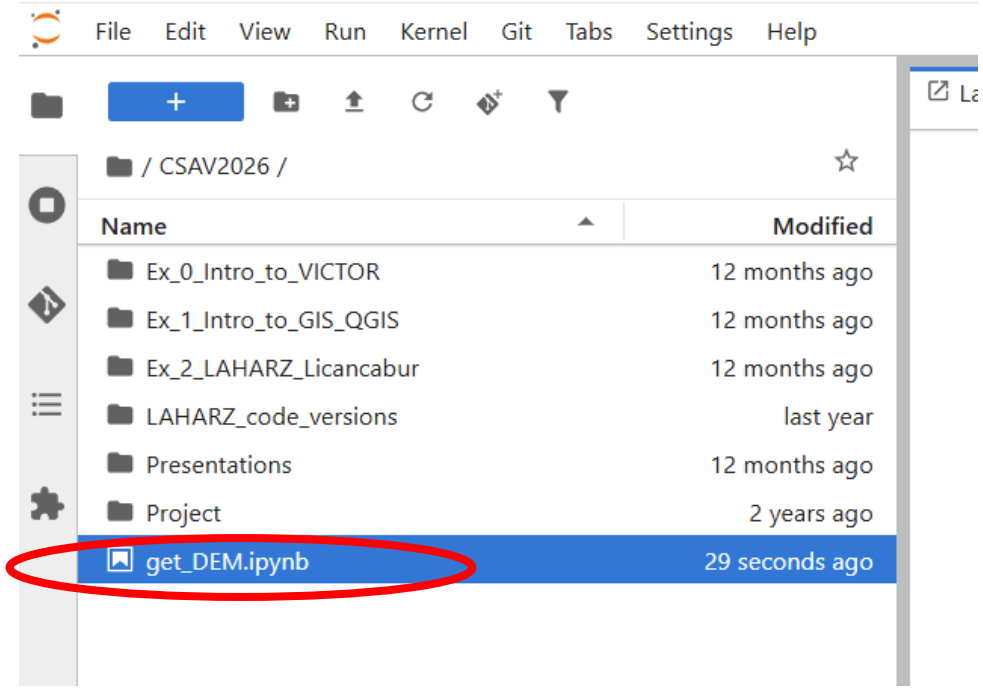
Or click the “New folder“ icon at the top



Name the new folder with your volcano name

Download a DEM

- Run the “get_DEM.ipynb” notebook to download a DEM
- If you plan to run TITAN2D, you might as well get both a larger DEM and a smaller DEM right now.
- Just name them so you can tell them apart and remember you will need to convert the smaller one to UTM at some point



Download a DEM

- Run the “get_DEM.ipynb” notebook to download a DEM
- Run the import libraries cells, then set the input parameters for your volcano

Setting input parameters

In the following cell we use the lat/lon of a target volcano’s summit, obtained from the Global Volcano Project database from the volcano’es name, to set the bounding box.

[8]:
```
#Choose a volcano and DEM options

volcano = victor.search_volcano('Licancabur') # Get volcano information from the GVP database

center_lat = volcano[1][0]  # Get volcano latitude
center_lon = volcano[2][0]  # Get volcano longitude

# Set the extent of the region of interest, in meters in each direction from the summit
eastwest_extent = 40000
northsouth_extent = 40000

# Set bounding box in lat/lon
utms = utm.from_latlon(center_lat, center_lon)
[north_lat, east_lon] = utm.to_latlon(utms[0]+eastwest_extent, utms[1]+northsouth_extent, utms[2], utms[3])
[south_lat, west_lon] = utm.to_latlon(utms[0]-eastwest_extent, utms[1]-northsouth_extent, utms[2], utms[3])

# Select a DEM format: Input "ascii" for ASCII format (.asc) and "tiff" for geotiff format
outputFormat = "tiff"

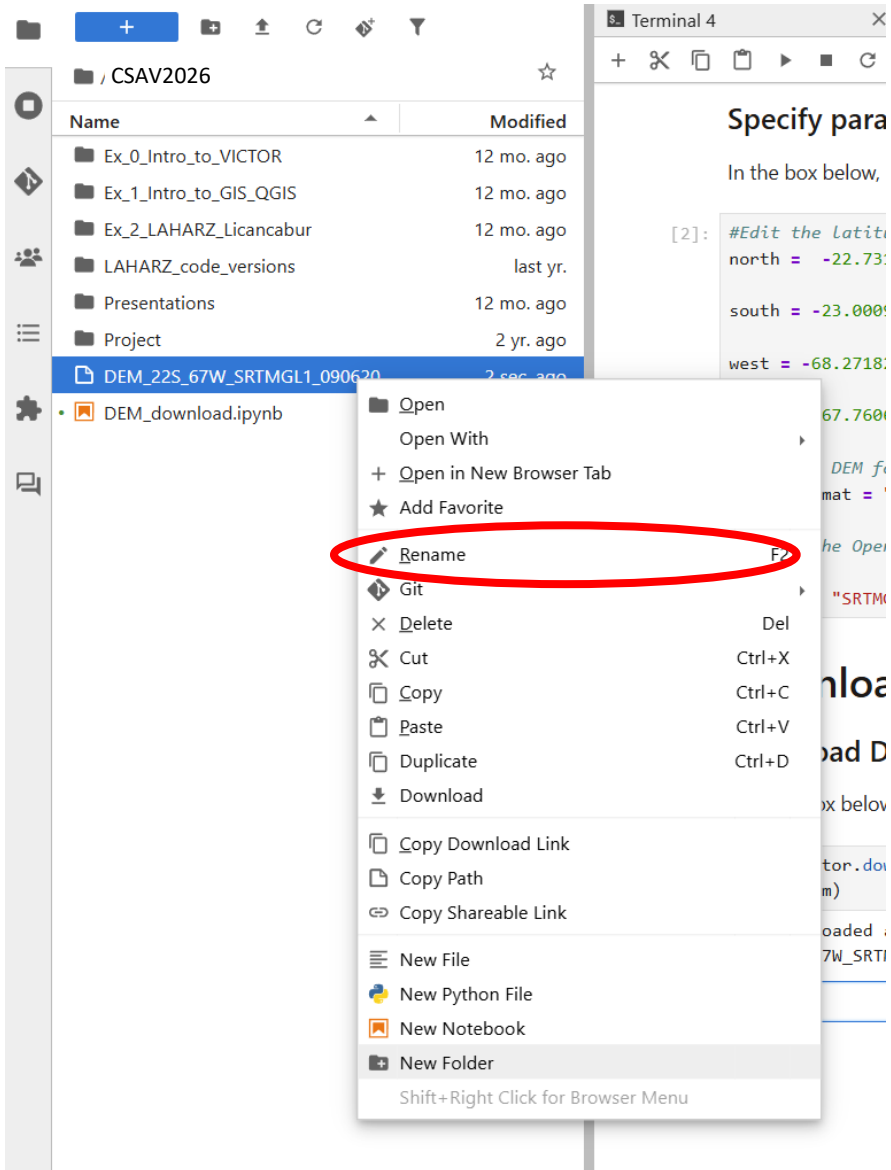
# Enter the name of the output file (without extension)
name = "Licancabur"

dataset = "SRTMGL1"

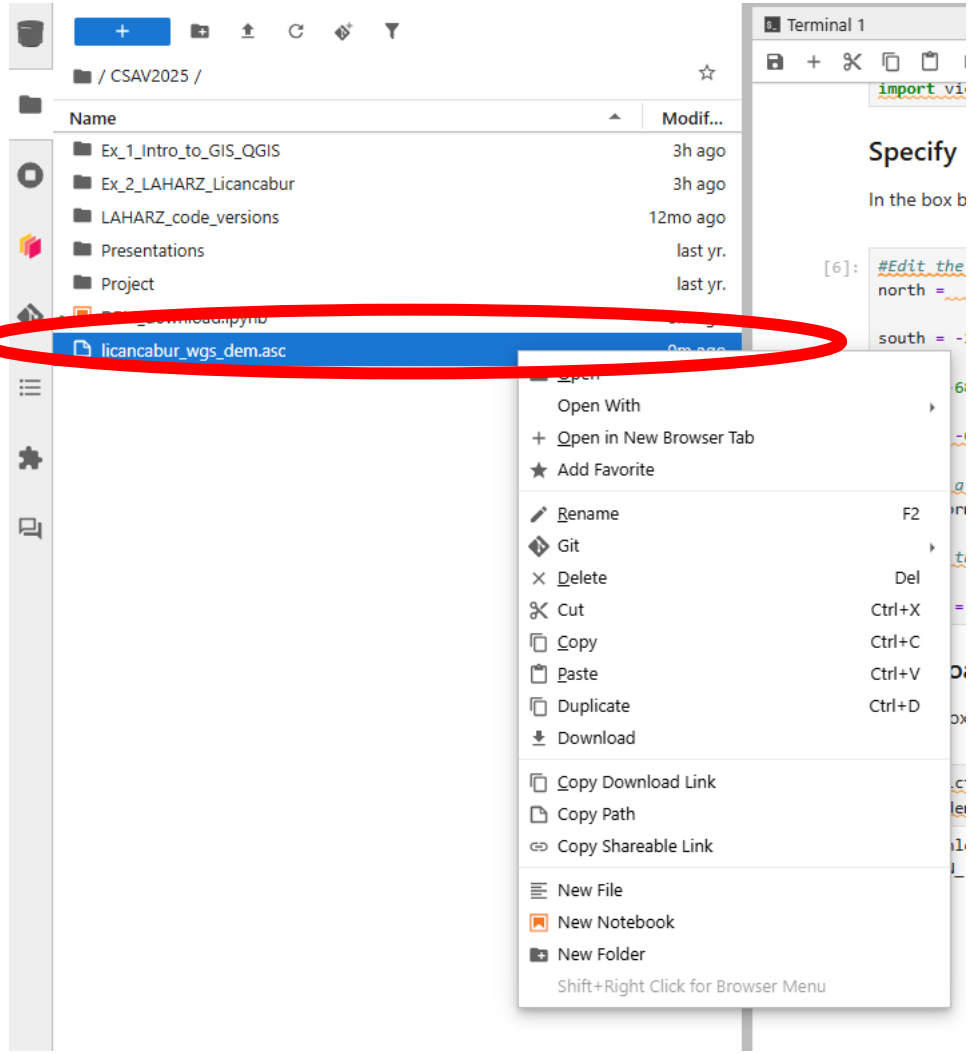
print(center_lat, center_lon)
```

```
-22.83 -67.88
```

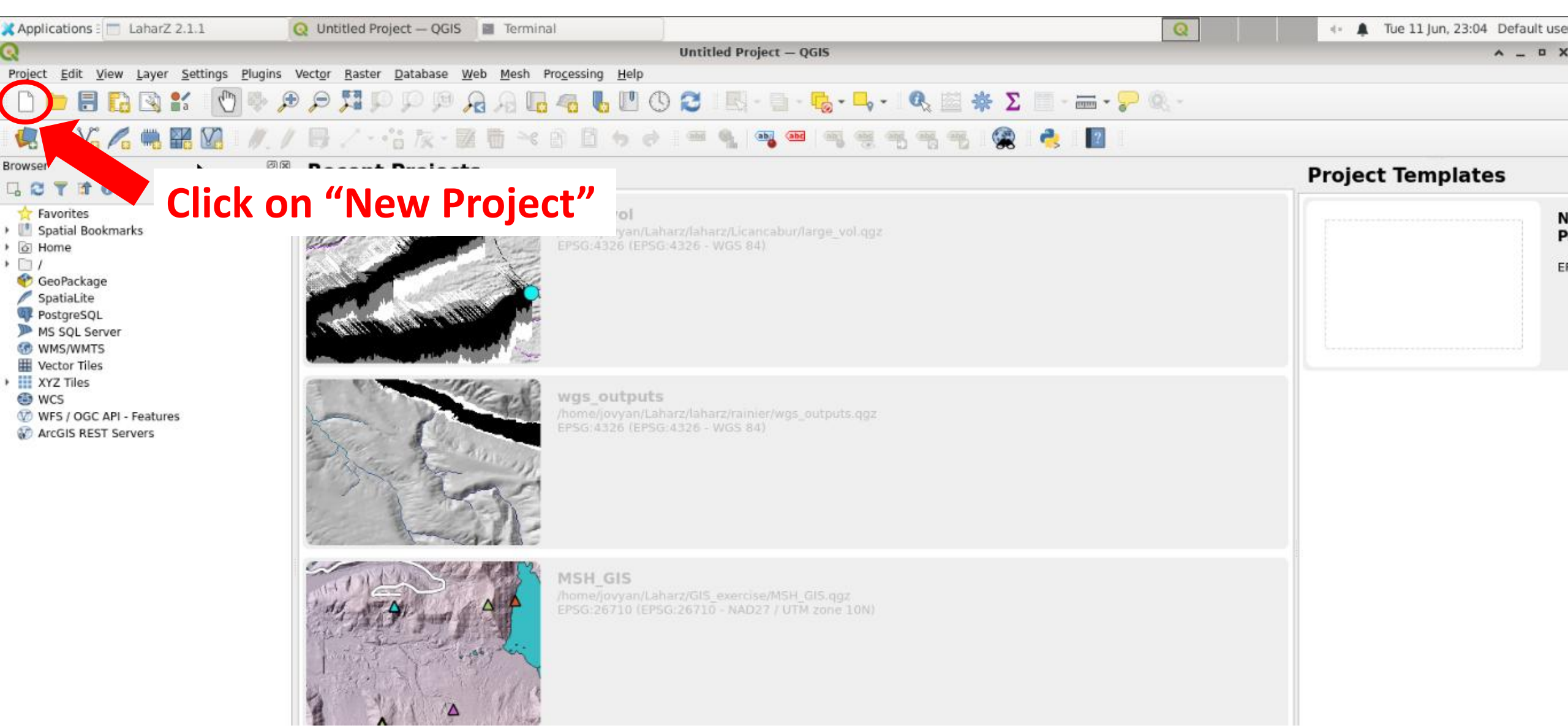
Rename the DEM



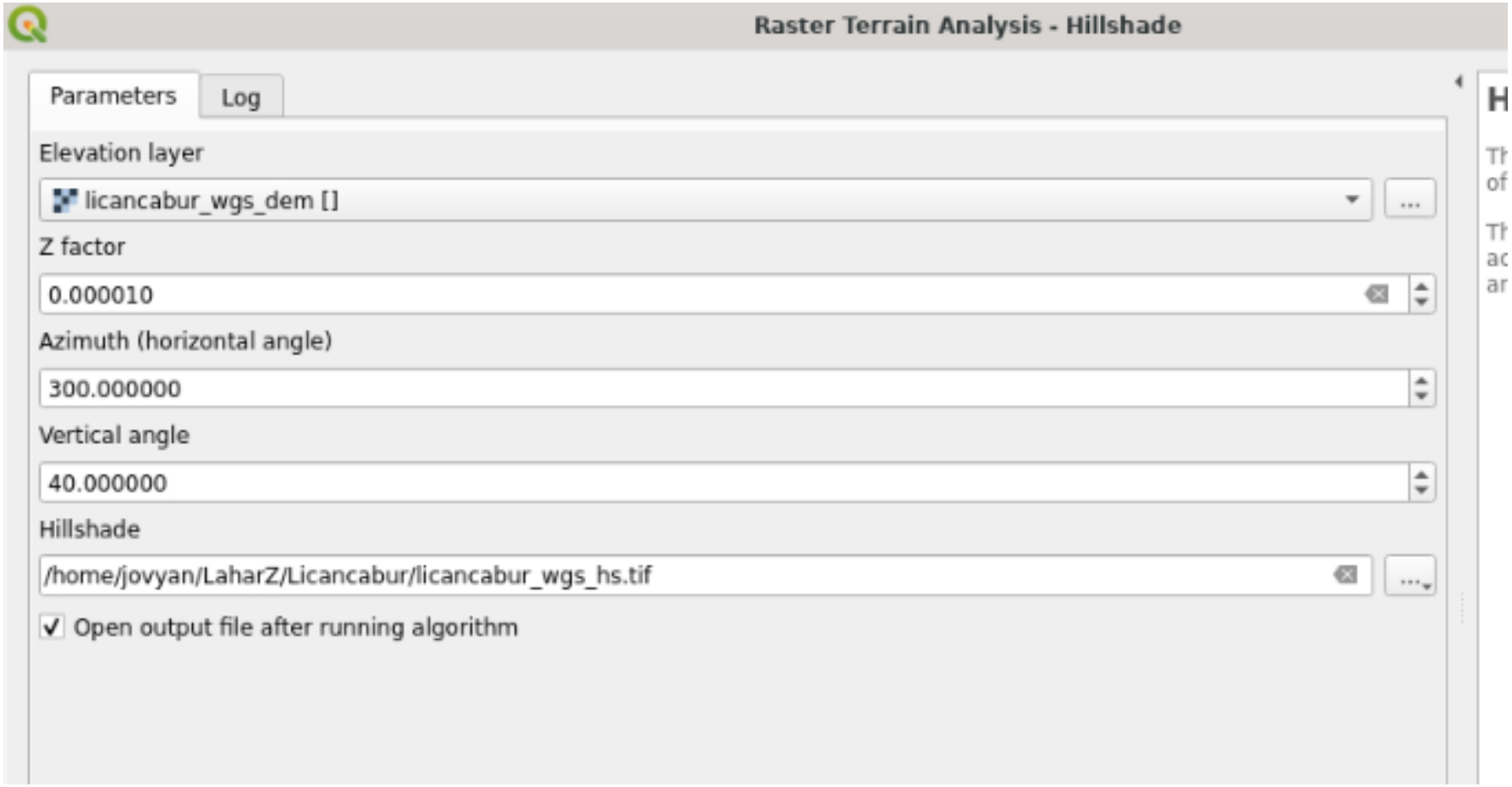
- The downloaded DEM should now be in your CSAV2026 folder
- Rename it by right clicking and choosing “Rename”
- Name it “yourvolcanoname_gcs_dem.tif”
- GCS=geographic coordinate system
- Make sure you change ‘geotiff’ to ‘tif’



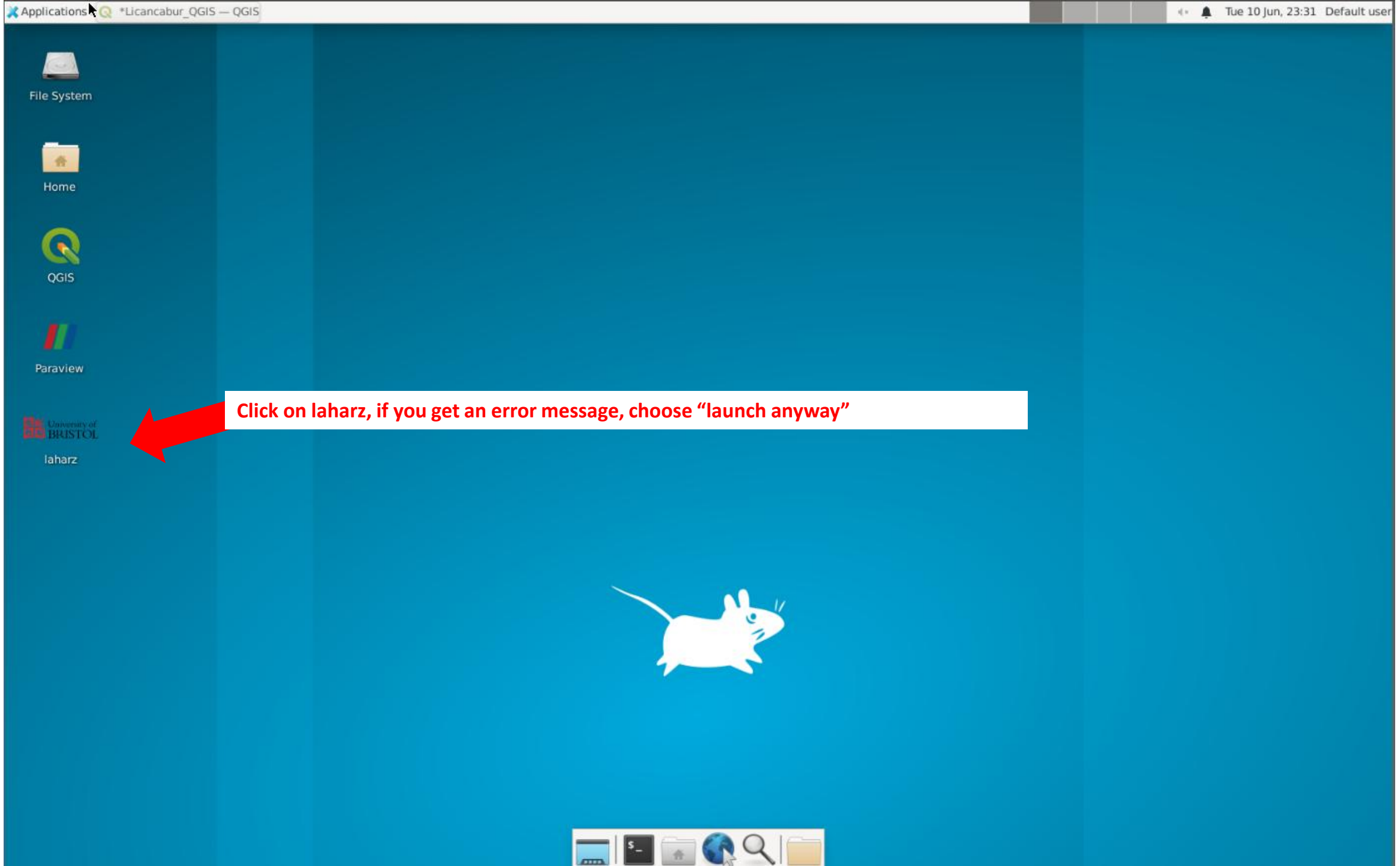
- Copy the new DEM to the Laharz/my_volcano folder



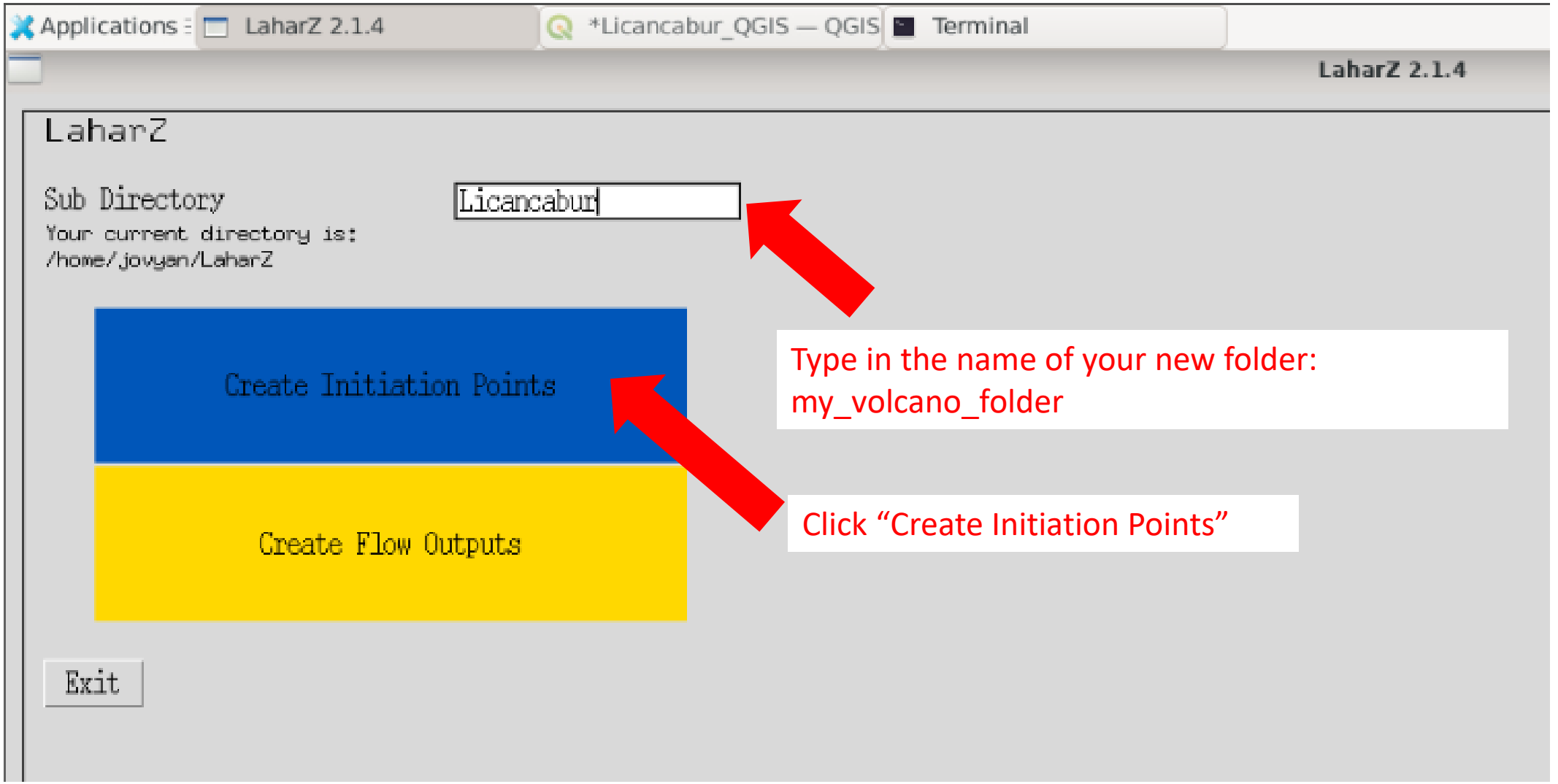
- Make a new QGIS project for your volcano
- Make a hillshade of your DEM

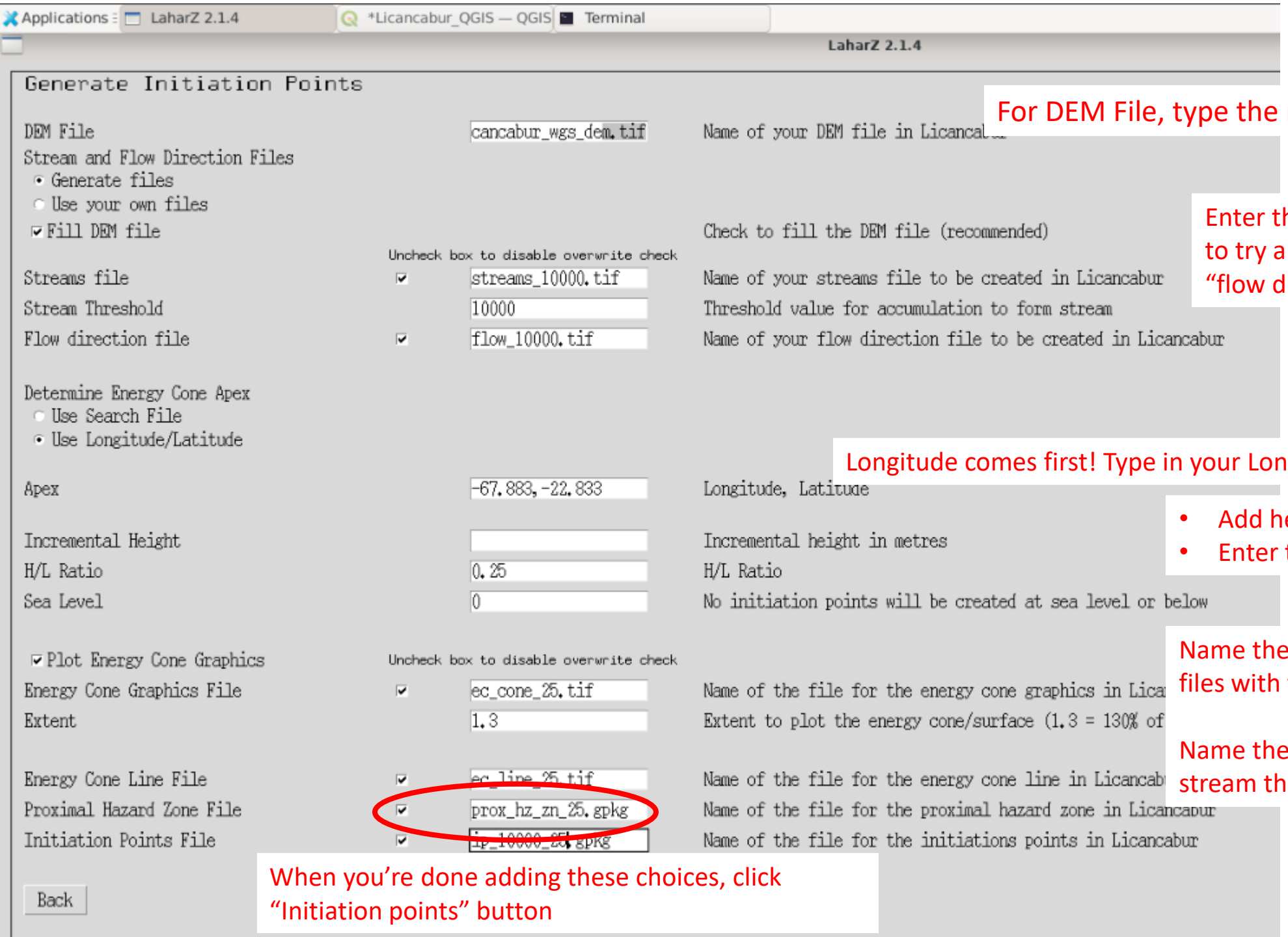


- **Now make a hillshade of the DEM**
- **Open the Raster Terrain Analysis →Hillshade tool**
- **Remember to change the Z factor to 0.00001 because the raster is in degrees lat/long**
- **Save the file in /LaharZ/my_volcano_folder as “my_volcano_hs.tif”**


Applications
*Licancabur_QGIS — QGIS
Tue 10 Jun, 23:31
Default user
File System
Home
QGIS
Paraview
laharz
Click on laharz, if you get an error message, choose “launch anyway”

Make a stream network, an H/L cone, and start points





For DEM File, type the name of your DEM (the tif)

Enter the stream threshold you decided to try and name the “streams file” and “flow direction” file using that number

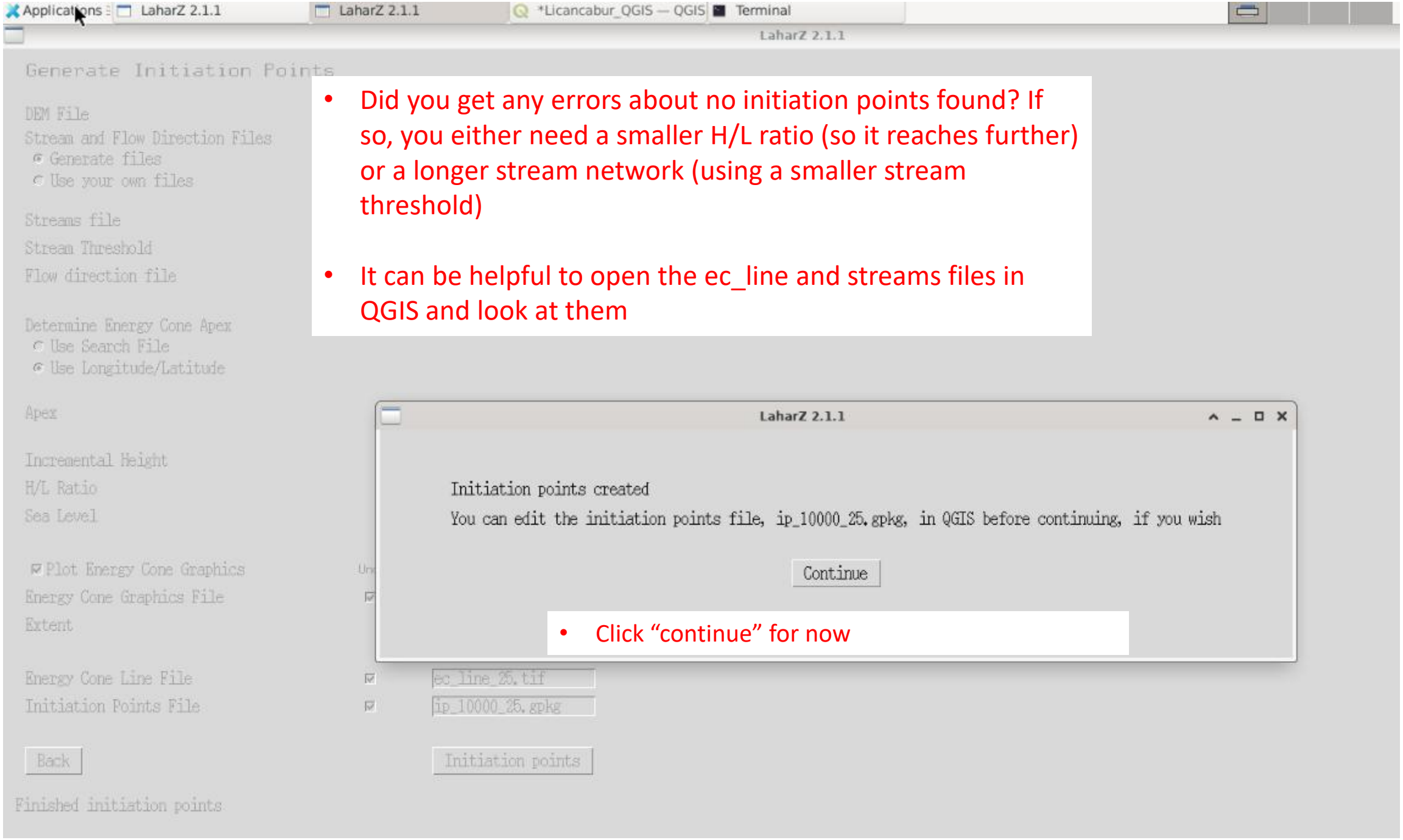
Longitude comes first! Type in your Longitude, Latitude

- Add height if you need to
- Enter the H/L ratio you decided to try

Name the energy cone and energy line files with the H/L ratio used

Name the initiation points with the stream threshold and H/L ratio you used

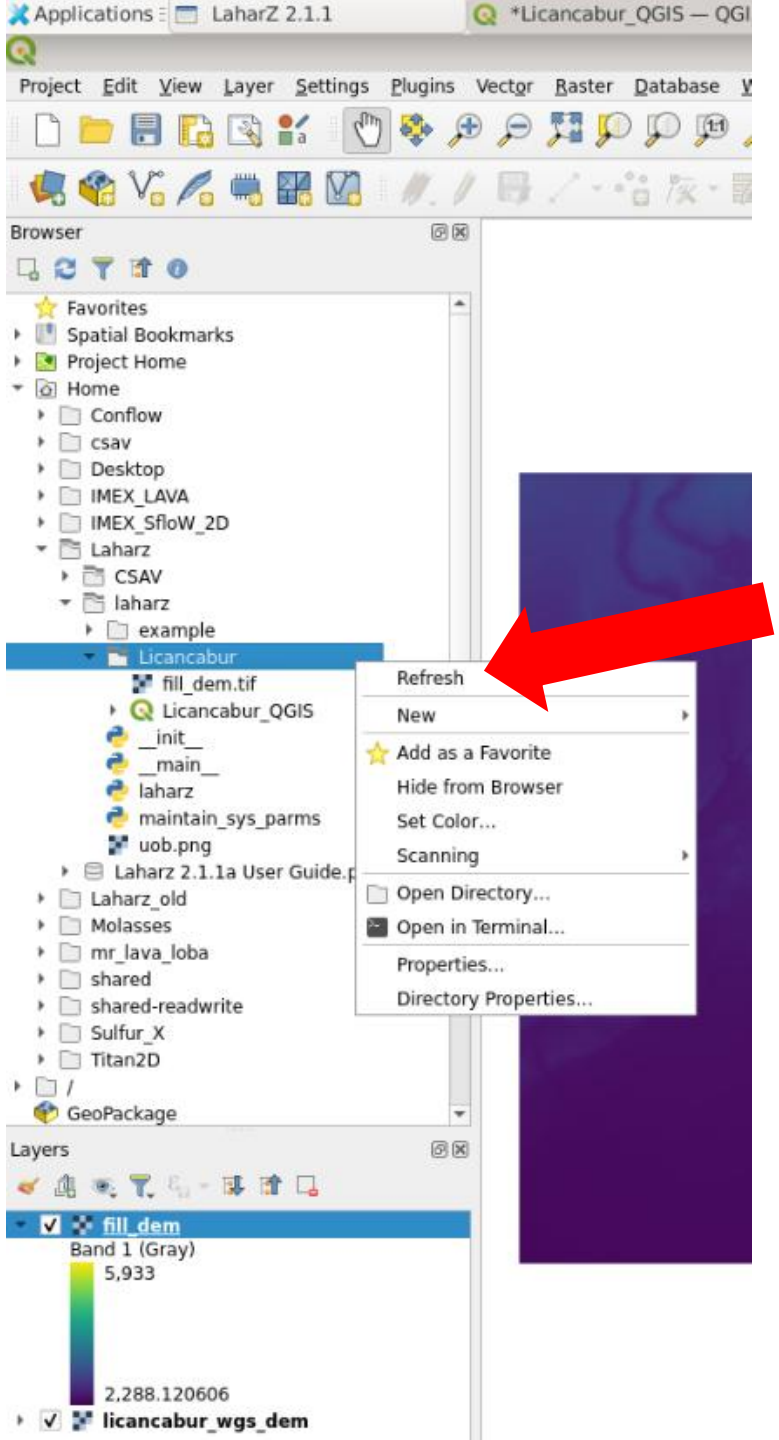
When you’re done adding these choices, click “Initiation points” button



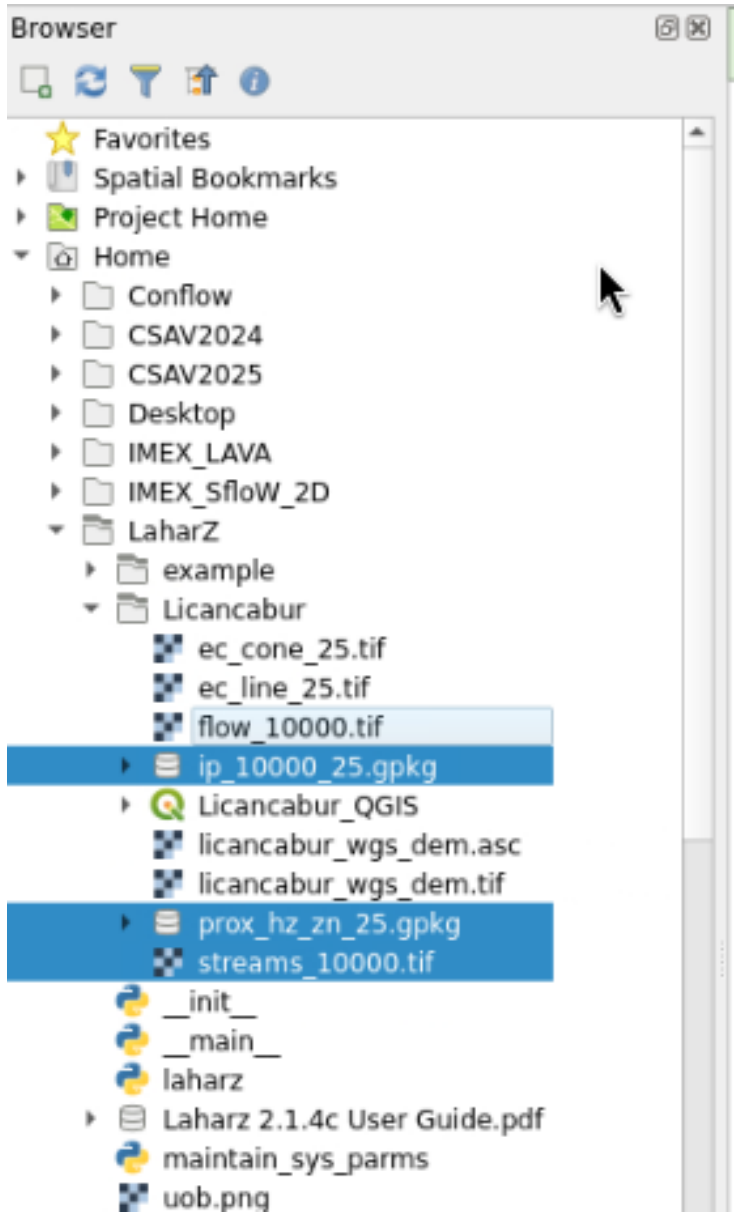
- Did you get any errors about no initiation points found? If so, you either need a smaller H/L ratio (so it reaches further) or a longer stream network (using a smaller stream threshold)

- It can be helpful to open the ec_line and streams files in QGIS and look at them

- Click “continue” for now



Run QGIS, right click on your volcano folder and click “Refresh” to see your new files



Look at the stream network, H/L cone (ec_line), and the initiation points you created!

Add these files to your map (for your new volcano):

streams_xx.tif

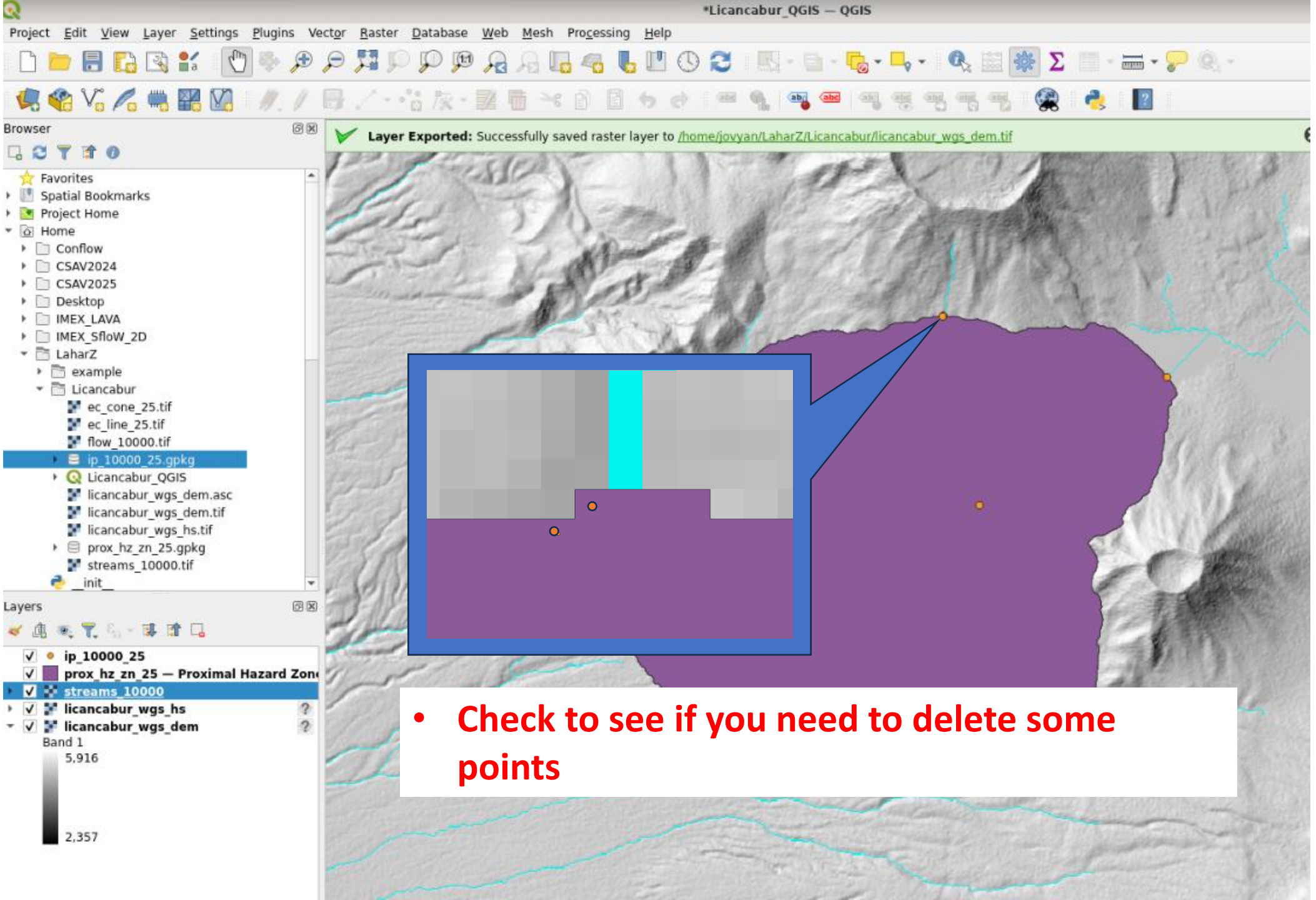
ec_line_xx.tif

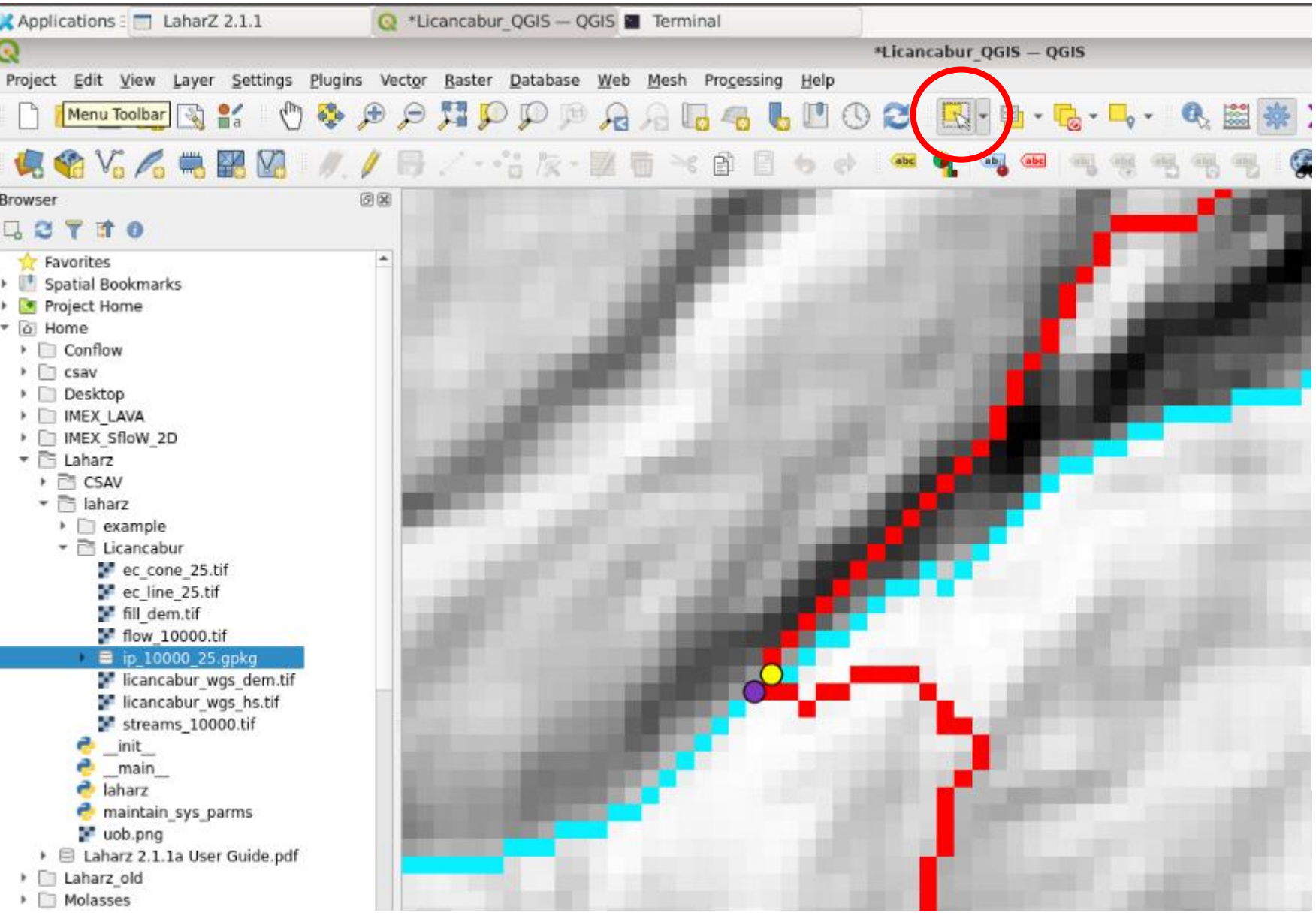
ip_10000_xx.gpkg

prox_hz_zn_xx.gpkg

Adjust if needed!

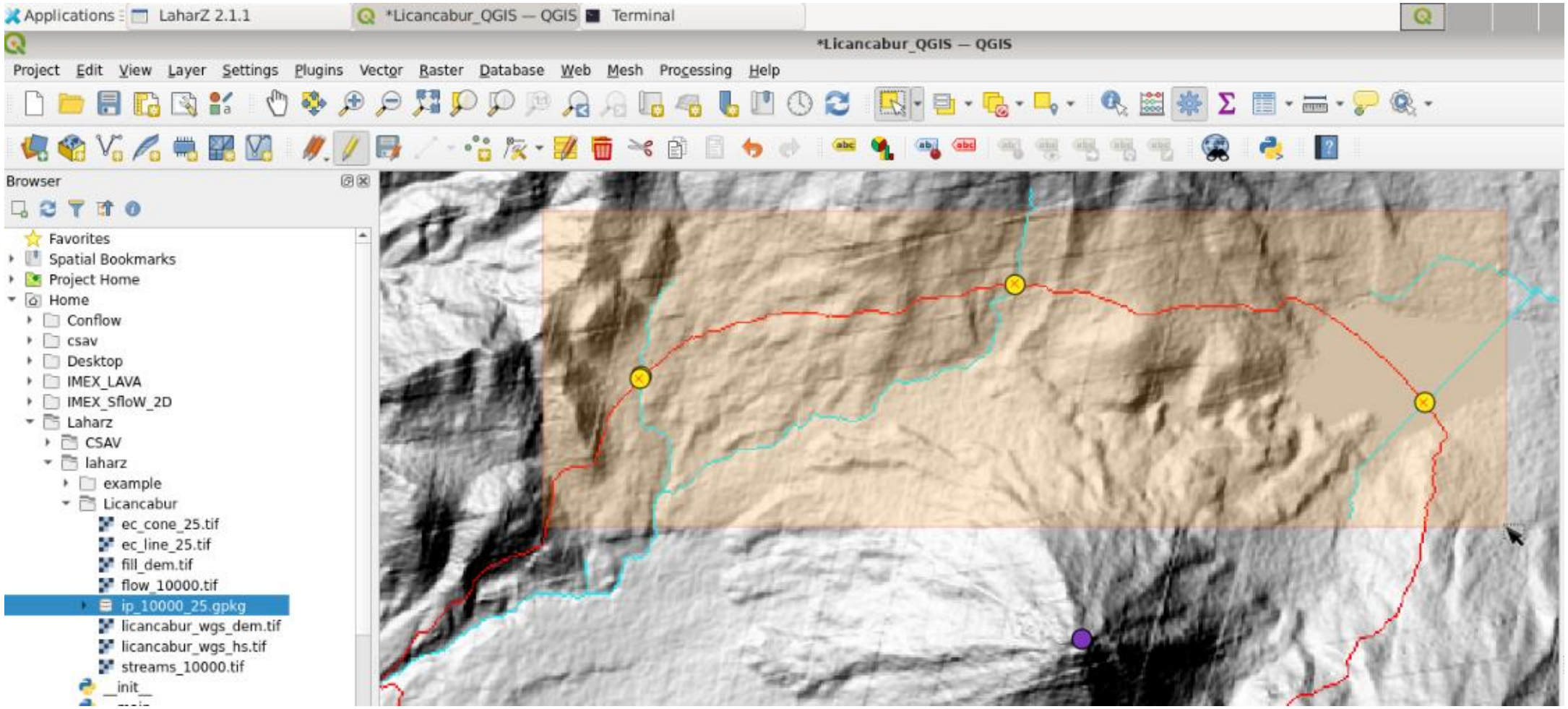
- Examine the H/L cone (ec_line), the stream network (streams), and the initiation points (ip)
- Did you get any errors about no initiation points found? If so, you either need a smaller H/L ratio (so it reaches further) or a longer stream network (using a smaller stream threshold)
- Compare the stream network to any topographical or geological maps you have
- Does the stream network look good?
  - Is it too short? If so, try a smaller stream threshold number
  - Is it too long with lots of crazy branches? Try a larger stream threshold number
- Are there too many initiation points? Are they clustered? Delete some!
- Ask me questions!


*Licancabur_QGIS — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Browser
Favorites
Spatial Bookmarks
Project Home
Home
Conflow
CSAV2024
CSAV2025
Desktop
IMEX_LAVA
IMEX_SfloW_2D
LaharZ
example
Licancabur
ec_cone_25.tif
ec_line_25.tif
flow_10000.tif
ip_10000_25.gpkg
Licancabur_QGIS
licancabur_wgs_dem.asc
licancabur_wgs_dem.tif
licancabur_wgs_hs.tif
prox_hz_zn_25.gpkg
streams_10000.tif
__init__
Layers
ip_10000_25
prox_hz_zn_25 — Proximal Hazard Zon
streams_10000
licancabur_wgs_hs
licancabur_wgs_dem
Band 1
5,916
2,357
Layer Exported: Successfully saved raster layer to /home/jovyan/LaharZ/Licancabur/licancabur_wgs_dem.tif
• Check to see if you need to delete some points



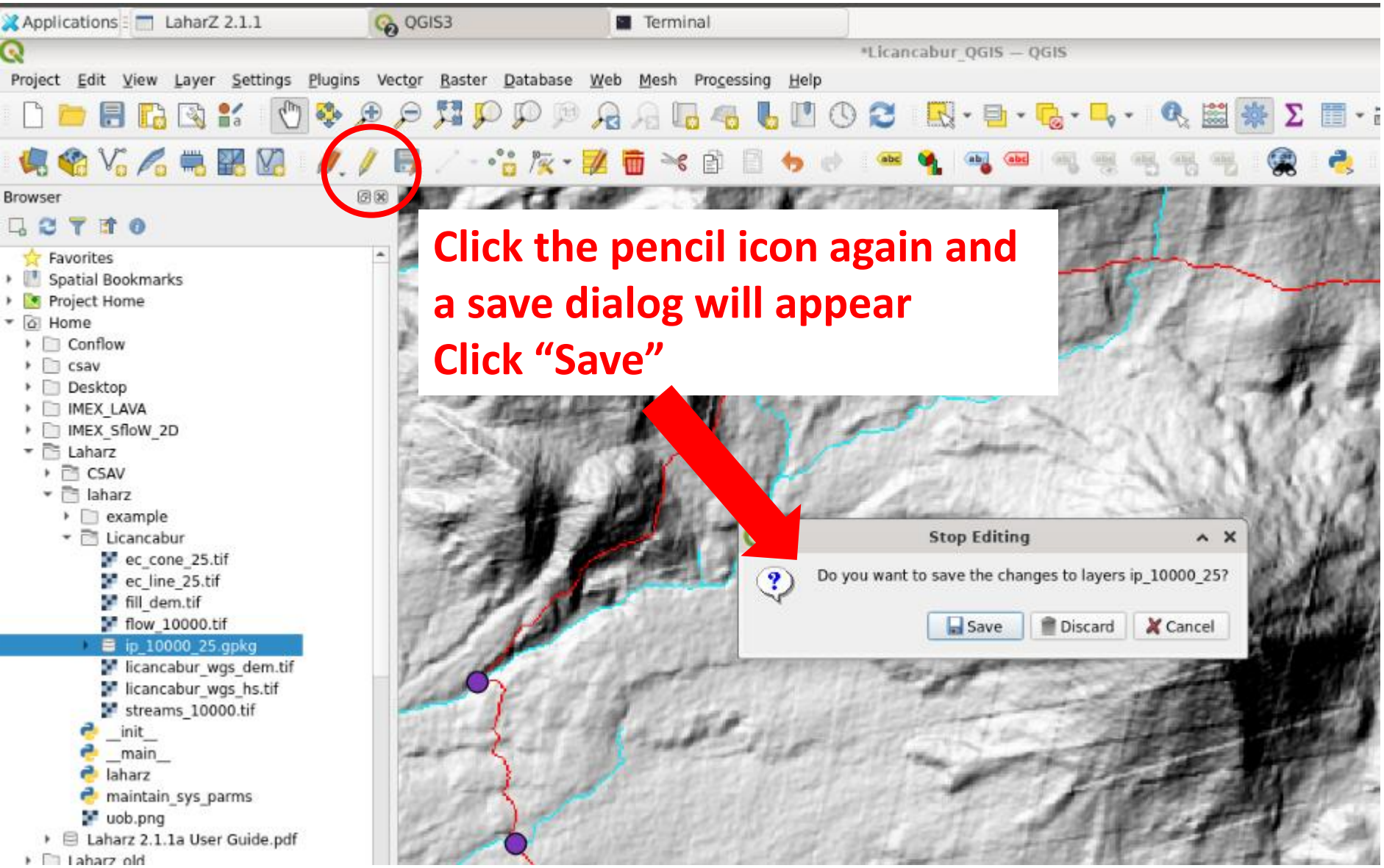
To delete some points

- Turn on the “Select features by area” tool and zoom in to each point cluster
- Click on the point you want to delete
- It will turn yellow
- The press the “delete” key on your keyboard

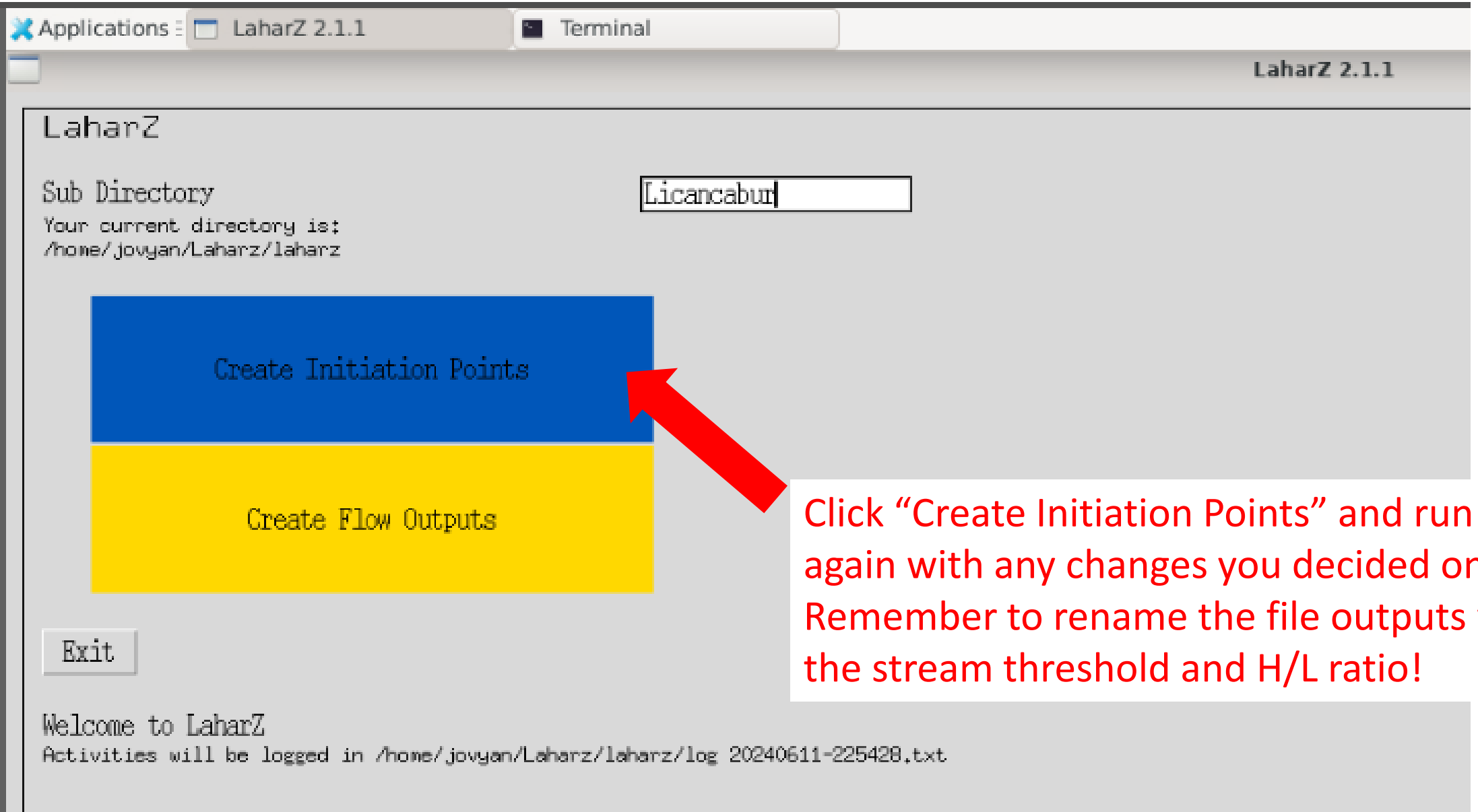


To delete some points

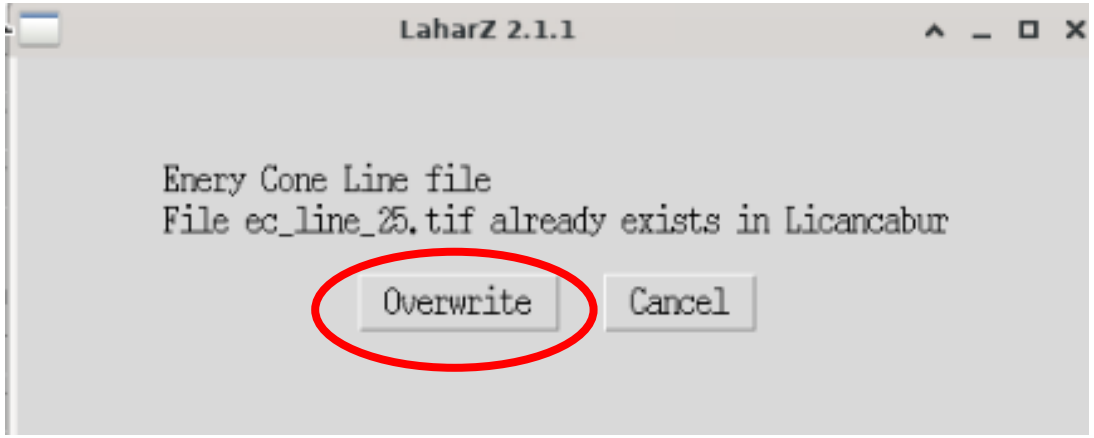
- You can also use the “Select features by area” tool to drag a selection box around multiple points
- They will turn yellow
- The press the “delete” key on your keyboard


Applications
LaharZ 2.1.1
QGIS3
Terminal
*Licancabur_QGIS — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Browser
Favorites
Spatial Bookmarks
Project Home
Home
Conflow
csav
Desktop
IMEX_LAVA
IMEX_SfloW_2D
Laharz
CSAV
laharz
example
Licancabur
ec_cone_25.tif
ec_line_25.tif
fill_dem.tif
flow_10000.tif
ip_10000_25.gpkg
licancabur_wgs_dem.tif
licancabur_wgs_hs.tif
streams_10000.tif
__init__
__main__
laharz
maintain_sys_parms
uob.png
Laharz 2.1.1a User Guide.pdf
Laharz_old
Click the pencil icon again and
a save dialog will appear
Click “Save”
Stop Editing
Do you want to save the changes to layers ip_10000_25?
Save
Discard
Cancel

If you aren't happy with the first try, you can create new start points.

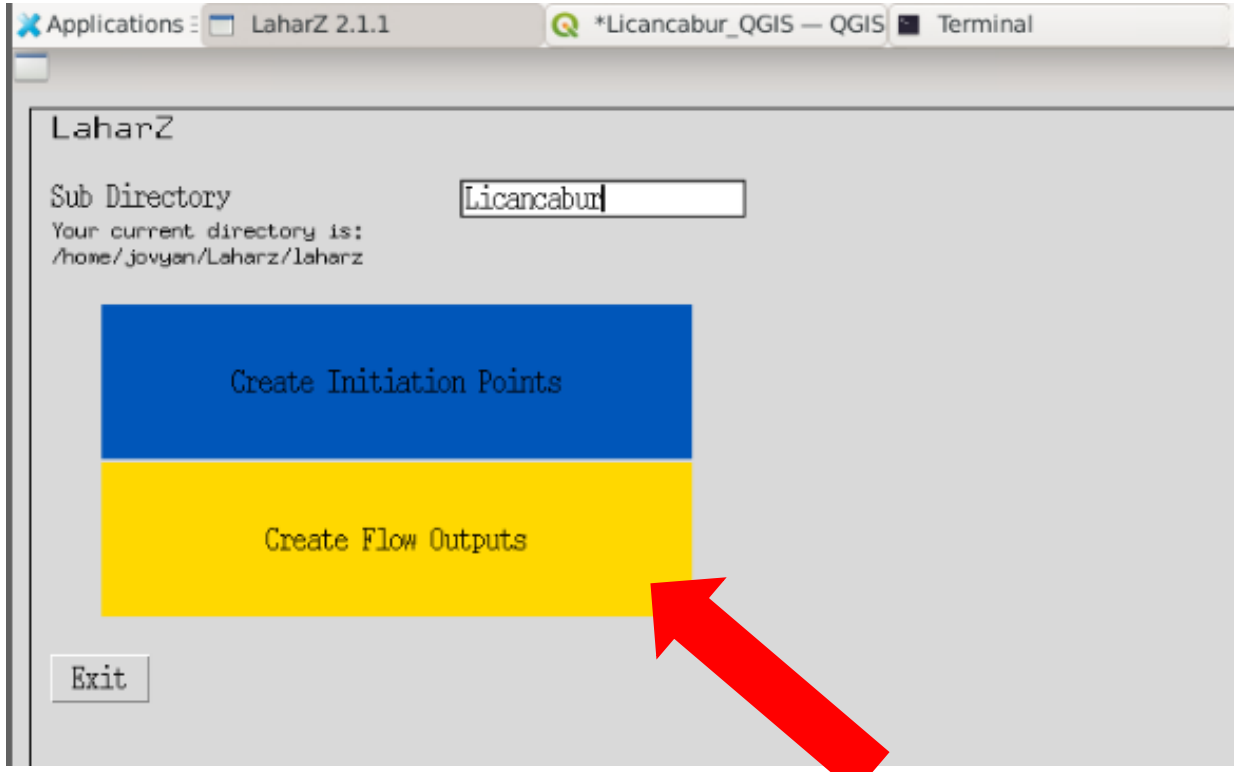


Click "Create Initiation Points" and run it again with any changes you decided on. Remember to rename the file outputs with the stream threshold and H/L ratio!

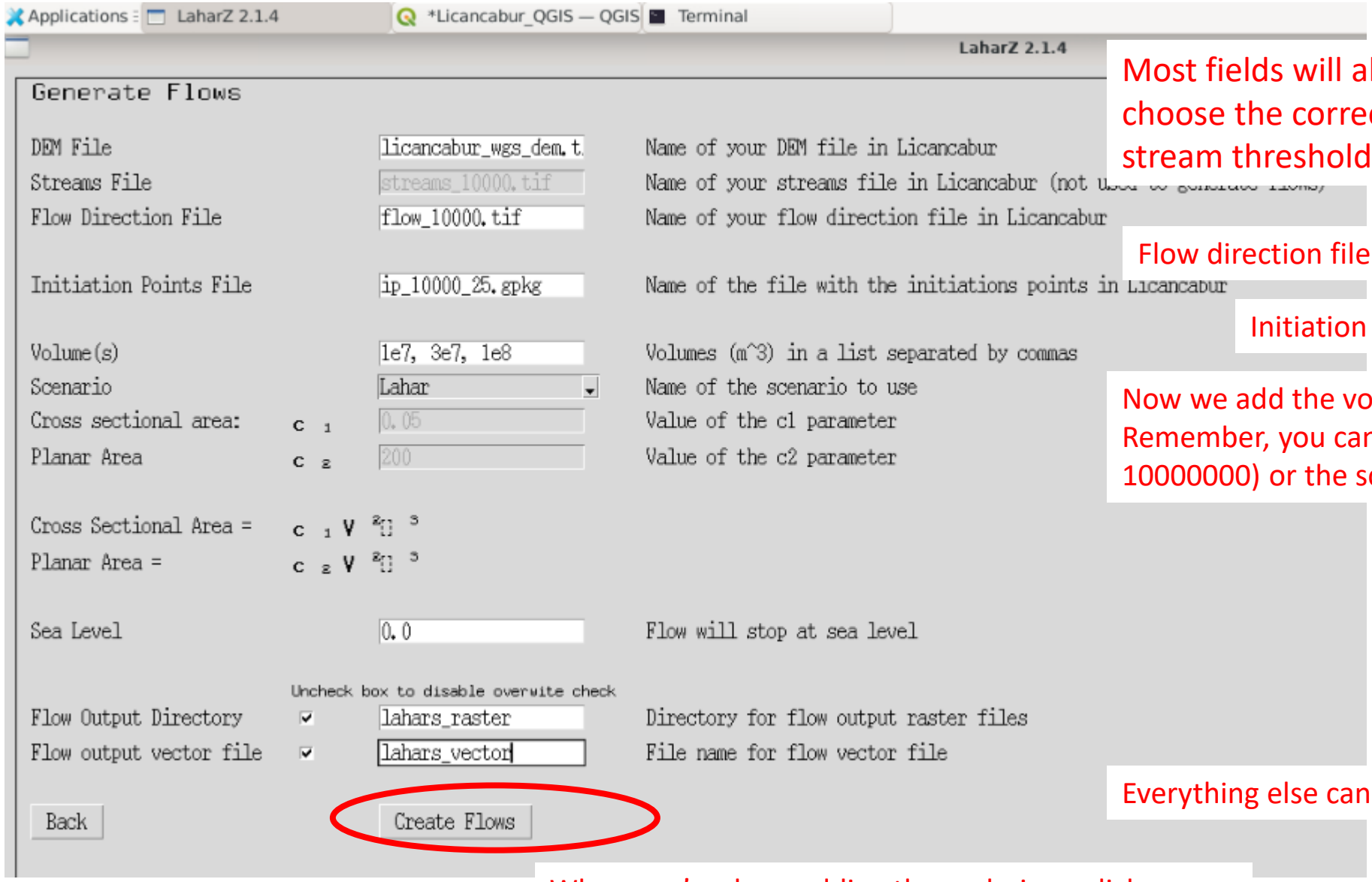


If you are running it again, choose “Overwrite” in the dialog boxes that pop up

Run LAHARZ!



Click “Create Flow Outputs”



Most fields will already be filled in, but you need to choose the correct files that correspond to the stream threshold and H/L ratio you decided on (x)

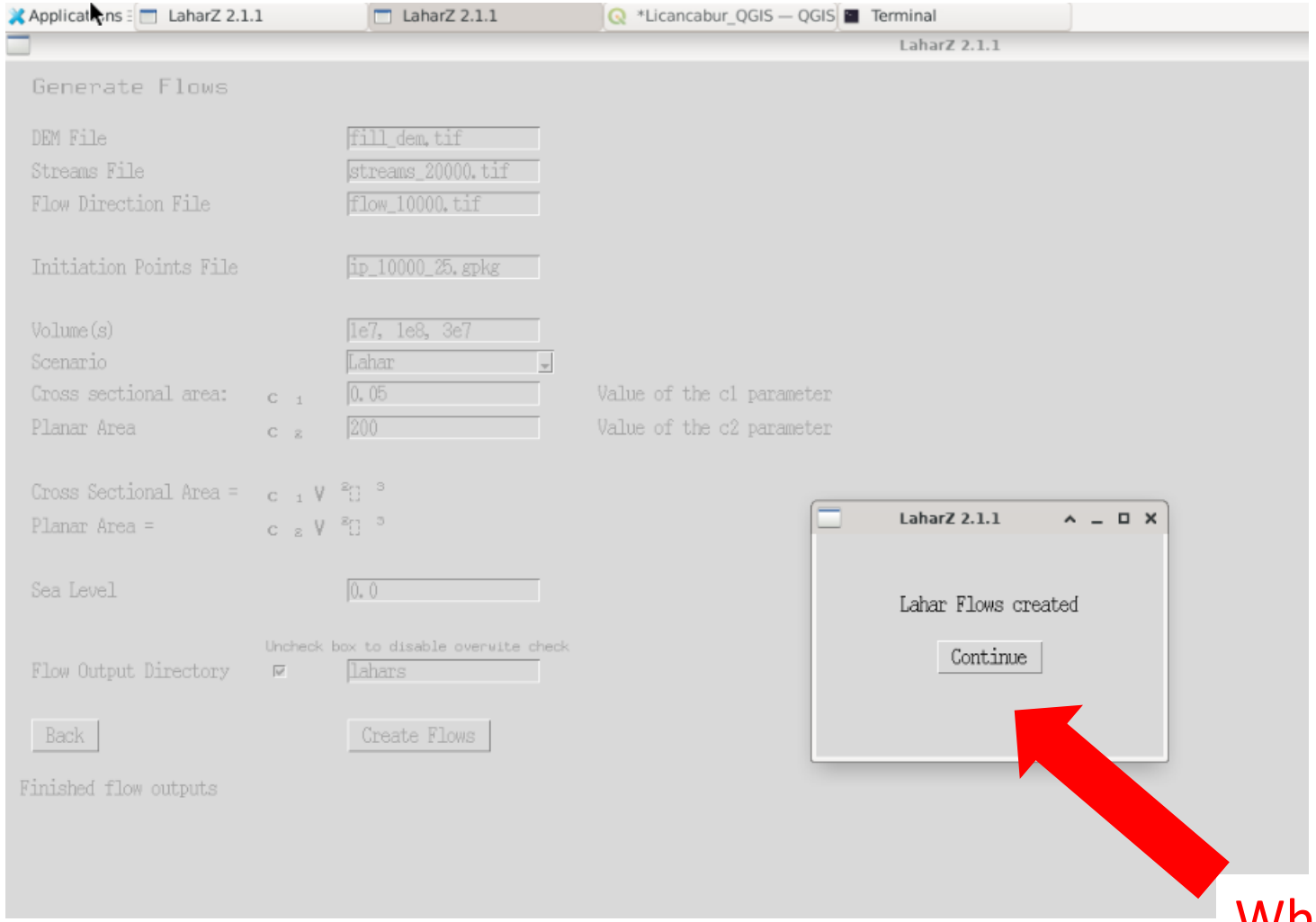
Flow direction file: flow_x.tif

Initiation points file: ip_x_x.gpkg

Now we add the volumes you decided on
Remember, you can type out the whole number (e.g., 10000000) or the scientific notation version (e.g., 1e7)

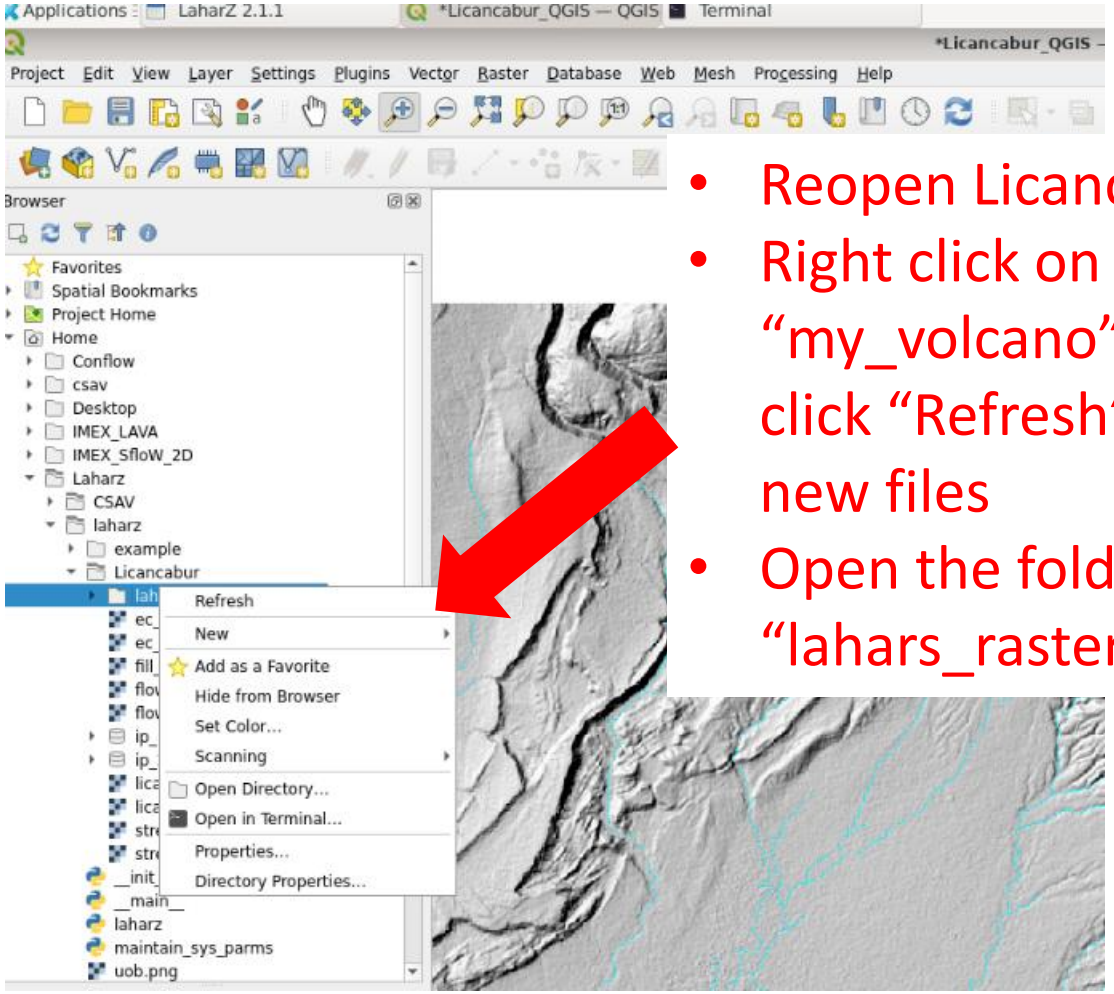
Everything else can stay as the default

When you're done adding these choices, click "Create flows" button

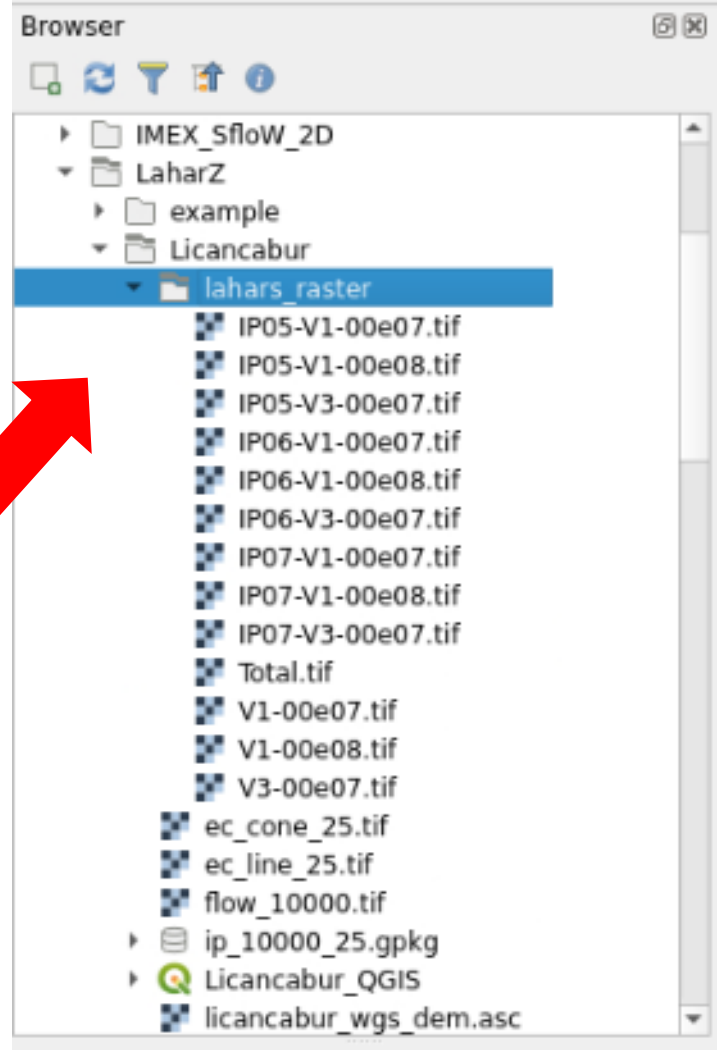


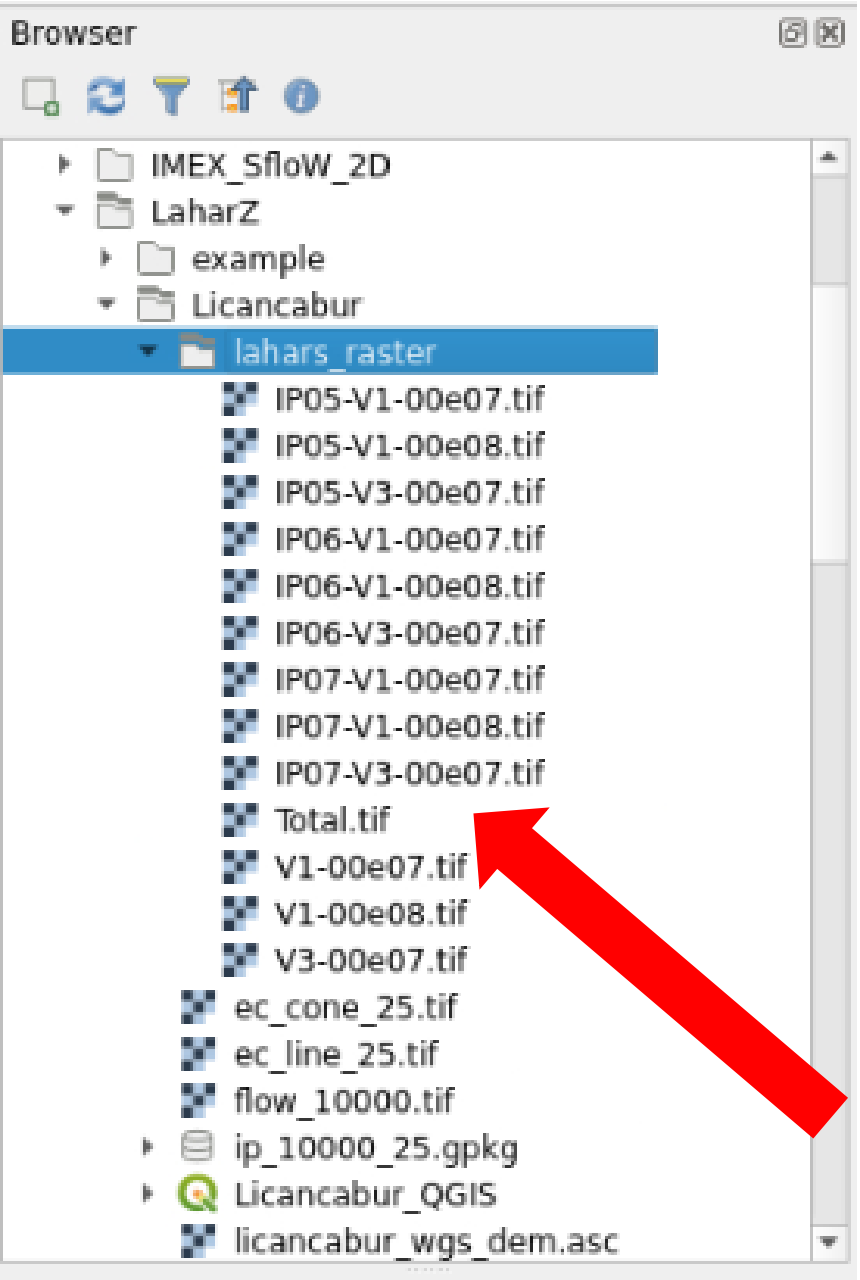
When it’s done, click “Continue”

Look at the new files.



- Reopen Licancabur_QGIS
- Right click on the “my_volcano” folder and click “Refresh” to see your new files
- Open the folder called “lahars_raster”





- LAHARZ made a bunch of different files
- There is a separate file for each volume/initiation point combination (these start with “IP”
- LAHARZ also made a file for each volume, merging all the flows from each initiation point (these start with “V”)
- And finally, there is a “Total” file that has all the volumes from all the initiation points in one file
- Add the “total.tif” file to your map

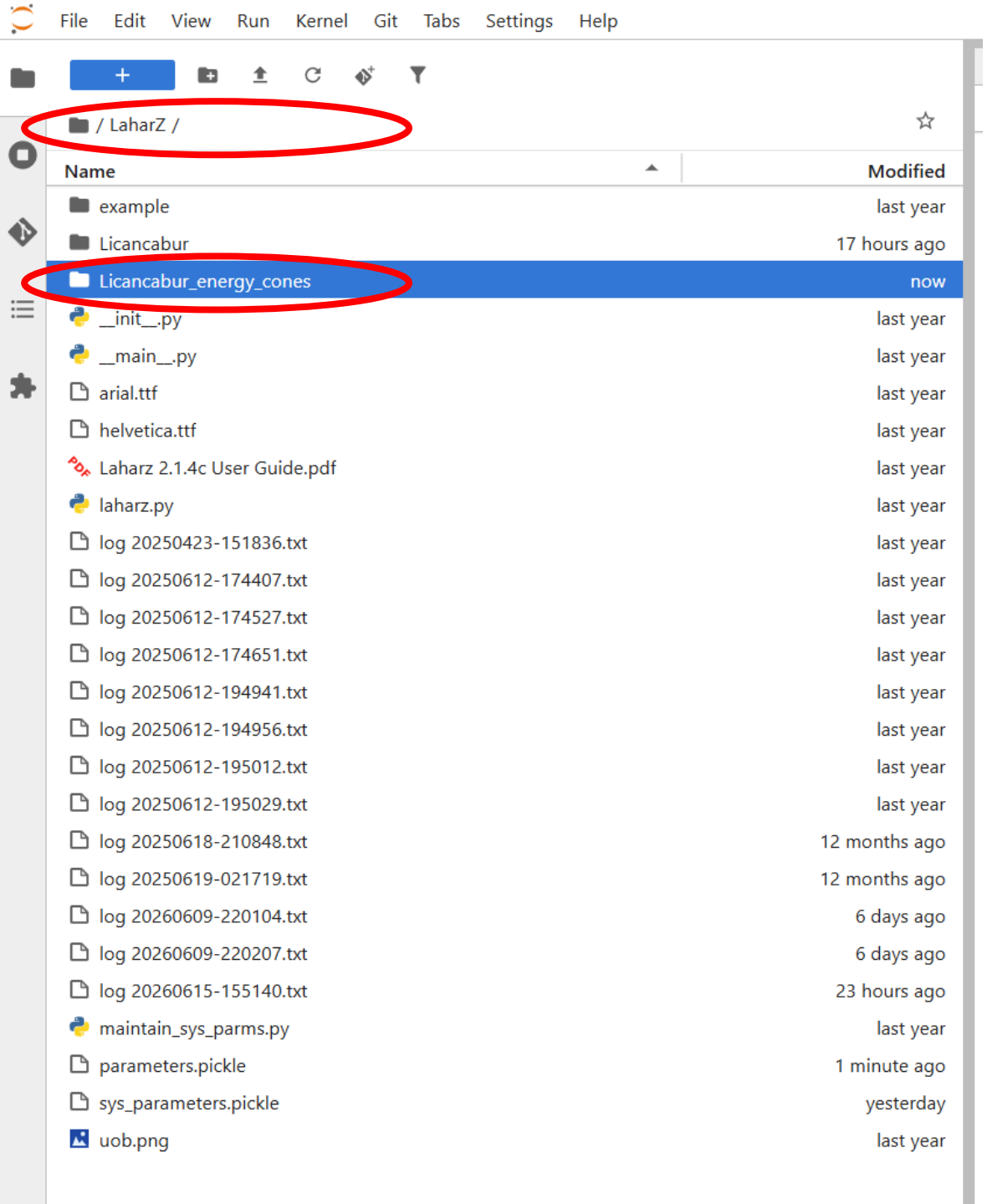
- There’s probably no reason to work with the colors or map layout until you have the rest of your layers (PDCs, ballistics)!
- Move on to modeling PDCs

PDC modeling

- Are you going to make energy cones?
- Are you going to use TITAN2D?

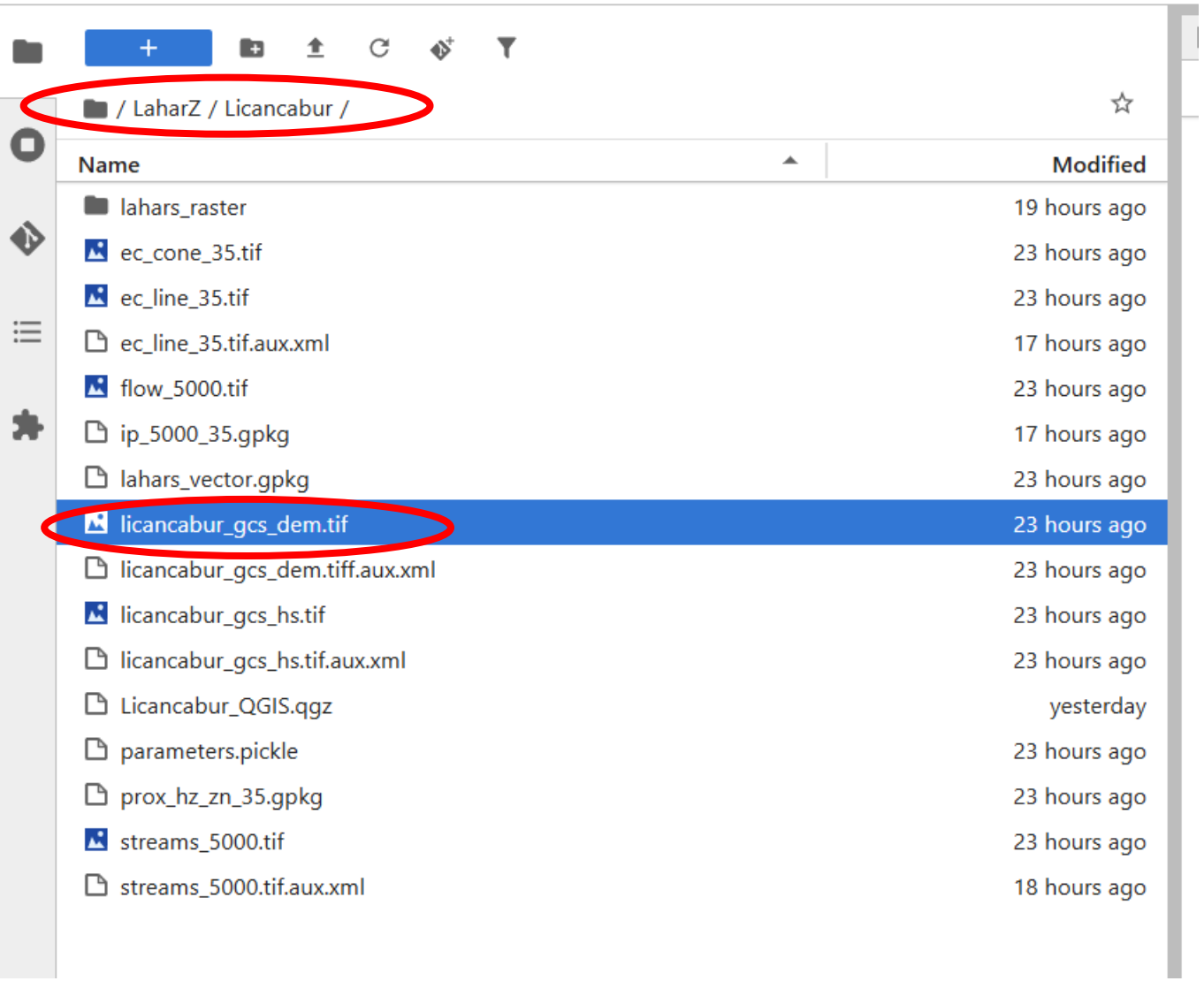
Depending on your strategy, follow along with the slides that apply to you!

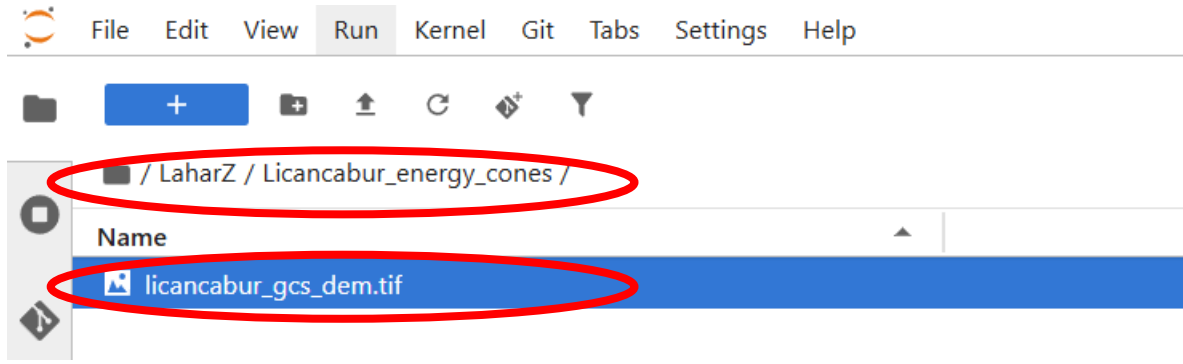
PDC modeling using energy cones



- Use LaharZ to make some more energy cones, but this time, use ΔH/L values from FlowDat for different PDC volumes
- First, go back to your LaharZ folder
- Make a new folder inside called “yourvolcano_energy_cones”

- Copy the DEM you used for modeling lahars into this new folder (for your volcano, not Licancabur)





ΔH/L ranges for dome- and column-collapse PDC energy cones from FlowDat regressions

- Use the regression equations or the table to decide on H/L values for the volumes you are going to model and the type of PDCs you are modeling

Volume (km^3)	Volume (10^6 x m^3)	ΔH/L best fit Dome collapse	ΔH/L best fit Dome collapse (degrees)	ΔH/L best fit Column collapse	ΔH/L best fit Column collapse (degrees)
0.0001	0.1	0.39	21.1	0.36	19.8
0.001	1	0.29	16.1	0.30	16.5
0.01	10	0.22	12.1	0.24	13.7
0.1	100	0.16	9.1	0.20	11.4

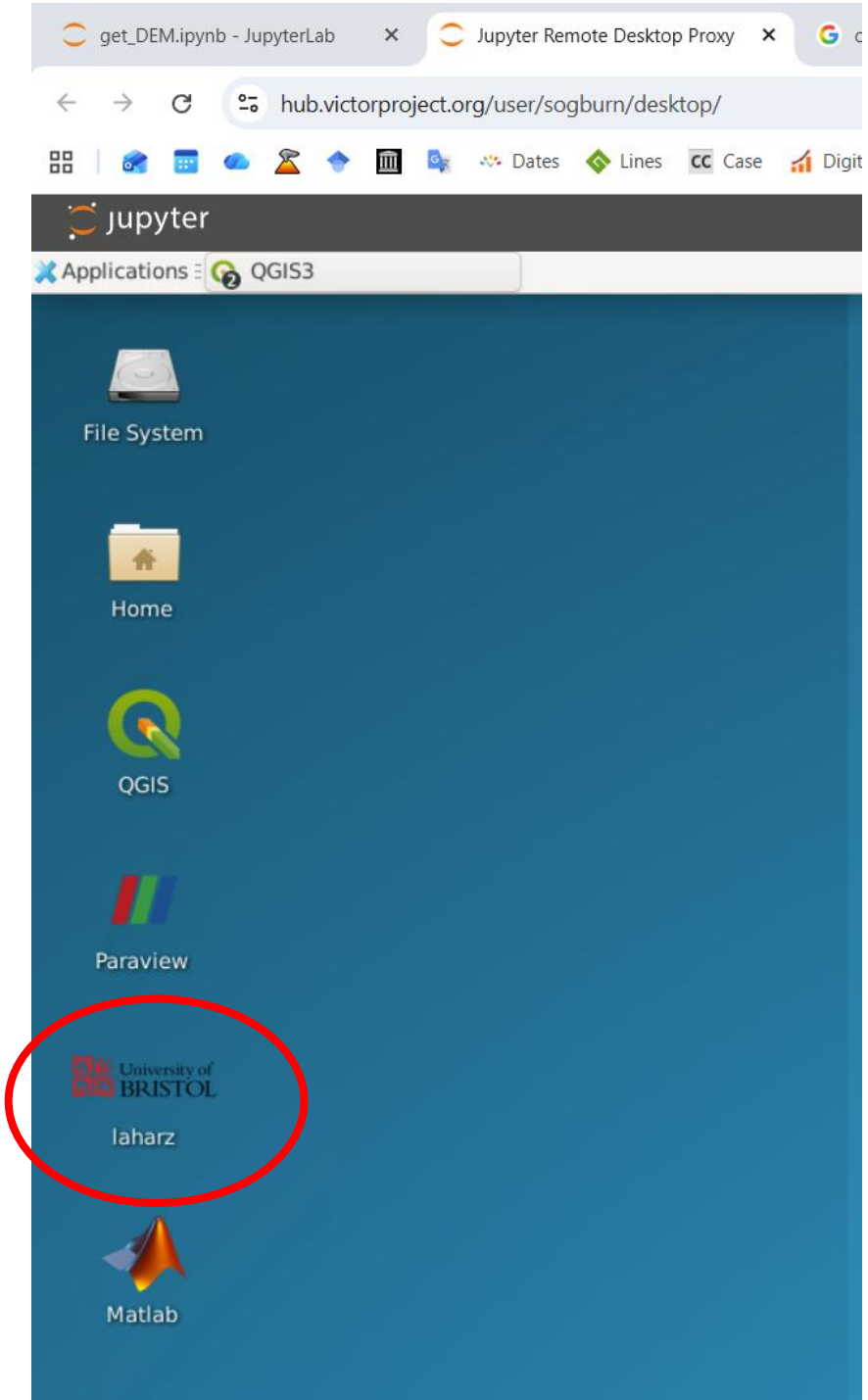
Dome-collapse PDCs (based on 121 flows): $\Delta H/L = 0.120(V^{-0.127})$

Column-collapse PDCs (based on 40 flows): $\Delta H/L = 0.166(V^{-0.084})$

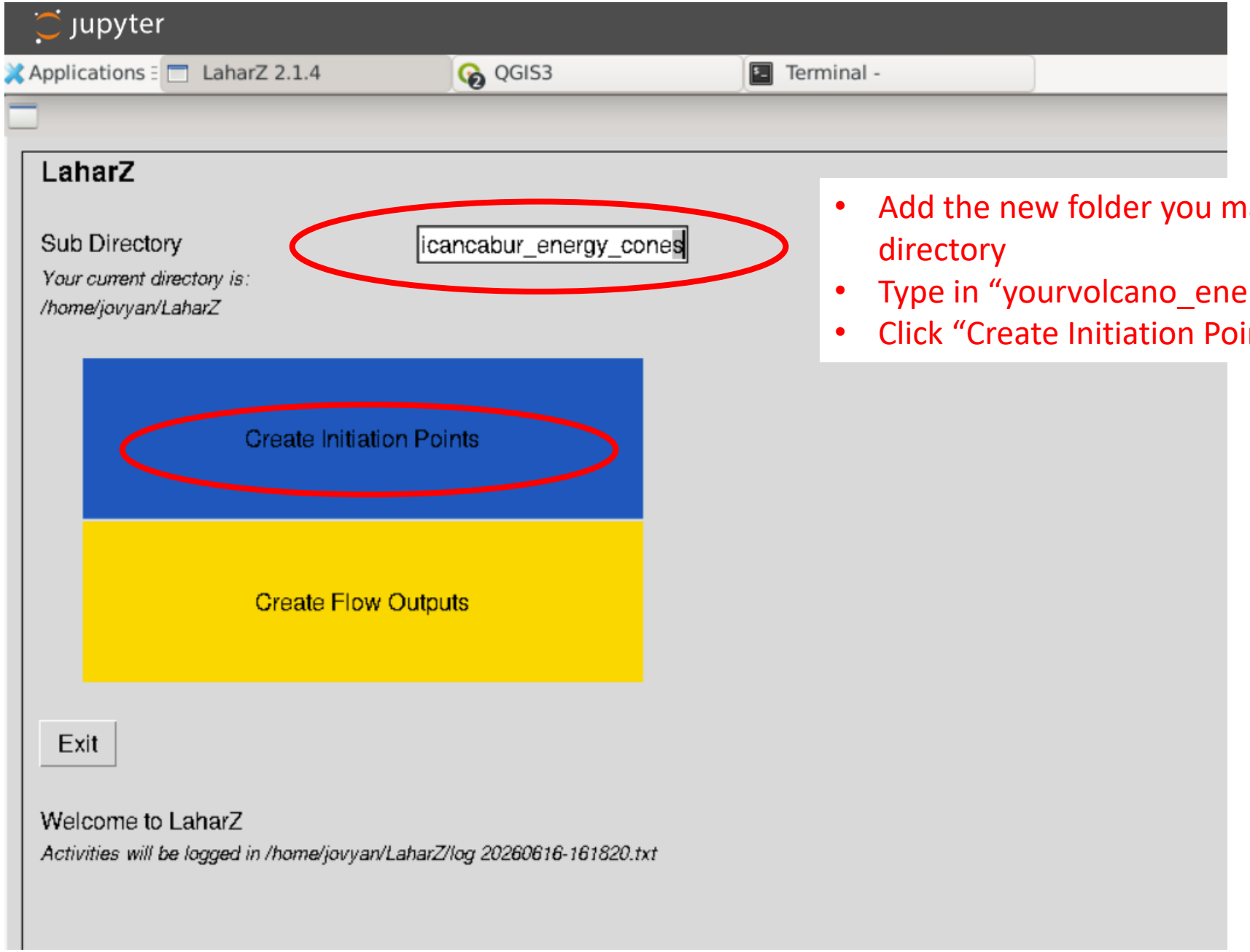
All PDCs (based on 188 flows): $\Delta H/L = 0.165(V^{-0.085})$

VDAP

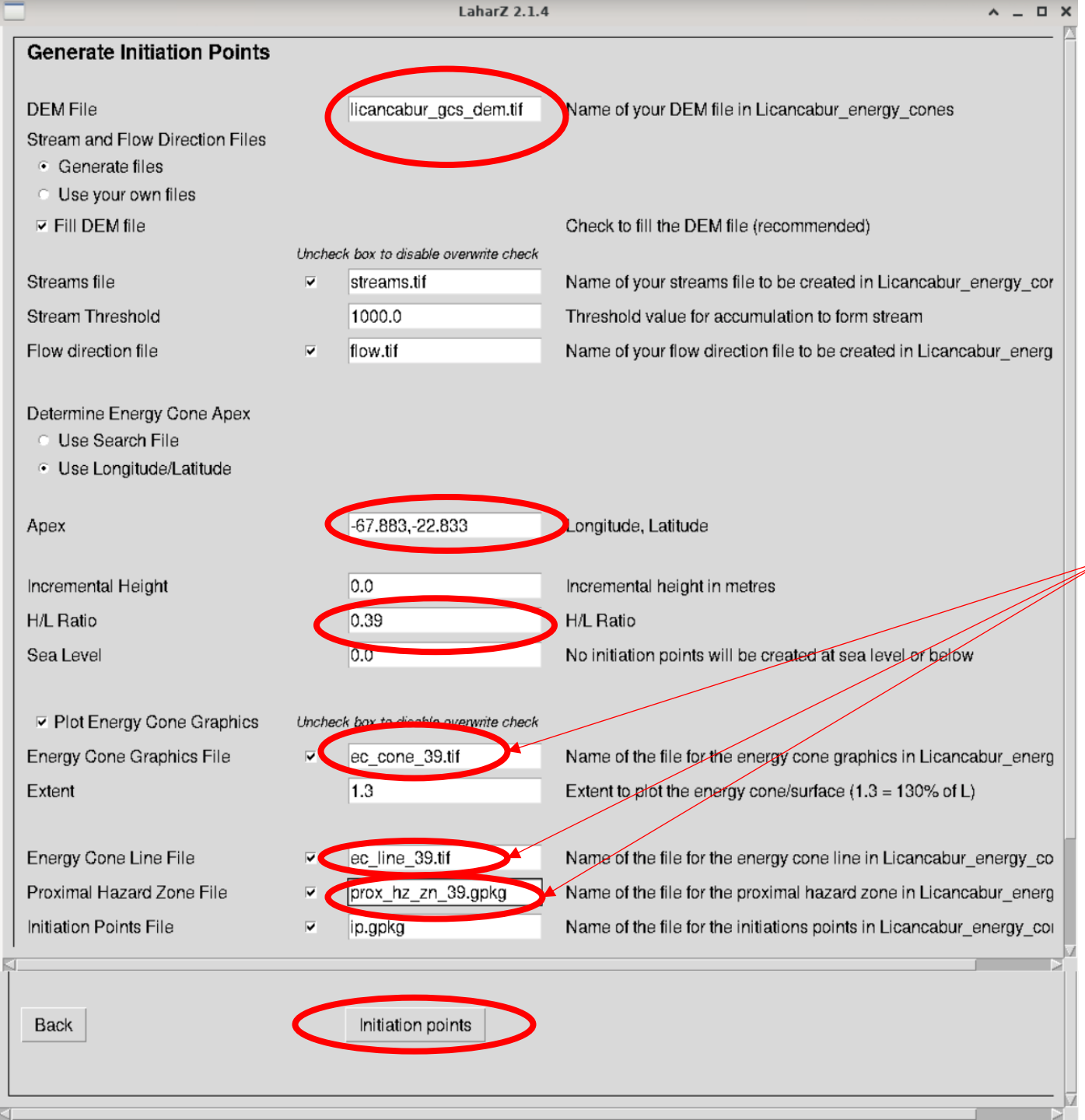
USGS



- Go back to your desktop and open laharz again



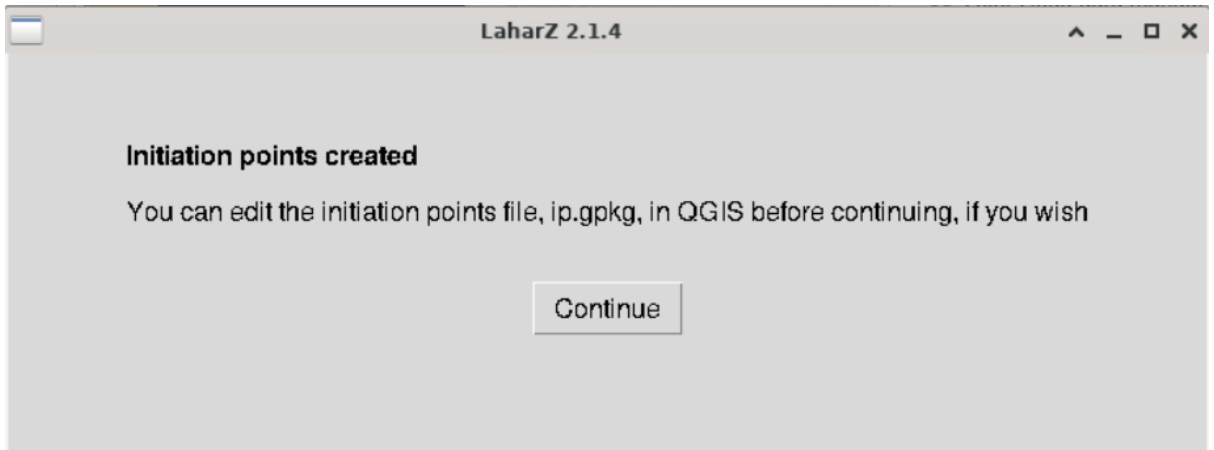
- Add the new folder you made for your volcano as the directory
- Type in “yourvolcano_energy_cones” as the folder
- Click “Create Initiation Points”



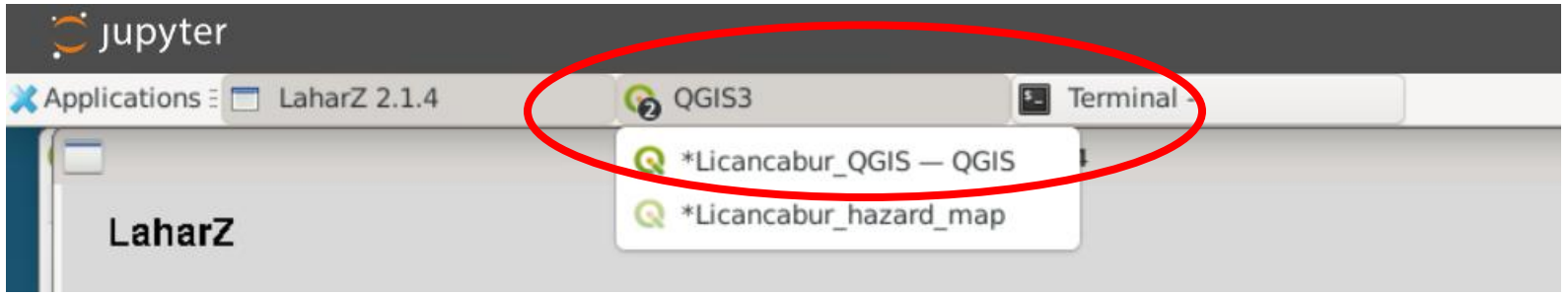
- For modeling PDCs, we don't care about the streams or the initiation points, so you can leave all those as default values

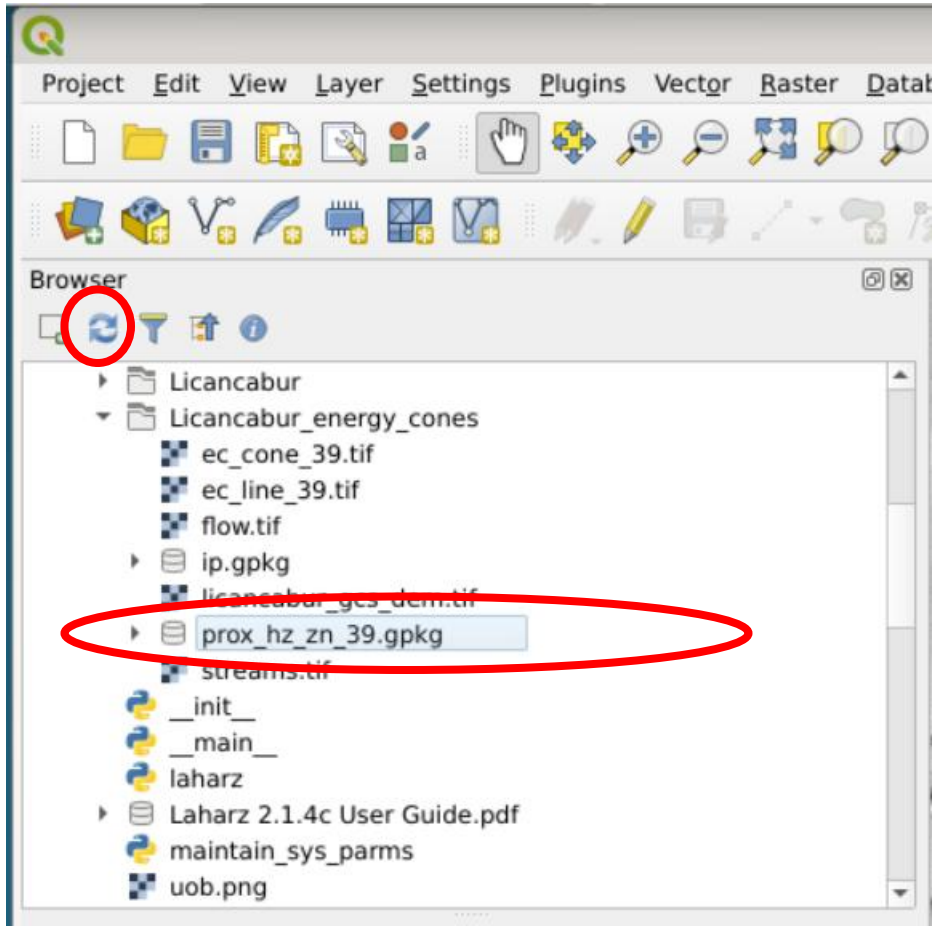
- Change the DEM File to the DEM you downloaded "yourvolcano_gcs_dem.tif
- Add the Long/Lat coordinates of the summit
- Change the ΔH/L ratio to the FlowDat value for the volume you are modeling (for energy cones it will be a decimal)
- Change all the energy cone files to include the value you are using

- Click "Initiation points" at the bottom

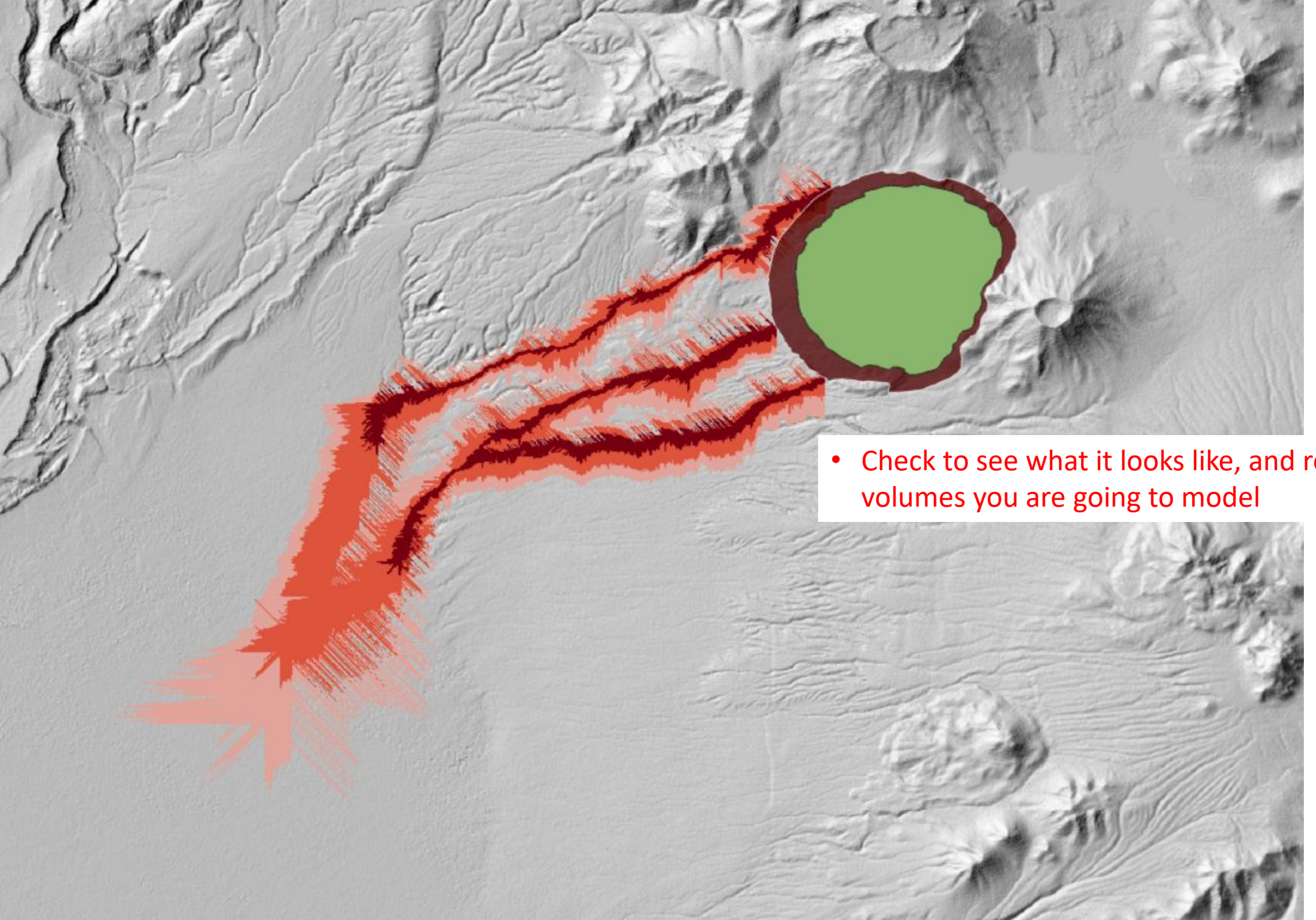


When it’s done, click continue





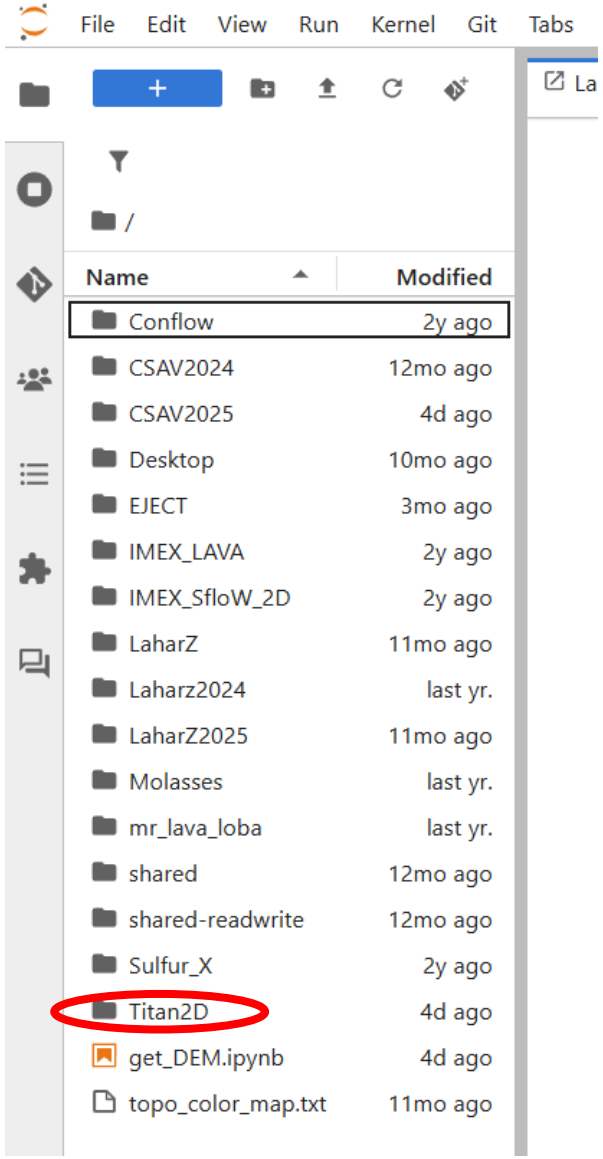
- Let's go look at the results!
- Navigate to QGIS
- Refresh the files if needed
- Add the "prox_hx_zn_39.gpkg" file to the map



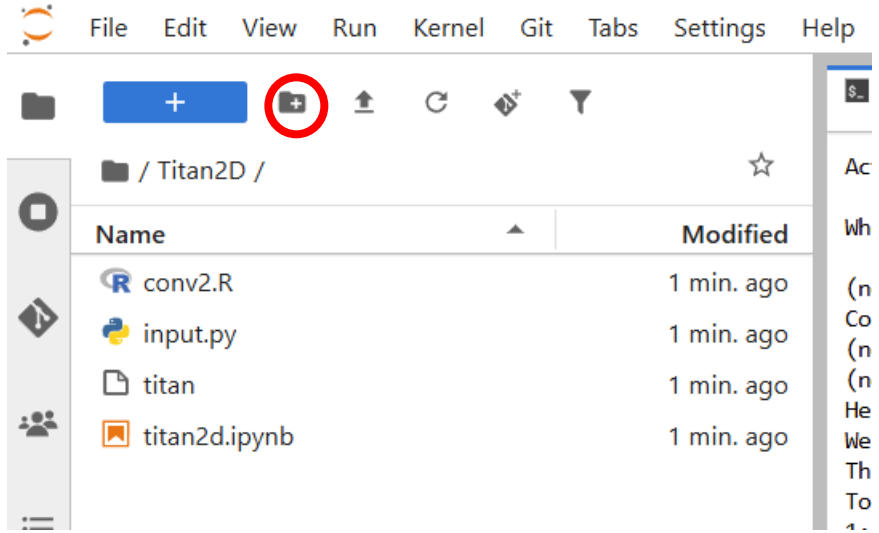
- Check to see what it looks like, and repeat for all volumes you are going to model

PDC modeling using TITAN2D

Make a folder to work in!

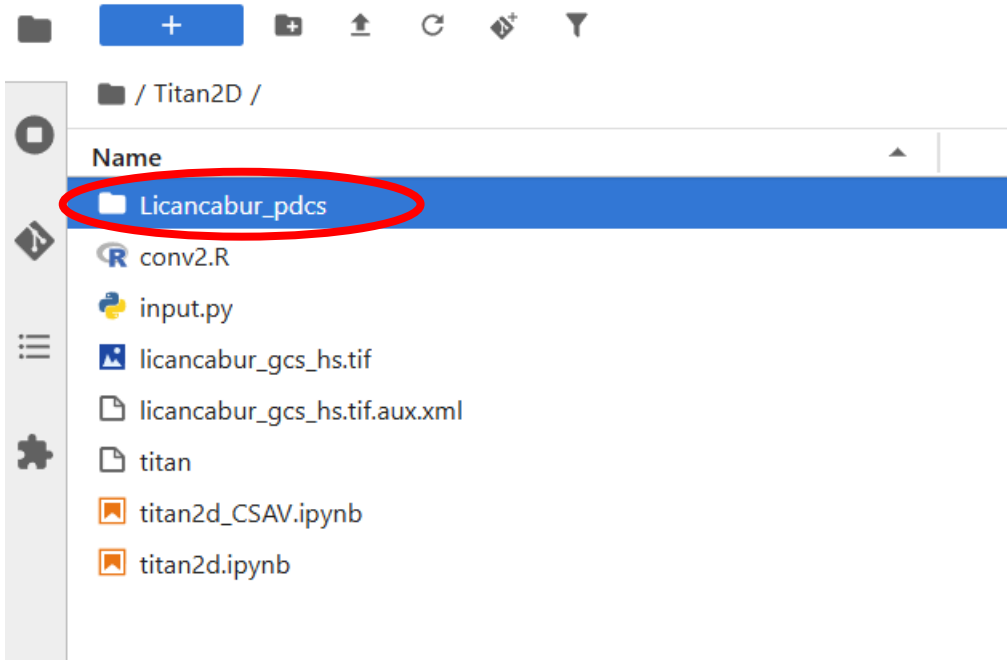


Double click on “Titan2D”

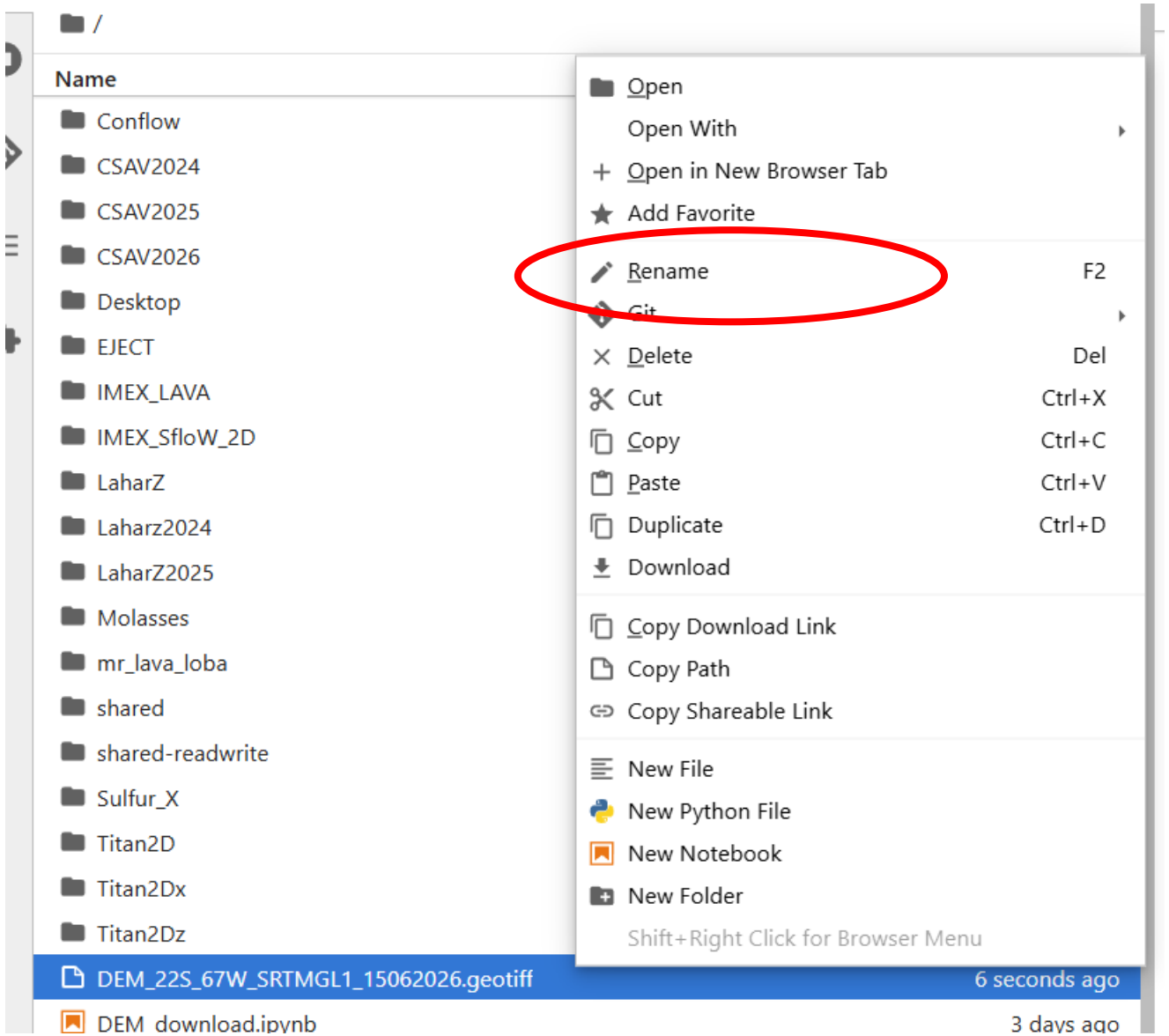


Right click in a blank space and choose “New Folder”

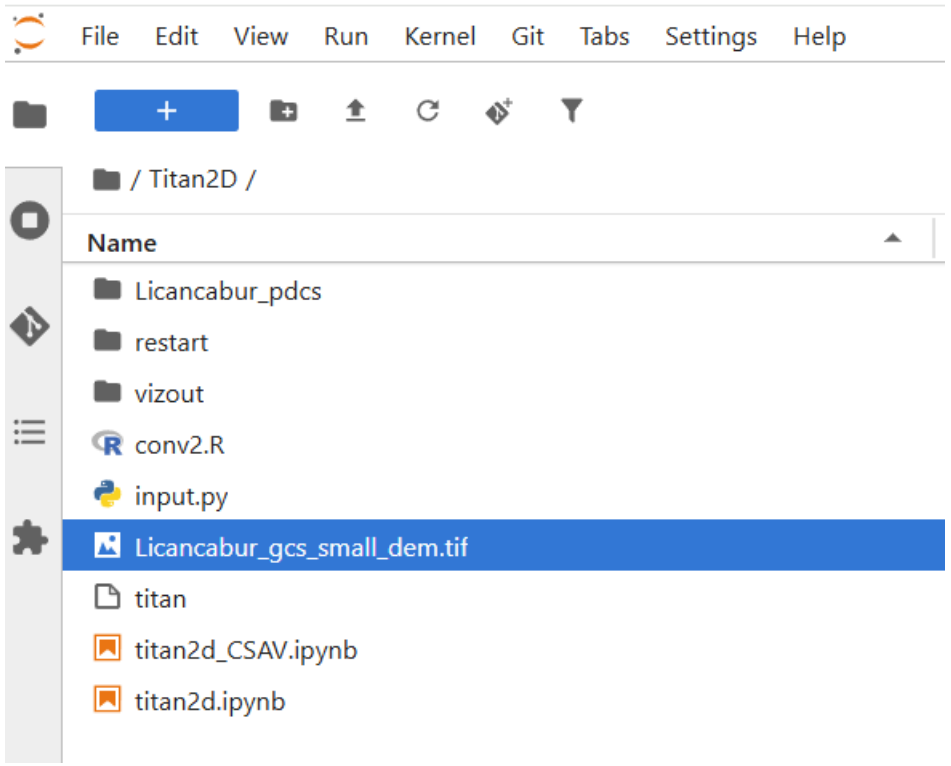
Or click the “New folder“ icon at the top



Name the new folder “yourvolcano_pdcs”

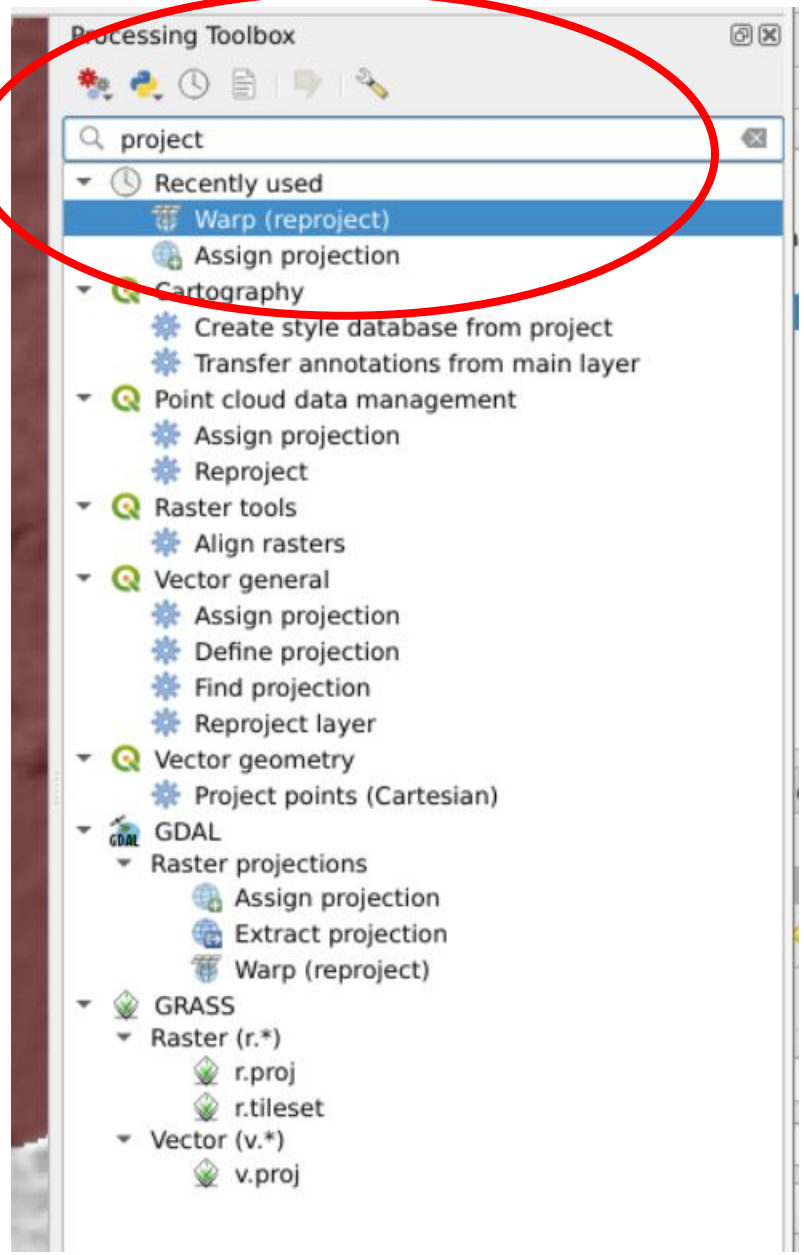


- Rename the file to:
- yourvolcano_gcs_small_dem.tif
- Move the file into your Titan2D folder

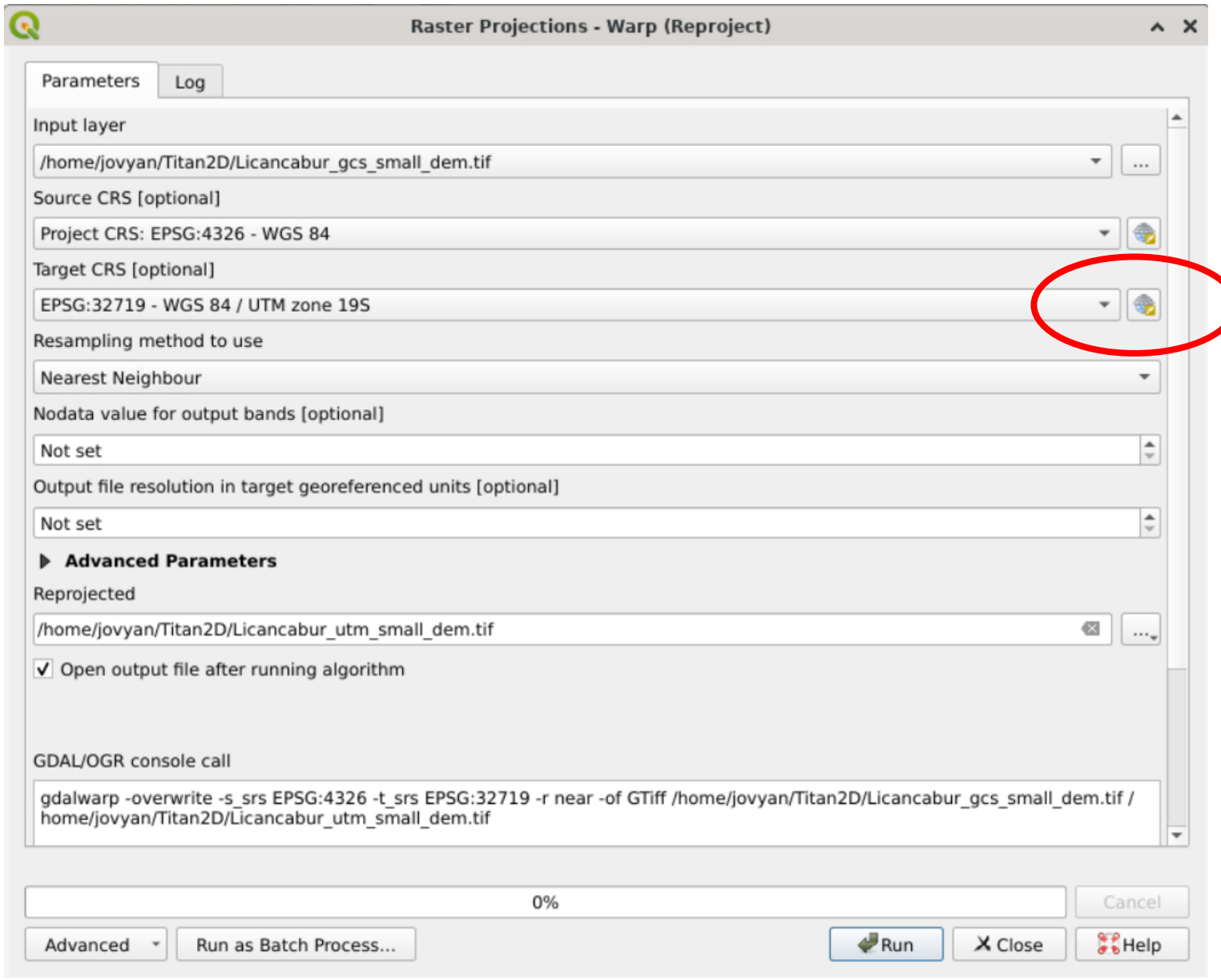


Projecting the DEM to UTM

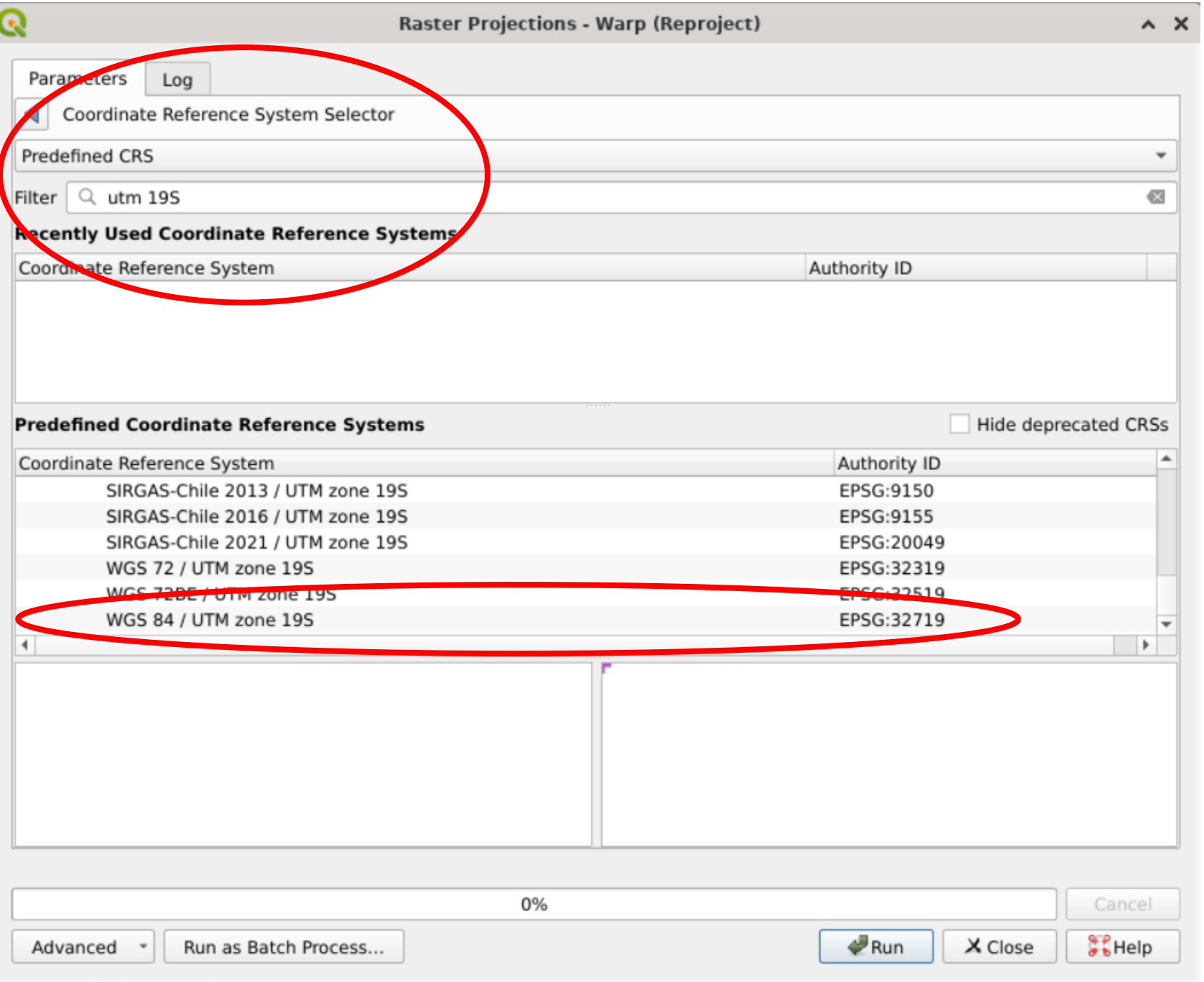
- The DEM we downloaded for LAHARZ was very, very big (80 km square)
- Using a DEM that big will slow down TITAN
- If you didn't already, get a smaller DEM using "get_DEM.ipynb" again
- The DEM that you used for LAHARZ was in geographic coordinate system (lat/long)
- The smaller DEM is also in GCS (lat/long)
- Titan2D uses UTM, so you need to use QGIS to project the DEM
- First, find the UTM zone for your volcano
- Go back to your QGIS project



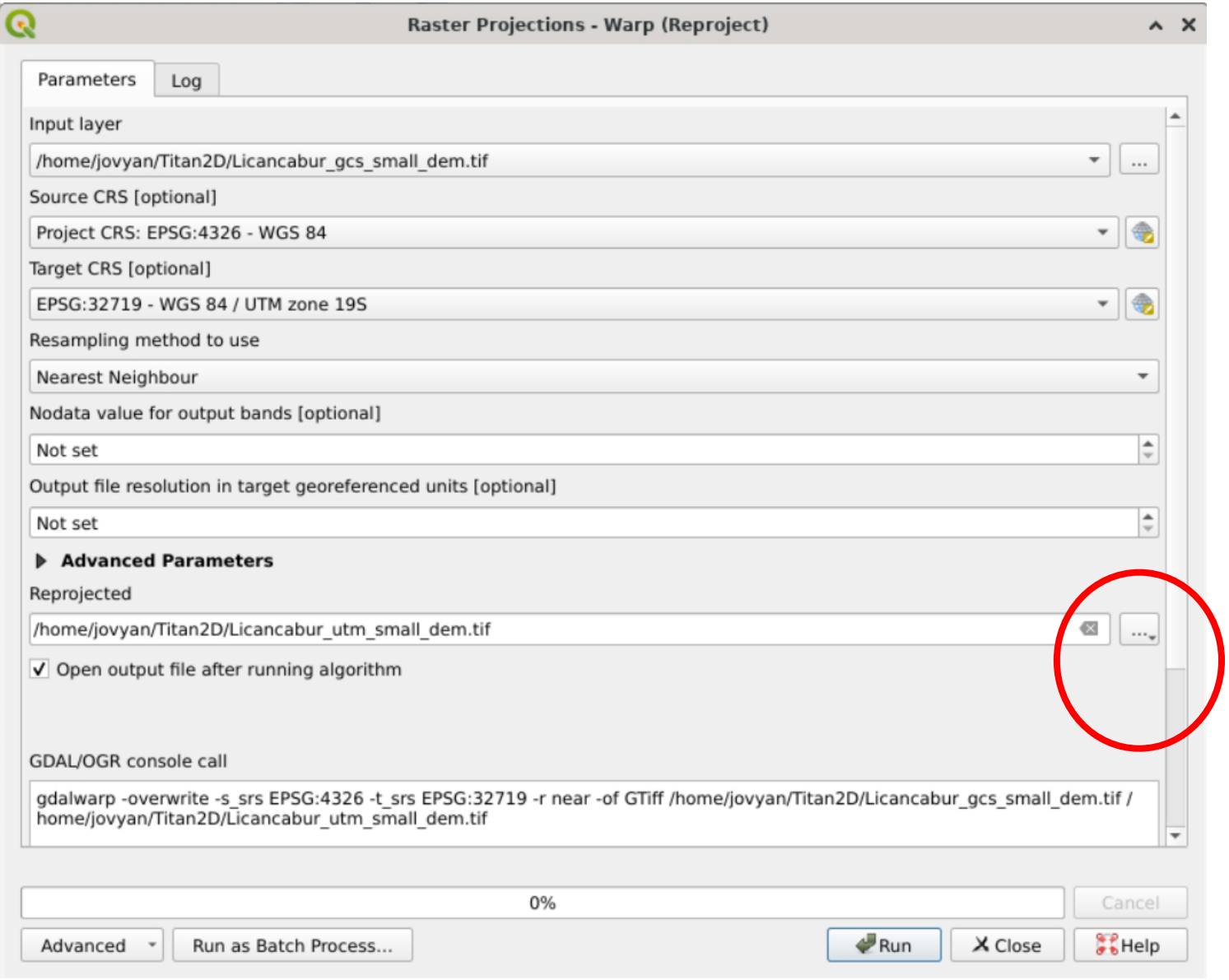
Find the tool called “Warp (reproject)” in the Processing Toolbox



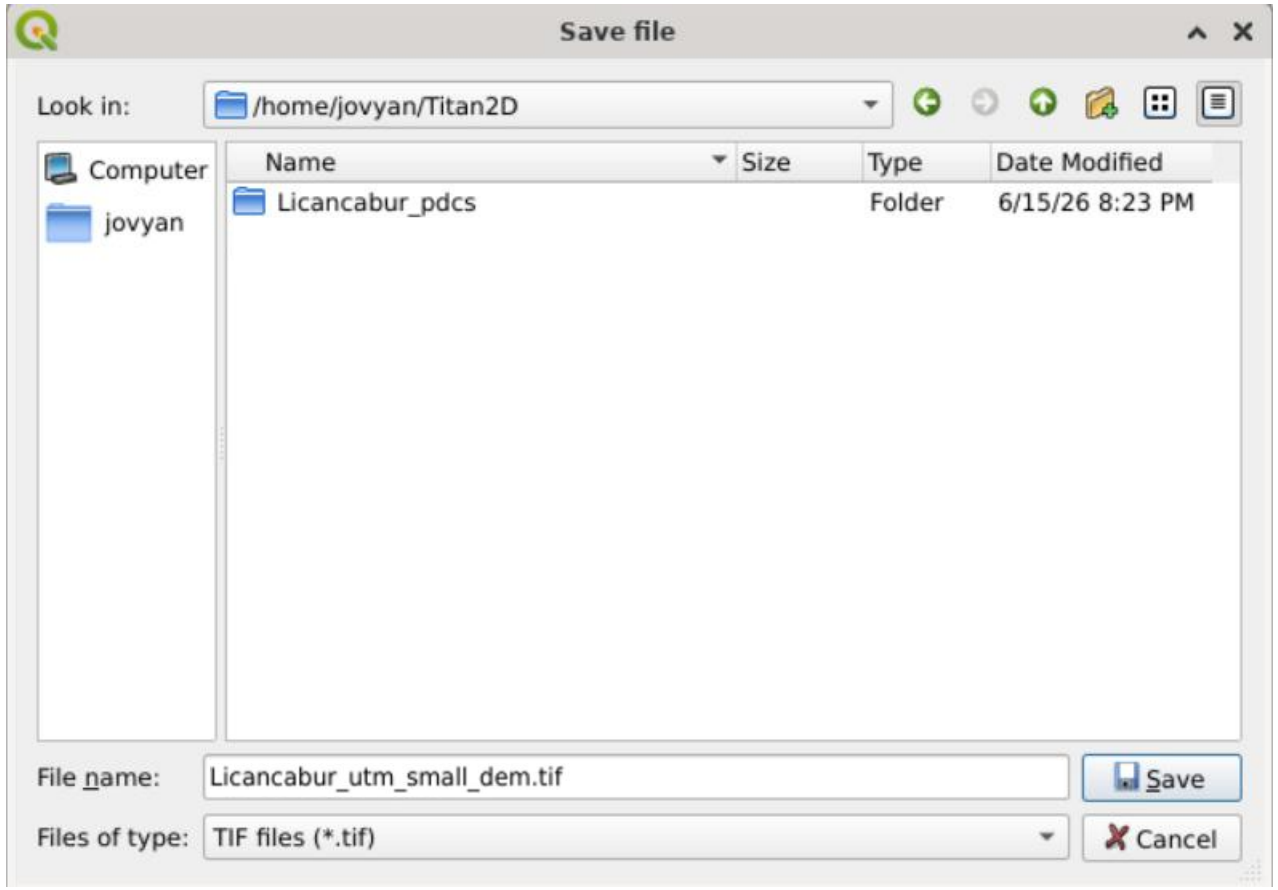
- The input is yourvolcano_gcs__small_dem.tif
- The source CRS is the Project CRS: EPSG: 4326 – WGS 84
- You will need to click “Select CRS” to find the Target CRS



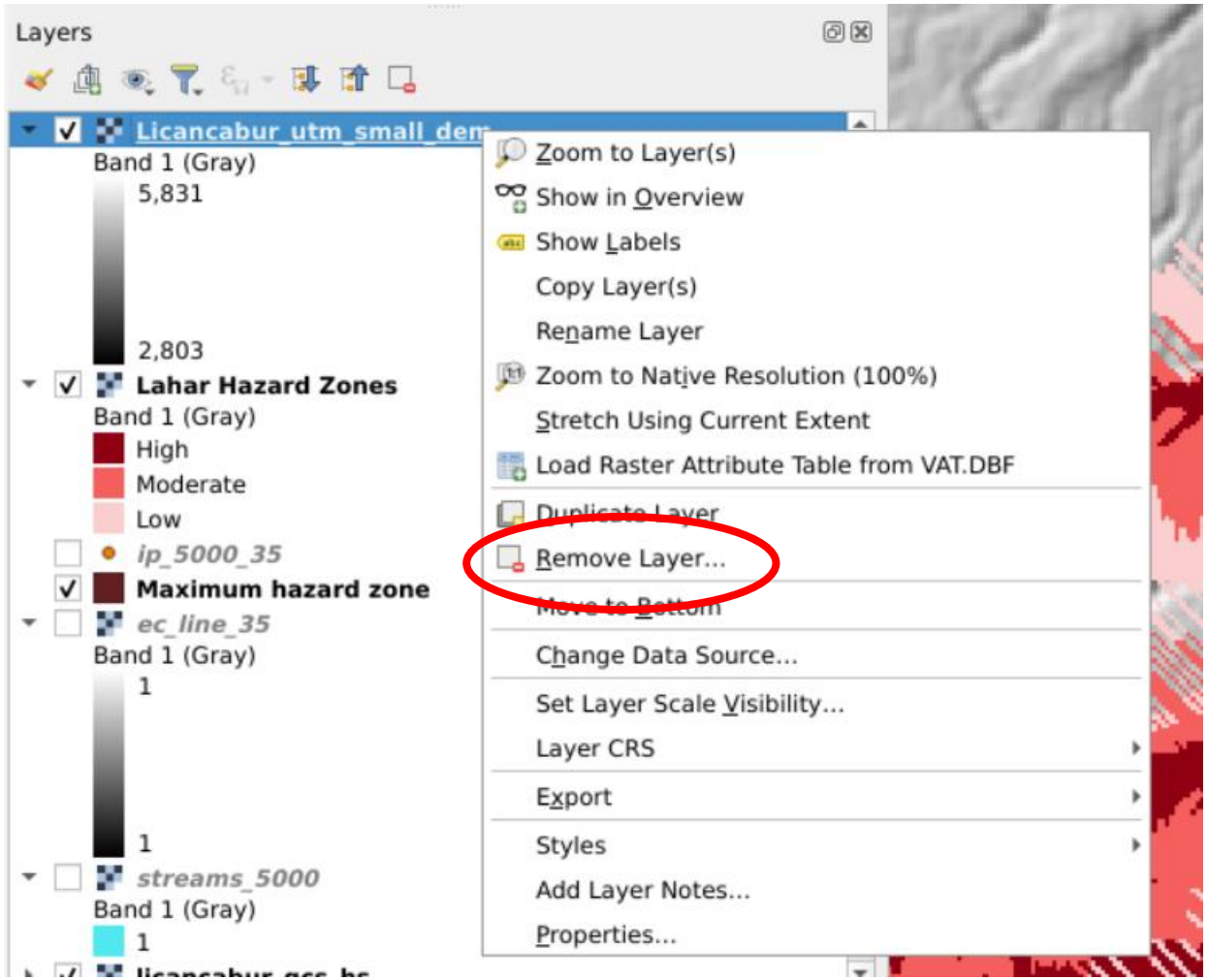
- Licancabur is in UTM zone 19S.
- **Use the UTM zone for your volcano!!**
- Don’t click Run, but click the blue arrow in the upper left to return to the previous screen



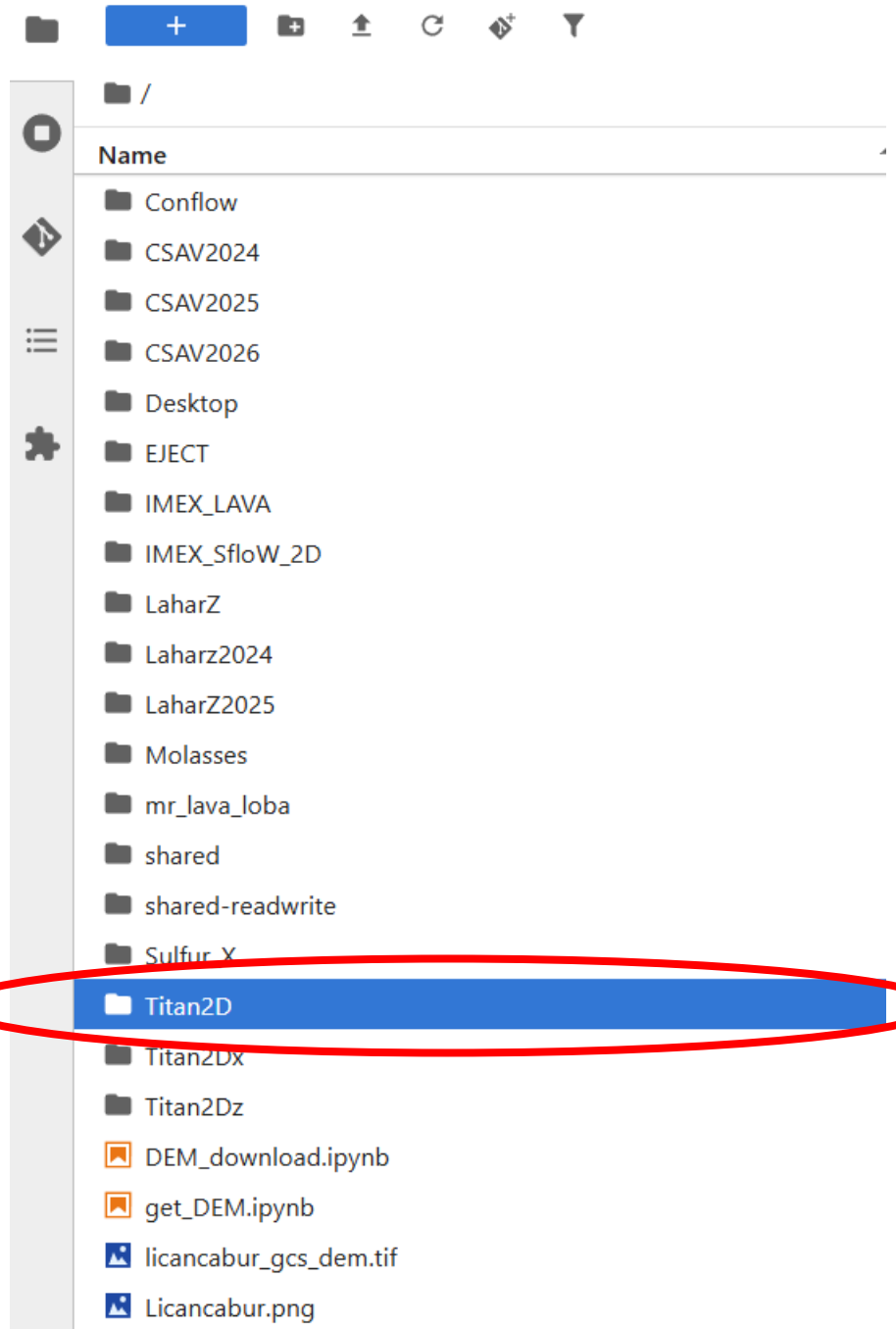
Make sure to save to a file and not a temporary layer



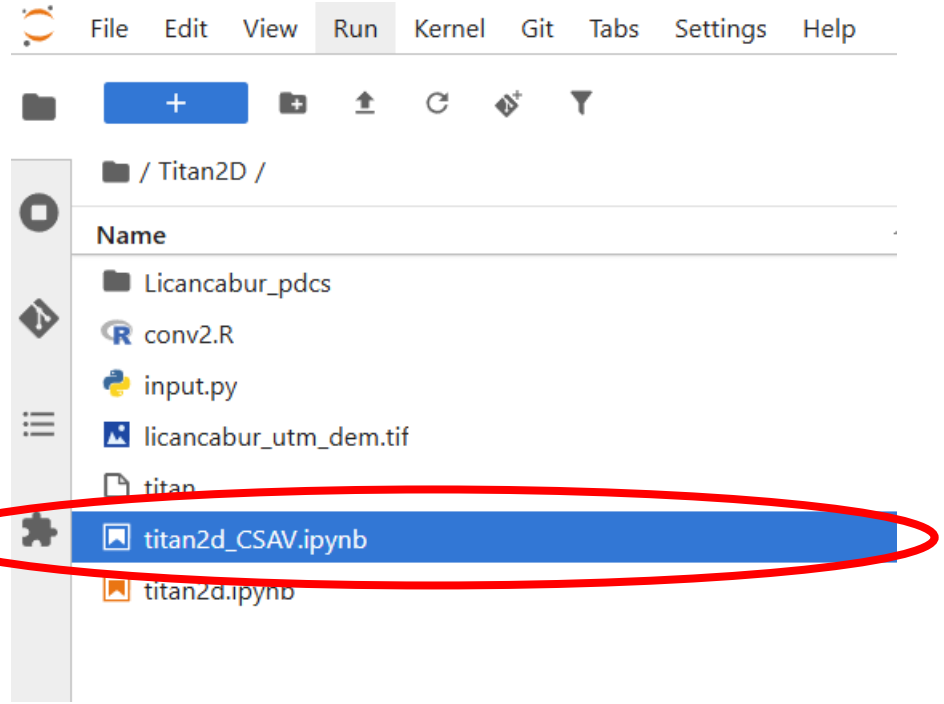
- Navigate to /home/jovyan/Titan2D
- Save the file as yourvolcano_utm_small_dem.tif



- Remove the result from the QGIS project so it's not open while we use it
- Leave QGIS and return to your VICTOR files

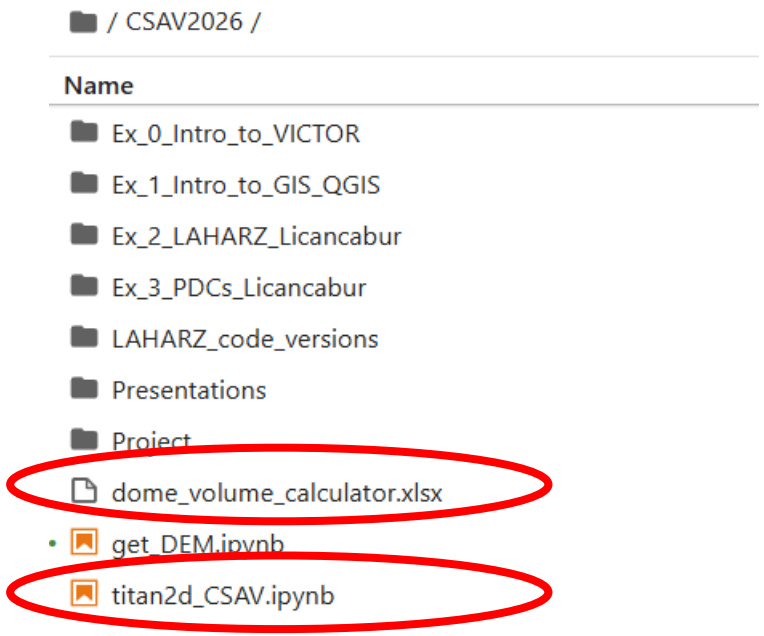


- Navigate to /home/jovyan/Titan2D
- Double click on titan2d_CSAV.ipynb



Decisions!

- Based on the types and volumes of PDCs you are going to model, you need to decide on basal friction angles for running TITAN2D (FlowDat values on the next slide)
- And the pile geometry, direction, and any initial velocity you will use
- For the pile geometry, you can use the dome volume calculator spreadsheet to help if you want
- To use TITAN2D, use titan2d_CSAV.ipnb, but change the values for your volcano



ΔH/L ranges for dome- and column-collapse PDC energy cones from FlowDat regressions

Note:

- Decimal values are used for energy cones
- Degrees are used for TITAN2D basal friction angles
- The regressions at the bottom are for decimal values
- Use the formula Degrees(atan(ΔH/L value)) to convert decimals to degrees in excel

Volume (km^3)	Volume (10^6 x m^3)	ΔH/L best fit Dome collapse	ΔH/L best fit Dome collapse (degrees)	ΔH/L best fit Column collapse	ΔH/L best fit Column collapse (degrees)
0.0001	0.1	0.39	21.1	0.36	19.8
0.001	1	0.29	16.1	0.30	16.5
0.01	10	0.22	12.1	0.24	13.7
0.1	100	0.16	9.1	0.20	11.4

Dome-collapse PDCs (based on 121 flows): $\boldsymbol{\Delta H/L = 0.120\left(V^{-0.127}\right)}$

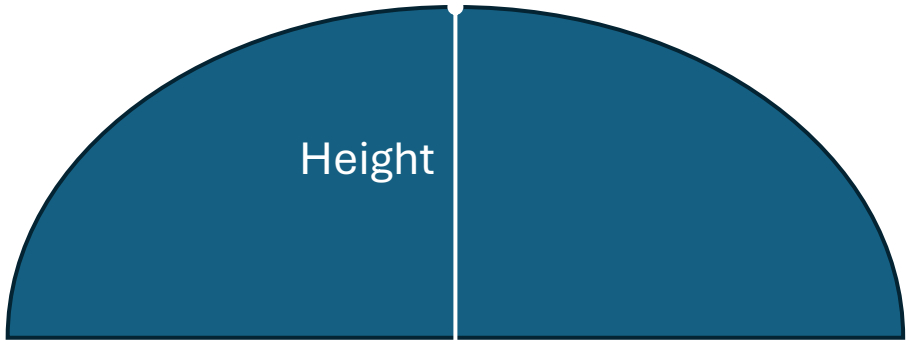
Column-collapse PDCs (based on 40 flows): $\boldsymbol{\Delta H/L = 0.166\left(V^{-0.084}\right)}$

All PDCs (based on 188 flows): $\boldsymbol{\Delta H/L = 0.165\left(V^{-0.085}\right)}$

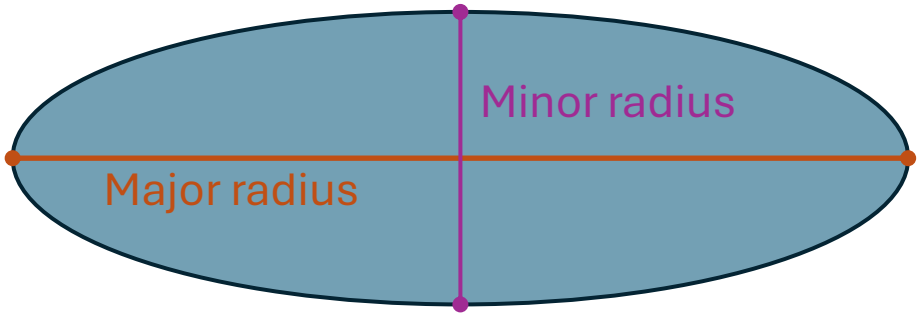
VDAP

USGS

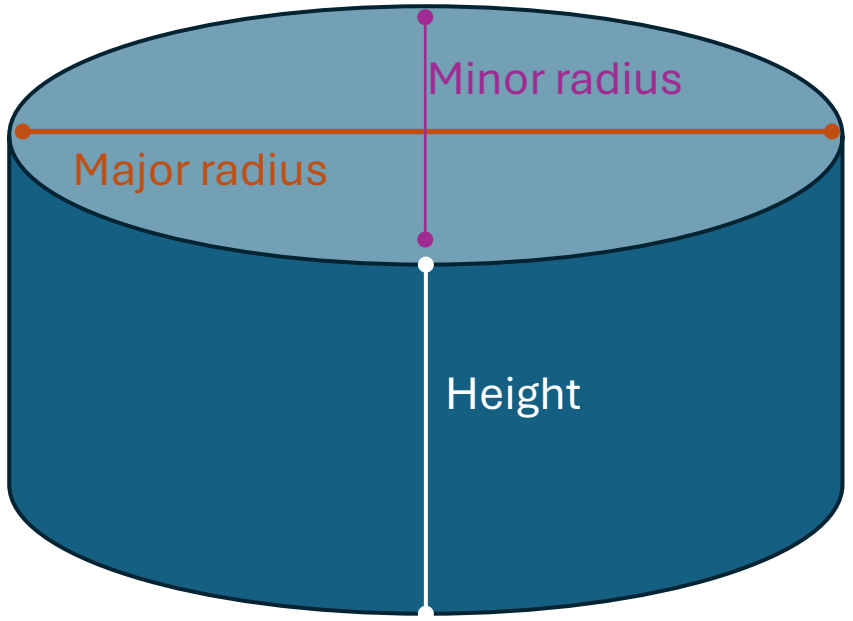
Source shape parameters: Pile size



Paraboloid (dome)



Paraboloid (dome)
Top view

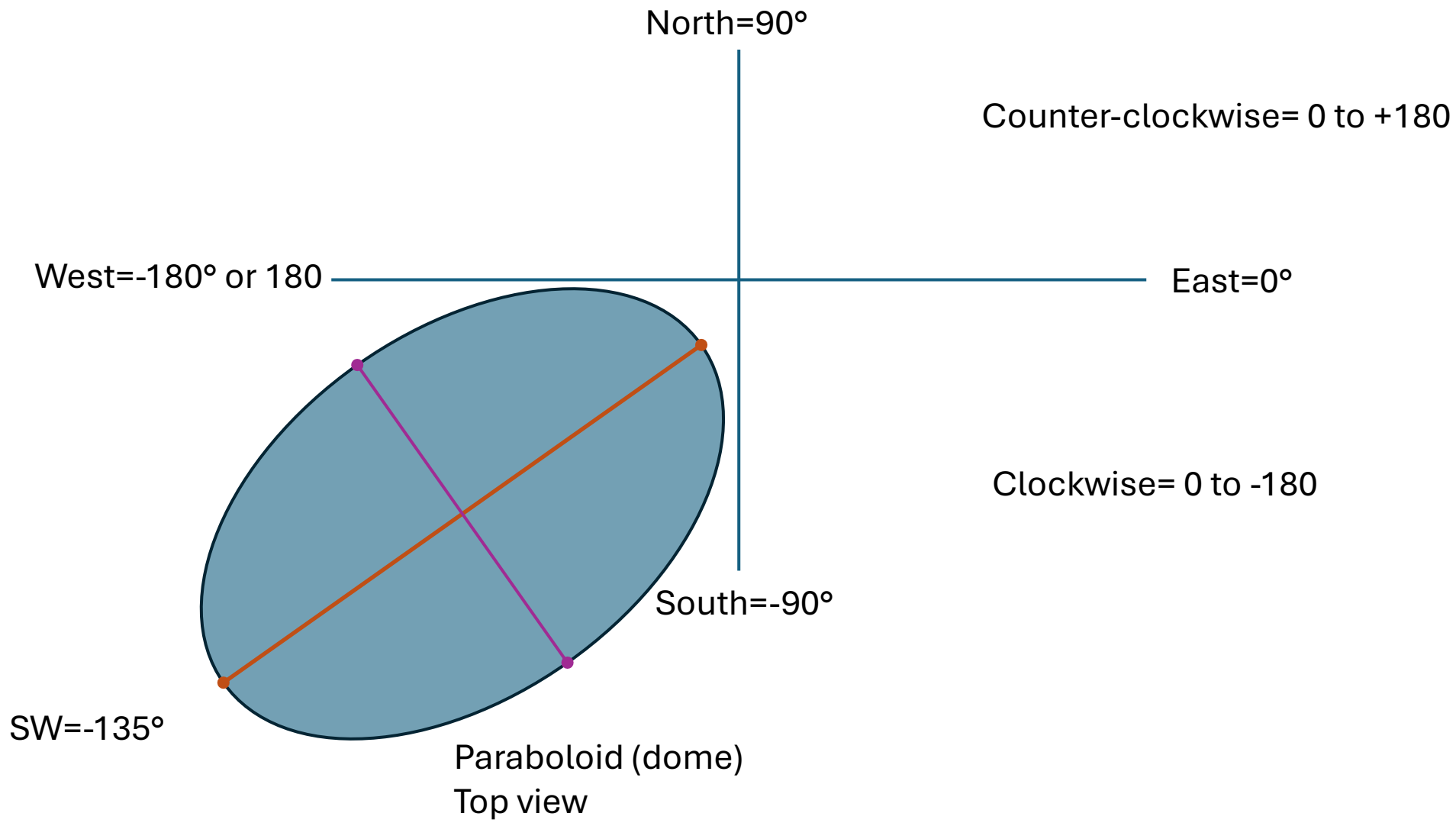


Cylinder (column)

dome_volume_calculator.xlsx

Volume=1/2*pi*major_radius*minor_radius*height

Source shape parameters: Pile orientation



Terminal 1 | Launcher | titan2d_CSAV.ipynb

Markdown

CSAV workflow for TITAN2D

TITAN2D is a geoflow simulation tool that, given initial parameters, will simulate granular flows over digital elev
guide is attached.

Importing Libraries [Run but don't change]

[1]:
```
#run_but_don't_change
import subprocess
import pandas as pd
import numpy as np
import sys
import matplotlib.pyplot as plt
import os
import rioxarray as rxr
import rasterio as rio
import h5py
import cartopy.crs as ccrs
import glob
```

[2]:
```
#run_but_don't_change
sys.path.insert(0, "/home/jovyan/shared/Libraries")
import victor
```

File Managment [Change these]

Notes from VICTOR team: Keep in mind that for the **GIS_GRASS** type, multiple arguments to specify file mapset/map locations are neccesary, and the DEM location fo
ascii shapefile. We highly recommend the **GDAL** format.

There are a variety of interesting but largely unecessary files that are output at the end of the simulation. Toggle the option below to remove them.

If you would like to create a gif of all the timesteps, please enable the option below. Keep in mind, this may take many minutes, depending on the number of iteration

[3]:
```
#The output files show things like momentum and thickness through time. You can set cleanup = False if you want to see these files.
#If you want a gif of the flow, you can set create_gif = True, but this will be slow. For the CSAV exercise, let's set it to False.
cleanup = False

create_gif = False
```

- Most of the rest of the instructions are in the Jupyter notebook itself!
- Read the title of the blocks to see if you will edit anything in the cells

- There is also text that starts with '#" that gives information about what parameters to choose for the exercise!
- Go to the next slide before starting!!

Run Titan here! [Run but don't change]

Titan outputs updates at **every** timestep, so we suppress the output here for clarity.

```
[*]: subprocess.run("export GDAL_DRIVER_PATH=''; ./titan -nt " + str(threads) + " input.py",shell=True, stderr=subprocess.DEVNULL)
```

```
Python home: /srv/conda/envs/notebook
Executable location: /home/jovyan/shared/Models/titan2d/bin/titan
python home: /srv/conda/envs/notebook
Titan2d 4.1.0
Executable location: /home/jovyan/shared/Models/titan2d/bin/titan
--------------------------------------------------------------------------------

Warning: This version was compiled without openmp support!
Set threads number to 1

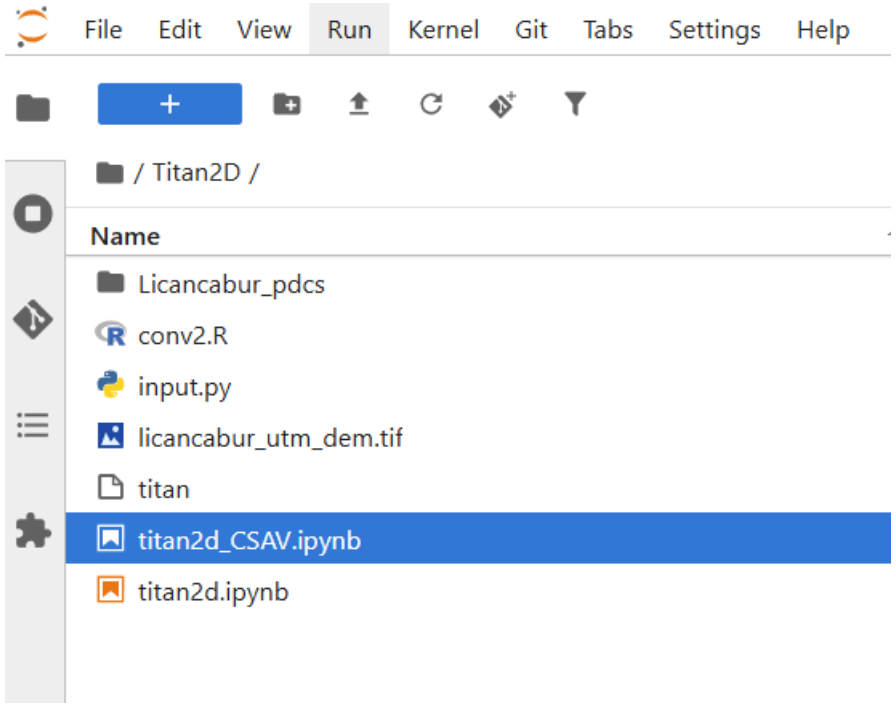

preproc...
The GIS region is:  x = 5.749191e+05 to 6.549142e+05   y = 7.434265e+06 to 7.515473e+06.
The simulation region is:  x = 5.749191e+05 to 6.549142e+05   y = 7.434265e+06 to 7.515473e+06.

GIS:
        gis_format: 2
        gis_main:
```

- Remember to wait for the cells to finish before moving on.
- When a cell is running, it will have an asterisk
- When you get to "Run Titan here!" it will take awhile!!!!
- Please let me know when you have started the Titan run!!!!
- We can take a break while these run.

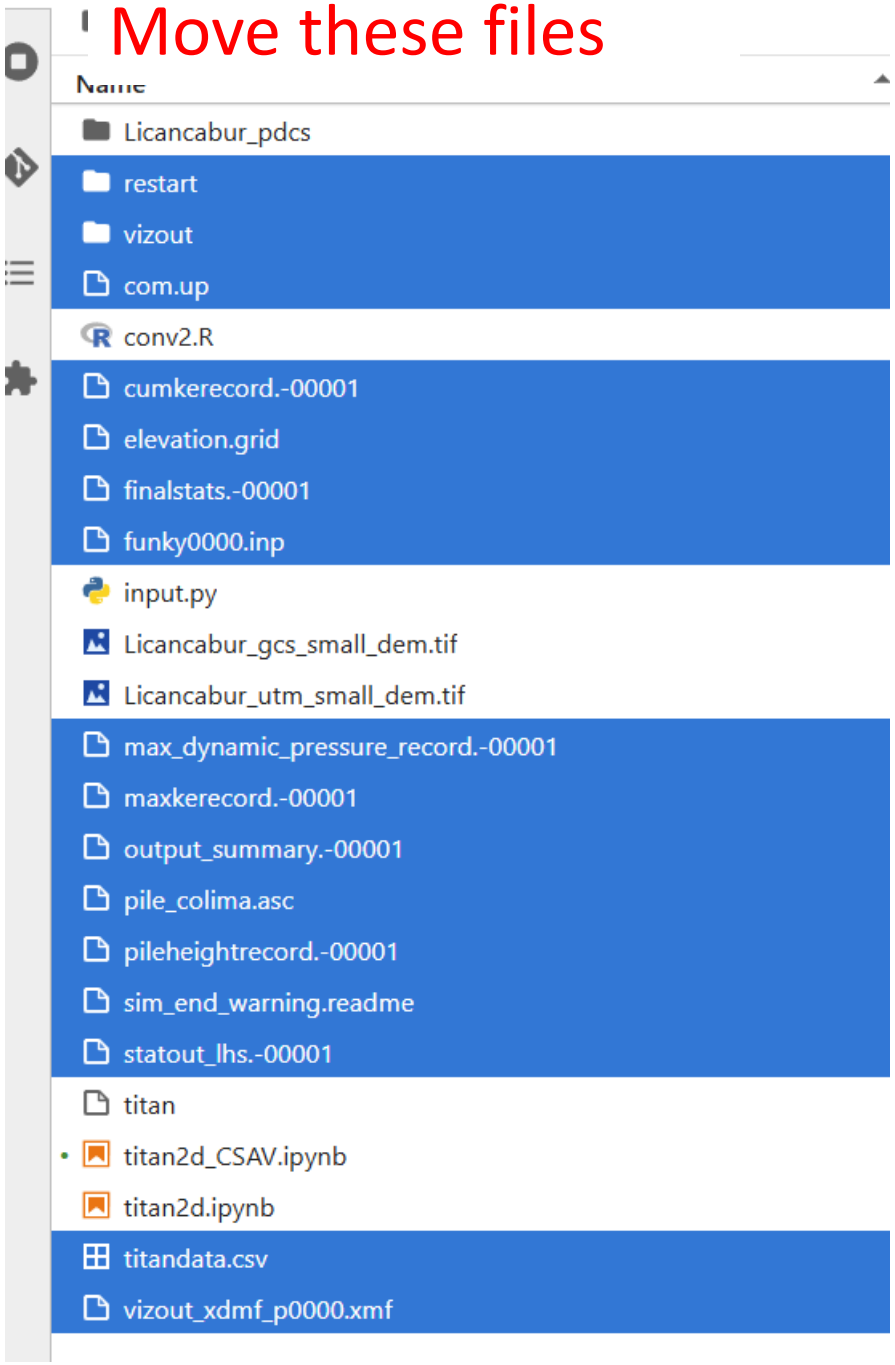
Clean-up

- To keep track of your Titan runs, move the results into your Licancabur_pdcs folder



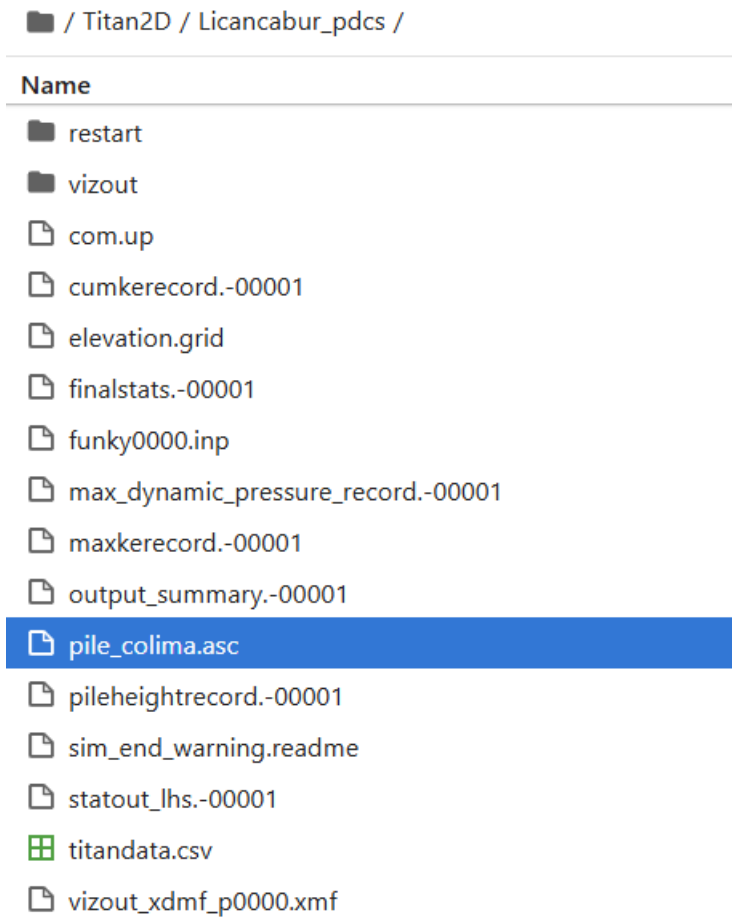
Keep these files where they are

Move these files



Clean-up

- Navigate into your yourvolcano_pdcs folder
- Rename “pile_Colima.asc” to “yourvolcano_hlvalue_volume.asc”
- This way, you have named it using the ΔH/L value and approx. volume
- You might also want to name the folders this way (with the h/l used and the volume) when you do your projects



Ballistics modeling using EJECT

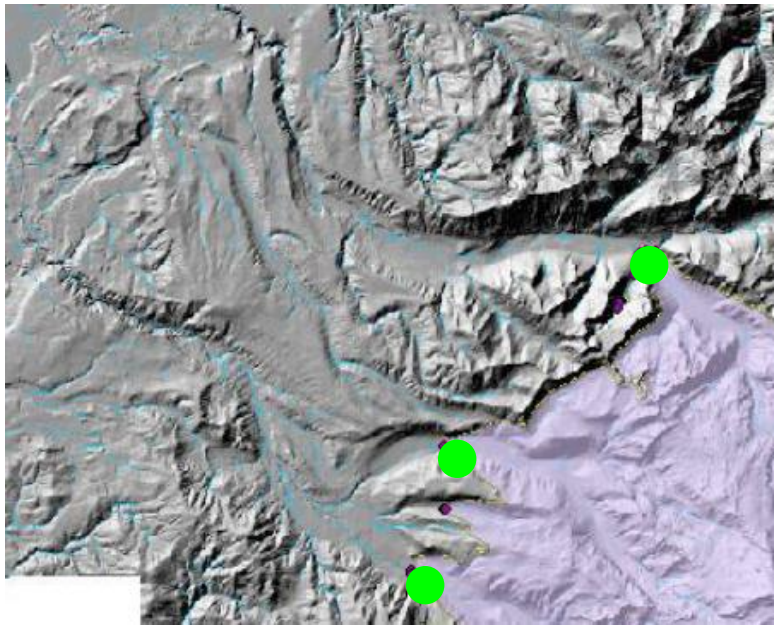
Make the map prettier!!

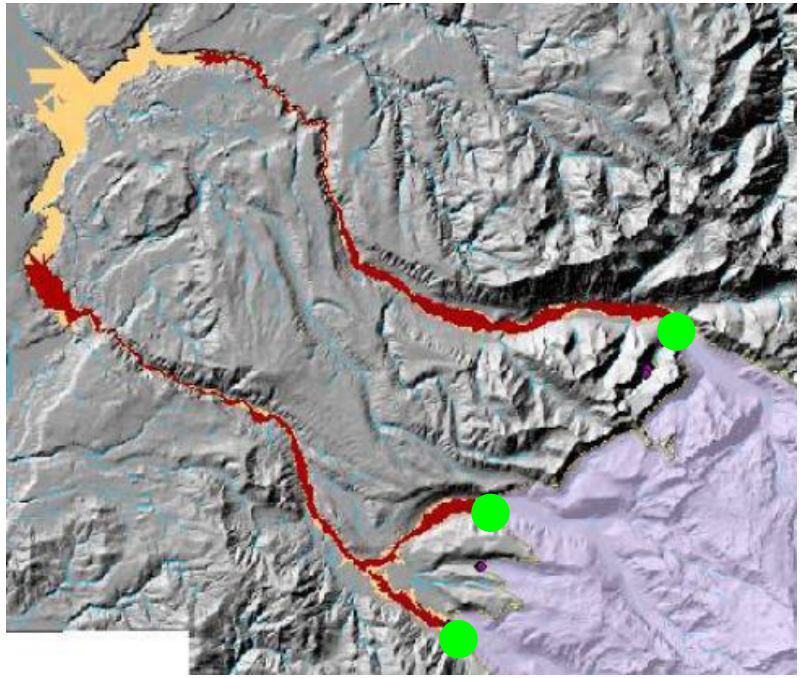
- You can open the exercise files to read through the QGIS instructions, but you don’t need to follow the same color schemes and design choices!

Reference Slides

What do you need to run LAHARZ?

- A starting location or locations:
  - Use the proximal hazard zone tool within LAHARZ to produce an H/L cone that defines the starting locations (can edit these to have fewer points)
  - Manually select starting locations



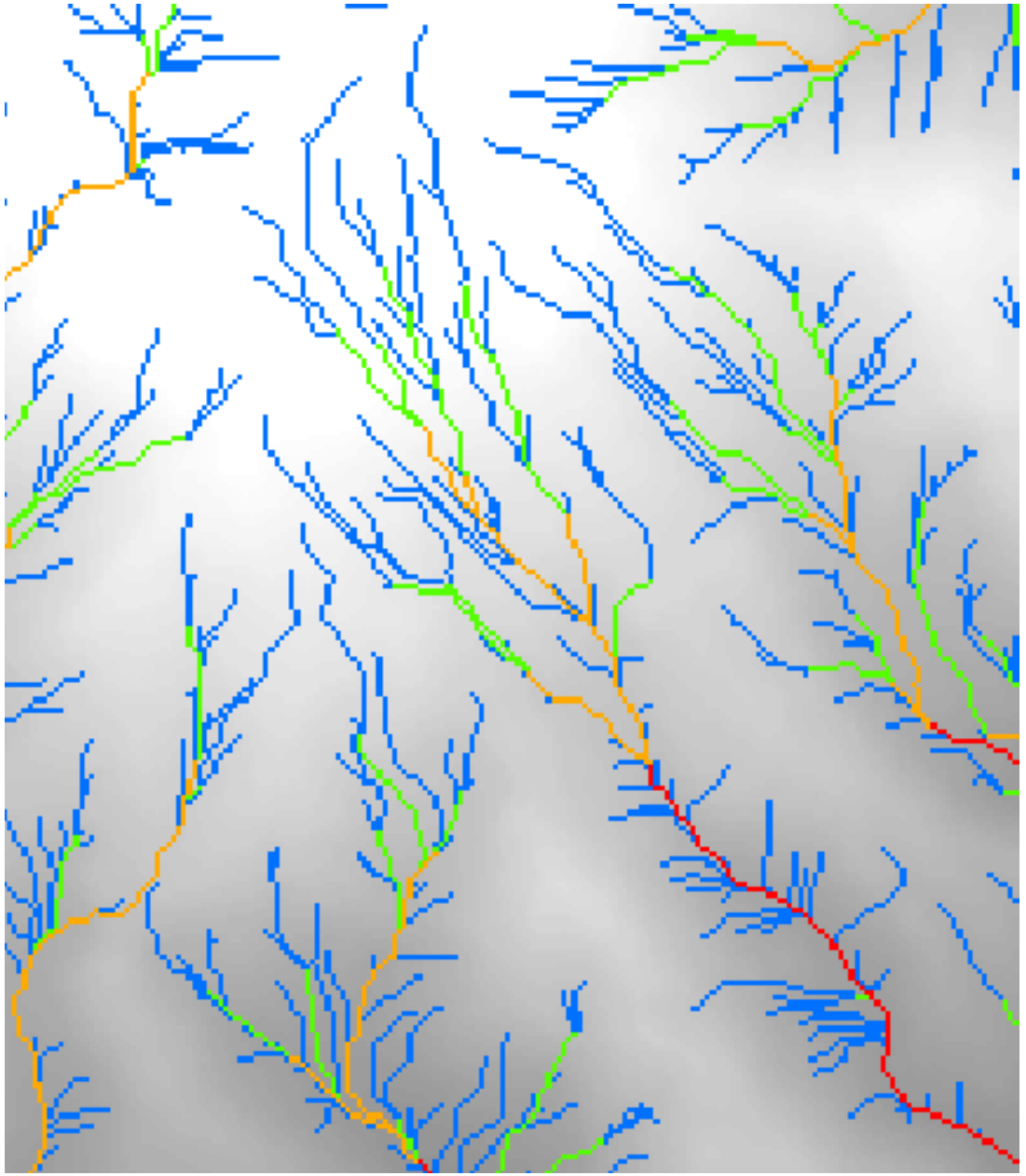


Stream Network

- LAHARZ generates stream network using the DEM, and a user defined stream threshold
- This is the number of upstream cells that have to flow into a cell to consider it a stream cell. A smaller number here will make a stream network with more branches and longer streams, a higher number creates a simpler stream network with fewer branches
- The network should generally match published maps, but should represent the *current* stream network

High stream threshold (6000): shorter stream network with fewer branches

Low stream threshold (50): longer stream network with more branches



How to decide on the stream threshold

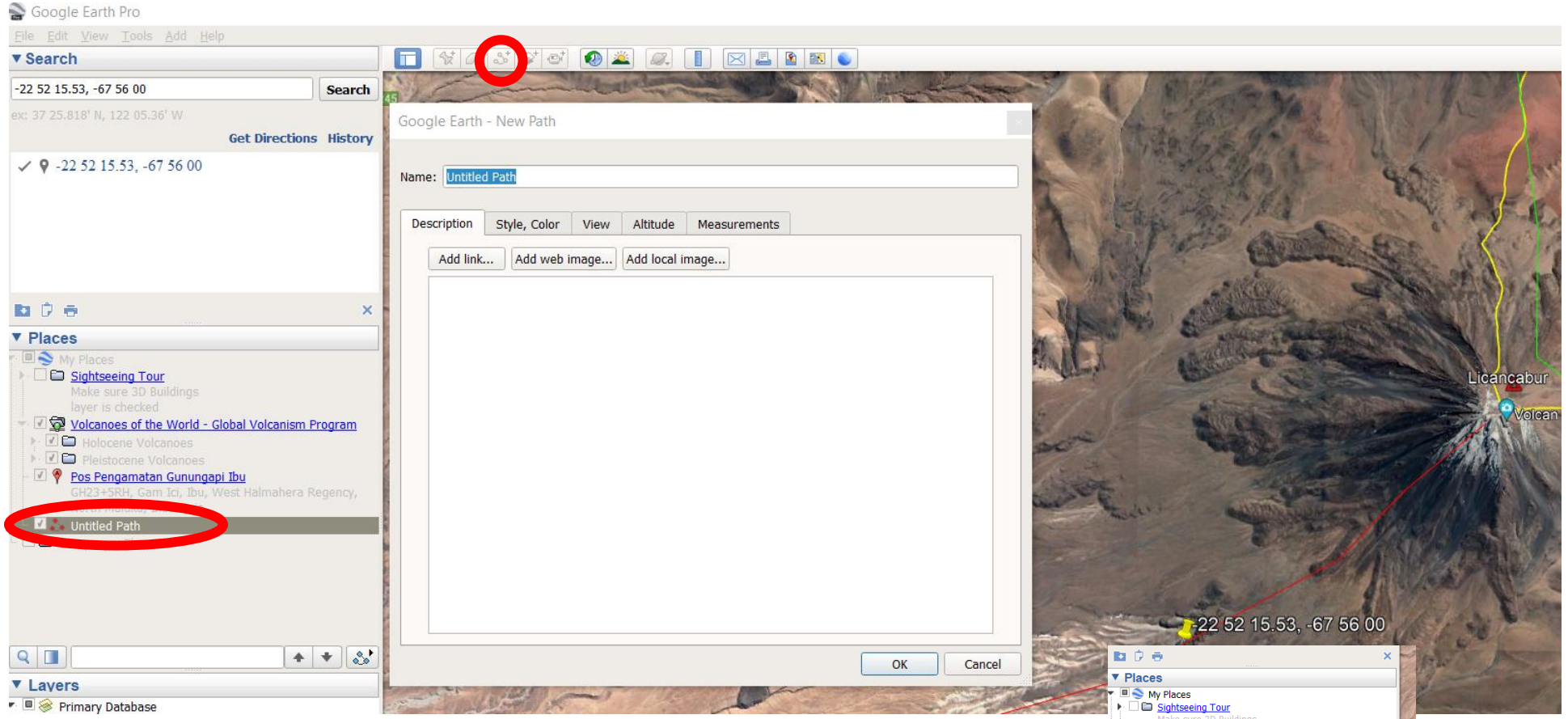
- Compare your stream network to a recent topographic map
- Compare your stream network to a recent satellite image
- Compare your stream network to field measurements
- Examine the stream network in locations where lahars/flash floods have been known to spill over into other channels
- If you are happy with you ΔH/L cone, but the streams don't intersect, you can try lowering the stream threshold, but this might also be an indication to think about your ΔH/L value!

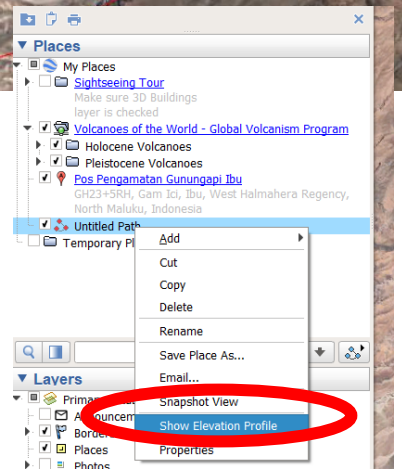
Okay, so how do we choose the H/L value for LAHARZ?

Changes in slope and the extent of PDC deposit fans (where most smaller PDC deposits are) are useful for choosing an H/L ratio

- **Visualize volcano in 3-D:**
  - Find photos (Google image search, GVP)
  - Use Google Earth to get oblique views, fly around, and make a topographic profile.
  - Look for change in slope
- **Look at the geomorphology:**
  - landforms, farming, land use
- **Look for PDC units in the field**
- **Examine satellite imagery**
- **Use global record of analog volcanoes (PDC H/L values)**
- **Start with H/L of 0.3 and refine**

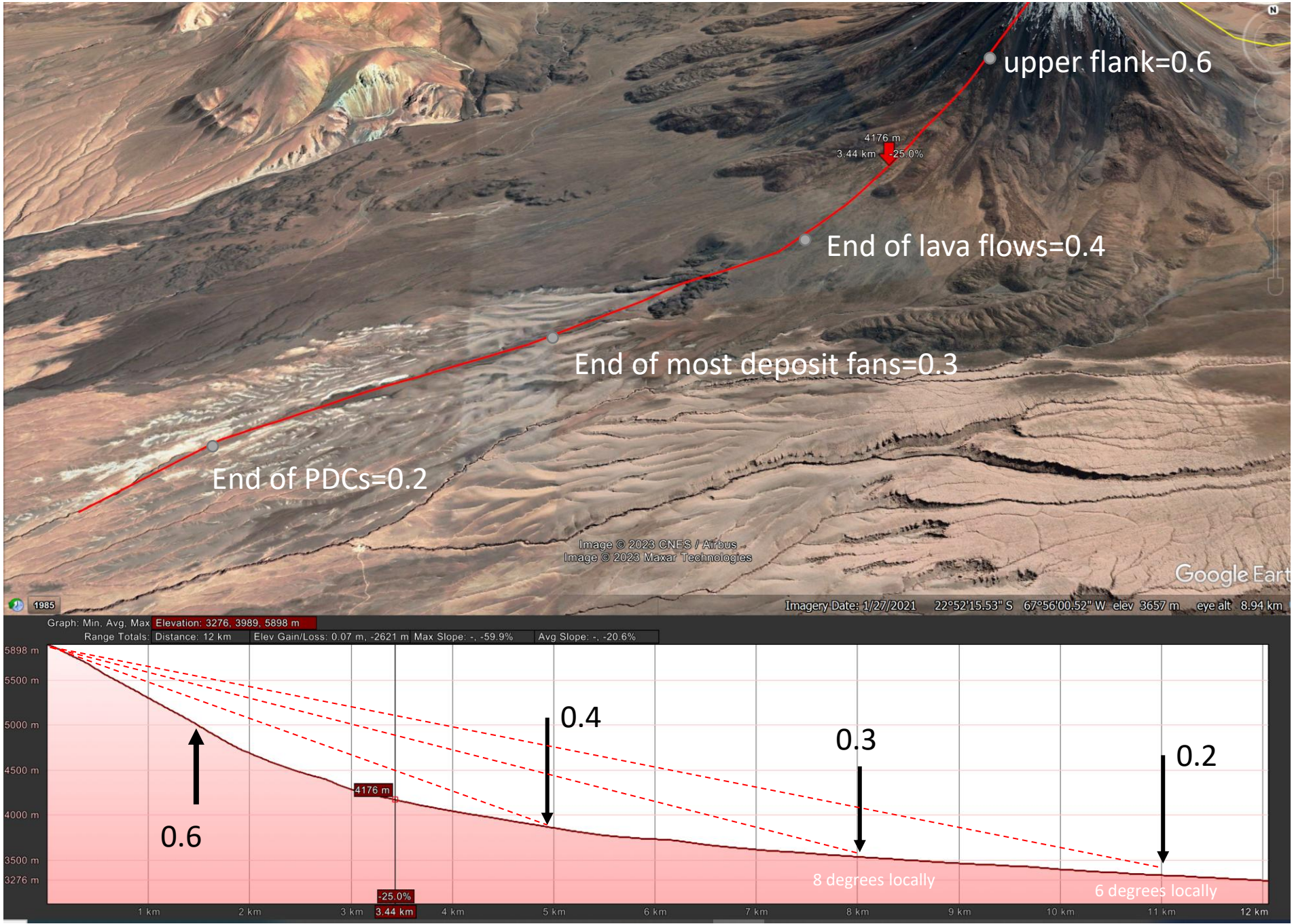
Visualize volcano in 3-D



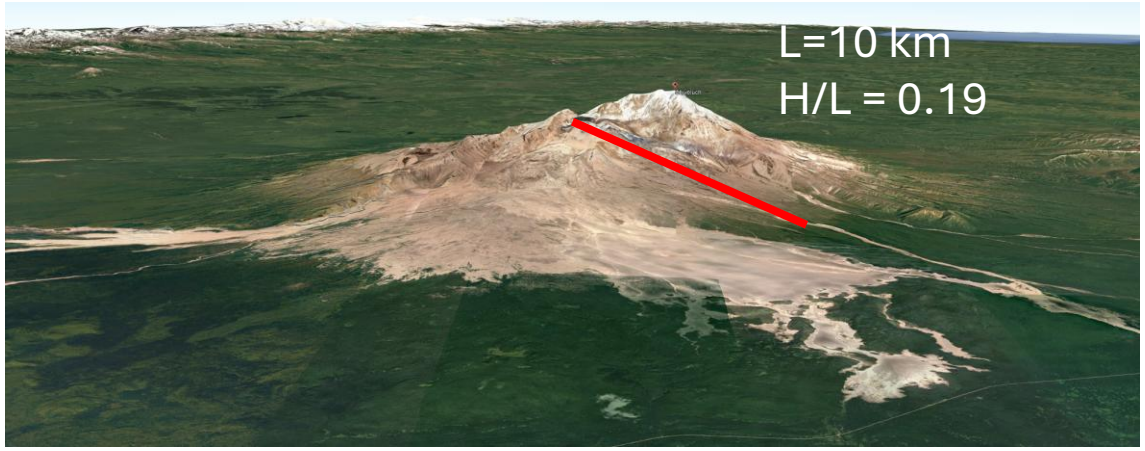


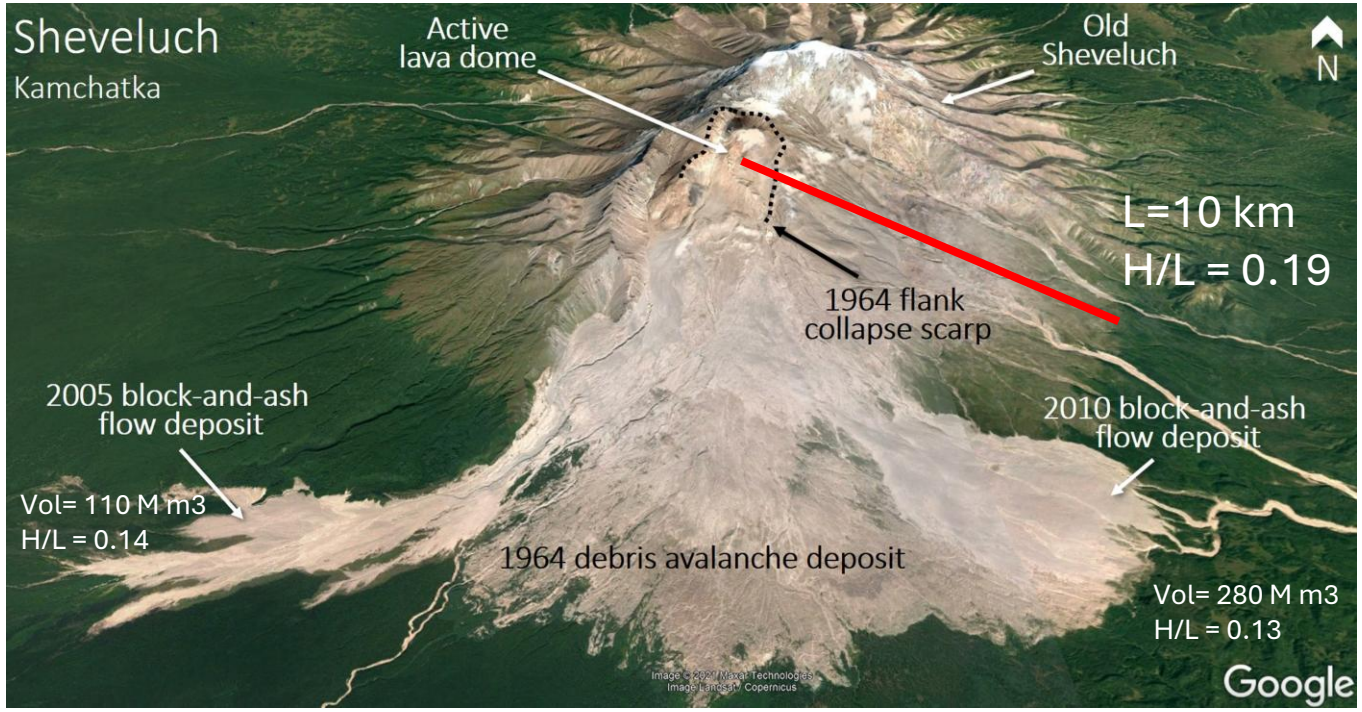
- Click on the “Add path” button
- Click on the map to make a path
- Name and save the path
- Right click on the path name under “Places”
- Choose “show elevation profile”
- Be careful: the slope values given in Google Earth are local, not H/L from the summit

Visualize volcano in 3-D



Sheveluch example



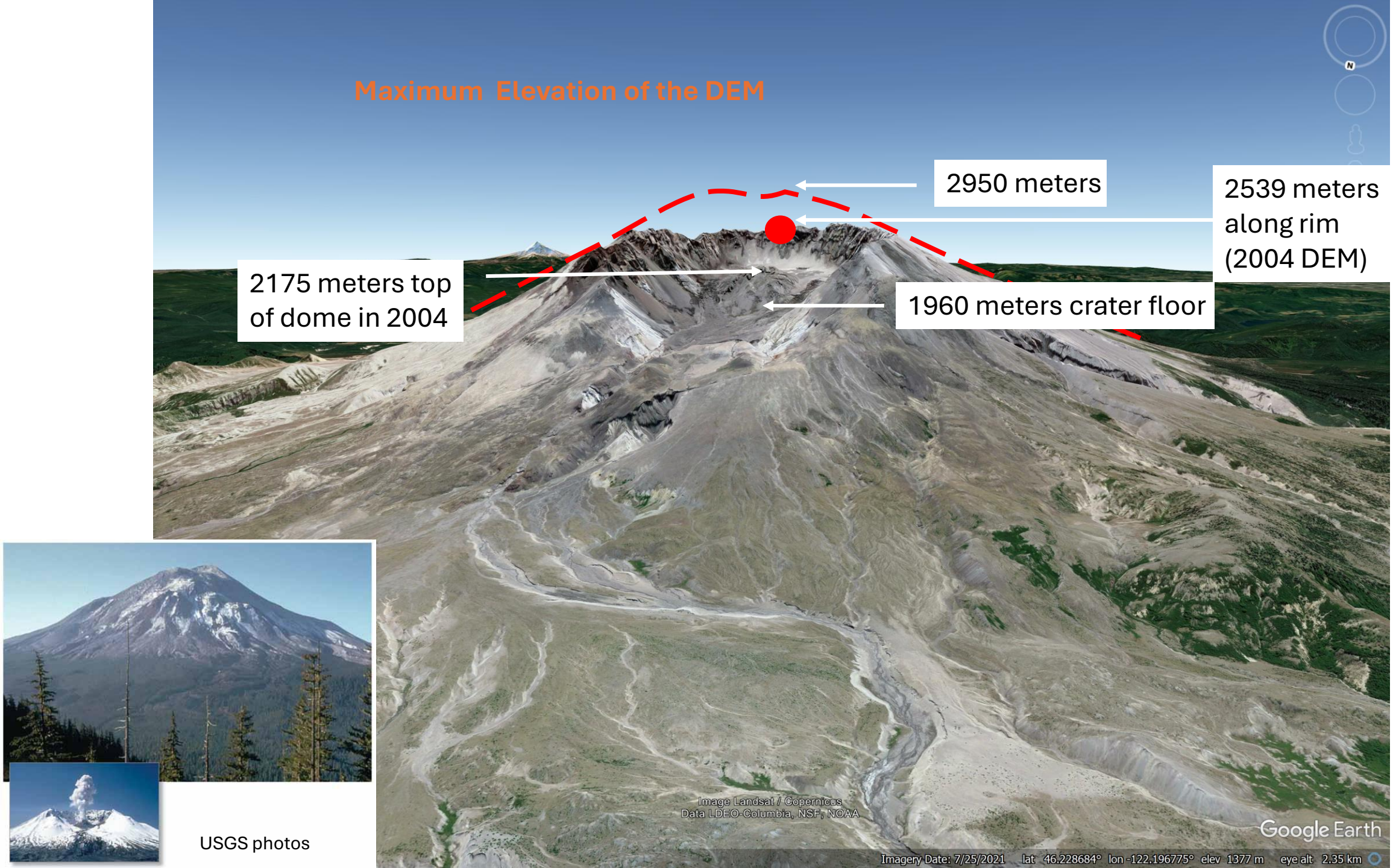




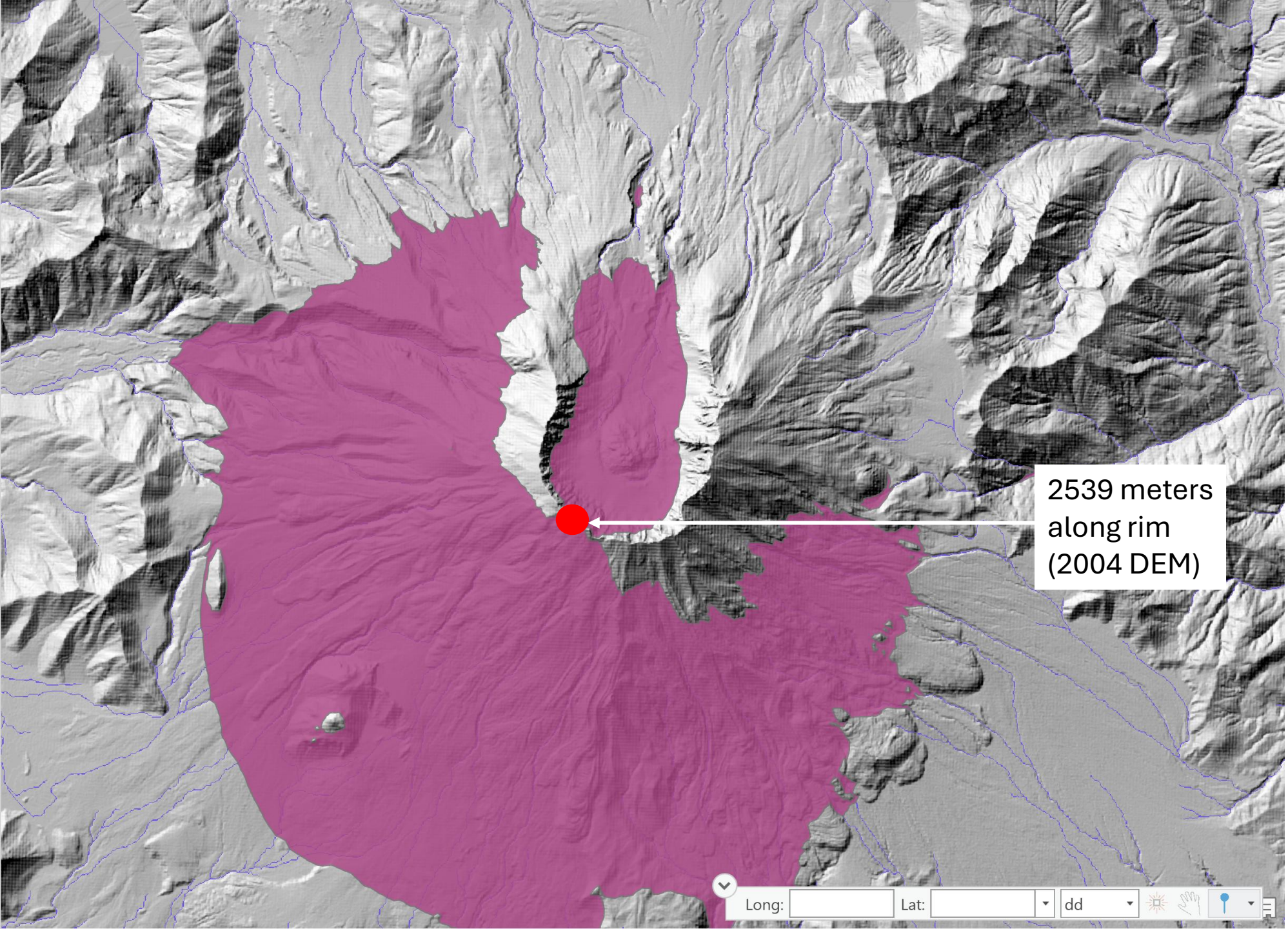
- **Look at the slope**
- **Look at majority of PDC deposits**
- **Look for landcover changes**

Special Cases: not a nice pointy volcano?
Add elevation in LAHARZ




Maximum Elevation of the DEM
2950 meters
2539 meters along rim (2004 DEM)
2175 meters top of dome in 2004
1960 meters crater floor
Image Landsat / Copernicus
Data LDEO-Columbia, NSF, NOAA
Google Earth
Imagery Date: 7/25/2021 lat 46.228684° lon -122.196775° elev 1377 m eye alt 2.35 km

USGS photos



This ΔH/L cone uses the highest point on the DEM, a point on the crater rim. This ΔH/L cone has holes and looks weird!

To-restablish the summit:
2950 m – 2175 m = add ~800 meters

2950 meters

2539 meters along rim (2004 DEM)

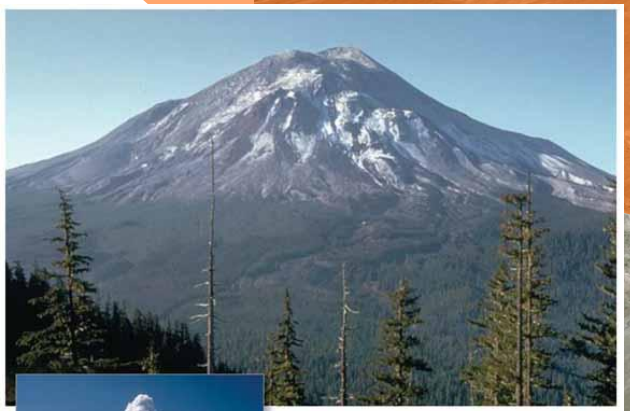
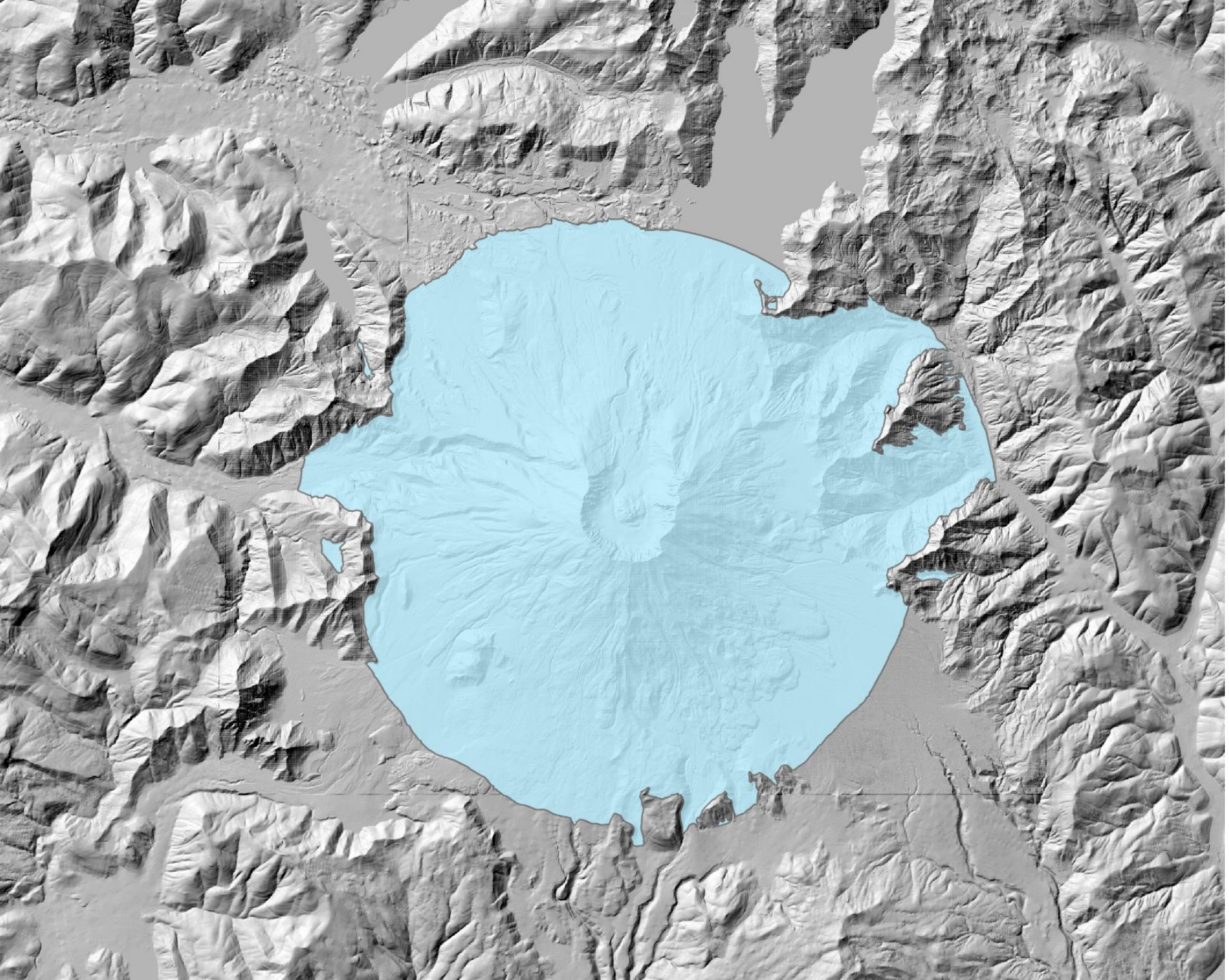




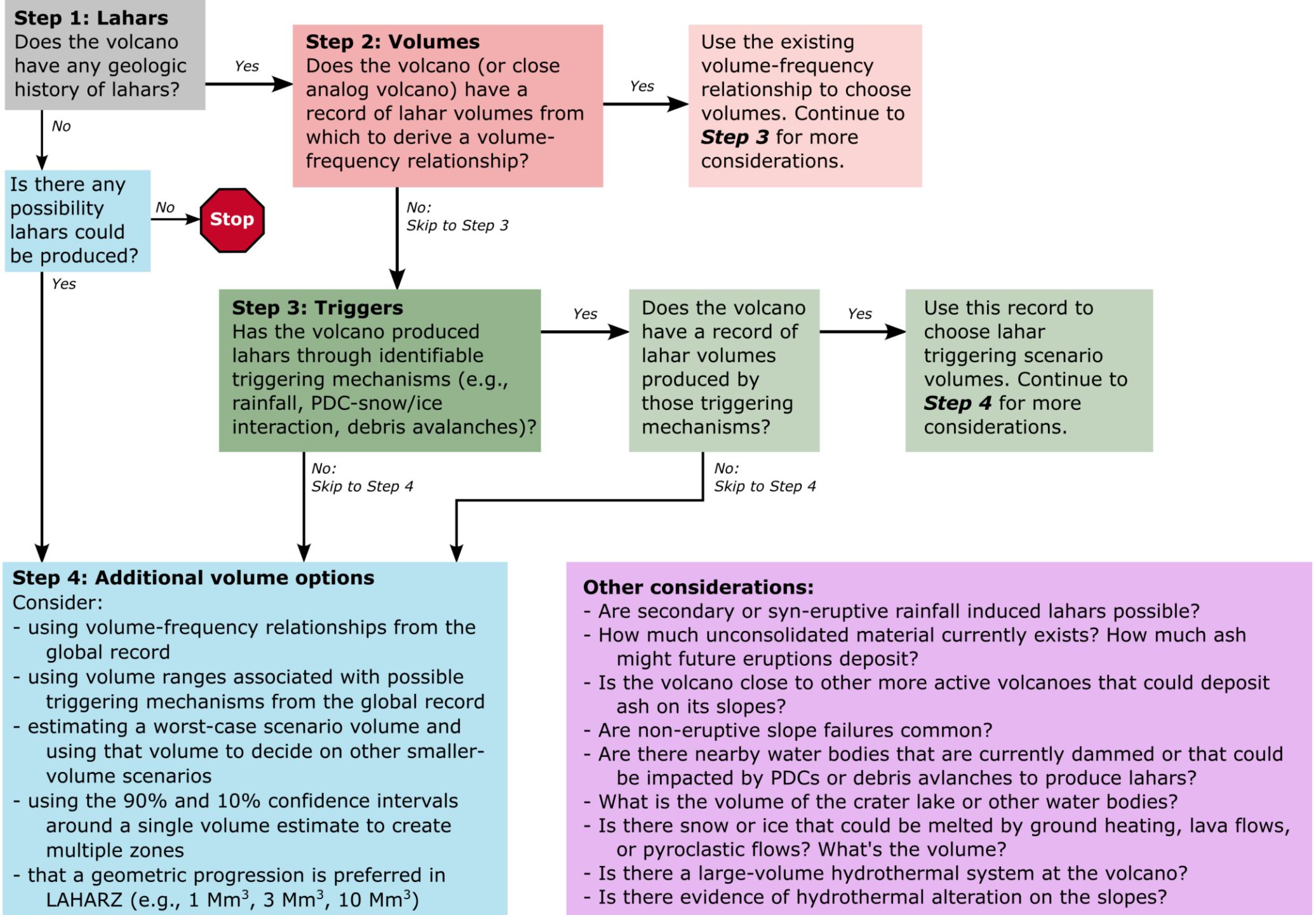
Image Landsat / Copernicus
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Google Earth

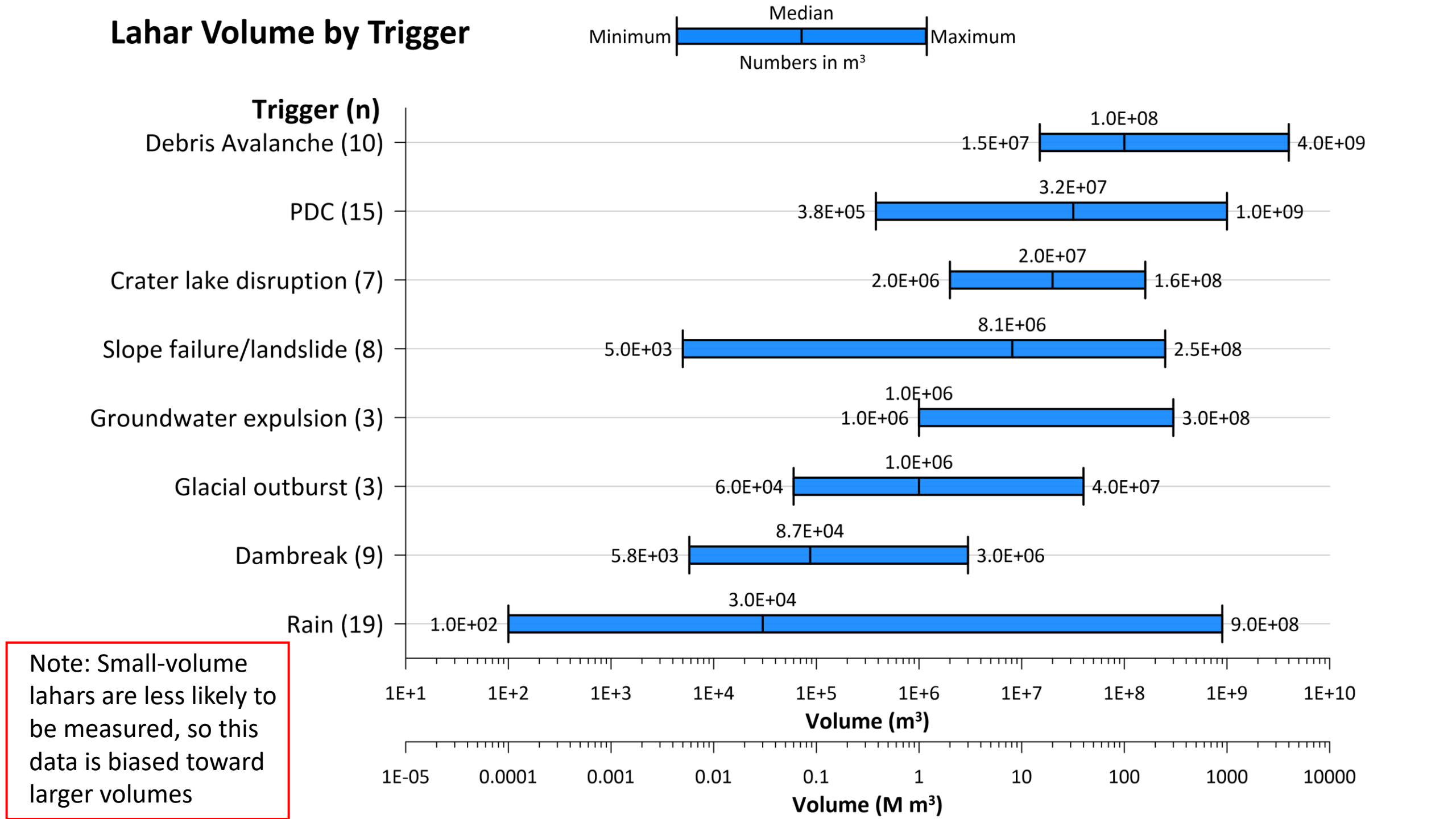
Imagery Date: 7/25/2021 lat 46.228684° lon -122.196775° elev 1377 m eye alt 2.35 km



This ΔH/L cone adds ~800 m above the center of the crater. This ΔH/L cone has looks pretty good!


Step 1: Lahars
Does the volcano have any geologic history of lahars?
Yes
No
Step 2: Volumes
Does the volcano (or close analog volcano) have a record of lahar volumes from which to derive a volume-frequency relationship?
Yes
Use the existing volume-frequency relationship to choose volumes. Continue to Step 3 for more considerations.
No:
Skip to Step 3
Is there any possibility lahars could be produced?
No
Stop
Yes
Step 3: Triggers
Has the volcano produced lahars through identifiable triggering mechanisms (e.g., rainfall, PDC-snow/ice interaction, debris avalanches)?
Yes
Does the volcano have a record of lahar volumes produced by those triggering mechanisms?
Yes
Use this record to choose lahar triggering scenario volumes. Continue to Step 4 for more considerations.
No:
Skip to Step 4
No:
Skip to Step 4
Step 4: Additional volume options
Consider:
- using volume-frequency relationships from the global record
- using volume ranges associated with possible triggering mechanisms from the global record
- estimating a worst-case scenario volume and using that volume to decide on other smaller-volume scenarios
- using the 90% and 10% confidence intervals around a single volume estimate to create multiple zones
- that a geometric progression is preferred in LAHARZ (e.g., 1 Mm3, 3 Mm3, 10 Mm3)
Other considerations:
- Are secondary or syn-eruptive rainfall induced lahars possible?
- How much unconsolidated material currently exists? How much ash might future eruptions deposit?
- Is the volcano close to other more active volcanoes that could deposit ash on its slopes?
- Are non-eruptive slope failures common?
- Are there nearby water bodies that are currently dammed or that could be impacted by PDCs or debris avlanches to produce lahars?
- What is the volume of the crater lake or other water bodies?
- Is there snow or ice that could be melted by ground heating, lava flows, or pyroclastic flows? What's the volume?
- Is there a large-volume hydrothermal system at the volcano?
- Is there evidence of hydrothermal alteration on the slopes?

Lahar Volume by Trigger



Note: Small-volume lahars are less likely to be measured, so this data is biased toward larger volumes

How to choose volumes for LAHARZ

- Check historical records of lahar volumes at your volcano
  - What types of lahars are produced at your volcano?
    - Rainfall-triggered?
    - Eruption-triggered (Snow melt? Ice melt? Groundwater expulsion? Damming of rivers by deposits? Flows impacting lakes or rivers? Crater lake expulsion?)
  - What is the most common lahar volume?
  - What is the worst-case scenario (biggest) lahar volume?
- Investigate the amount of snow and ice on your volcano
- Investigate whether any crater lakes are present and what volume they are
- Investigate the rainfall/typhoon/hurricane history at your volcano
  - Is there a rainy season?
- Check historical records at analog volcanoes
  - Volcanoes of a similar size
  - Volcanoes with similar amounts of snow/ice
  - Volcanoes with similar eruptive histories (VEI and PDCs)
  - Volcanoes with similar crater lakes
  - Volcanoes in similar climates

Possible Lahar and PDC Energy Cone Workflow

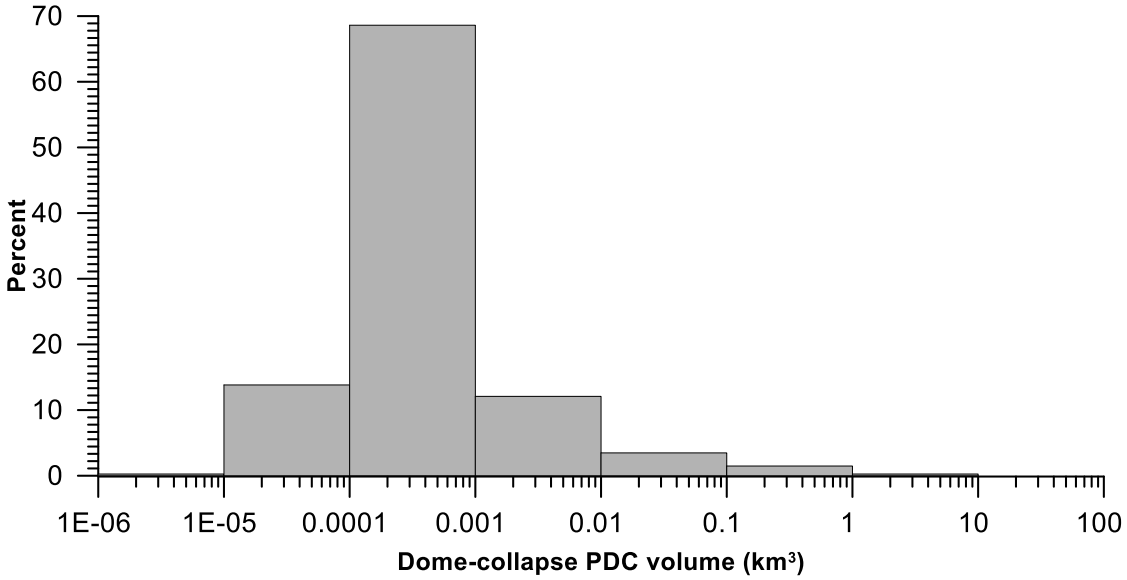
For volcanoes with small to moderate PDCs:

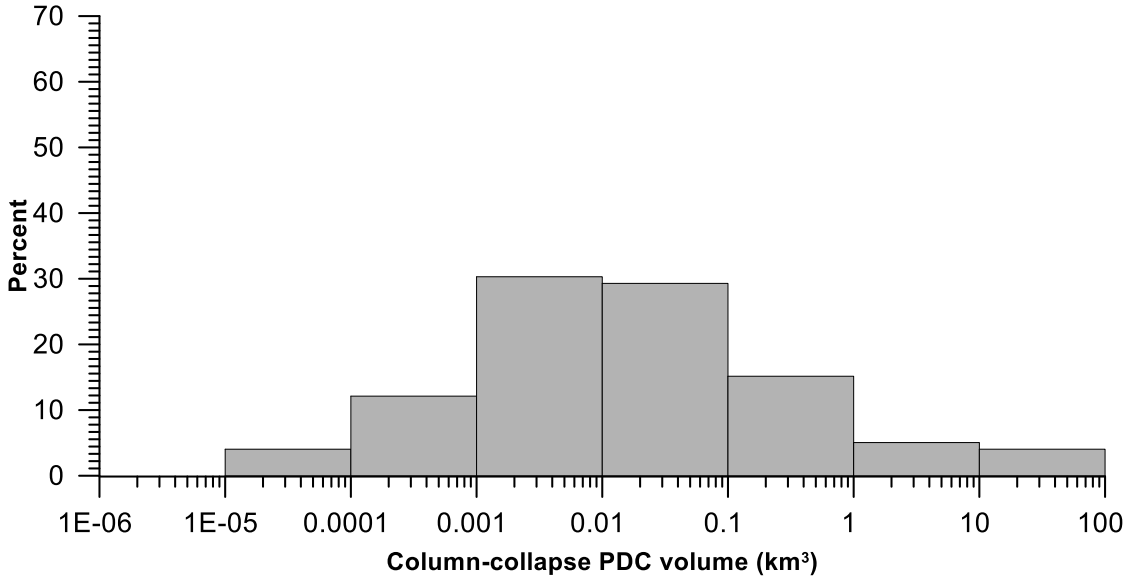
1. Run LAHARZ using an ΔH/L cone that identifies where lahars will switch from deposition to erosion. Remember, you look at where the break in slope is and where most PDCs have gone at your volcano. It is usually between 0.1-0.3 and will depend a bit on the size of the volcano.
2. Use the starting points to run LAHARZ for your desired range of lahar volumes (you can always edit the starting point list down to one point per stream, it usually identifies too many points).
3. Display this H/L cone on the map and call it the 'proximal hazard zone' or 'maximum hazard zone' for PDCs and ballistics. Done!

For volcanoes with larger PDCs:

4. Complete steps 1 & 2 above.
5. Create more PDC hazard zones instead of just one proximal hazard zone. Choose some other PDC volumes, maybe 0.001, 0.01, 0.1 km^3 (depends on your volcano).
6. Using these volumes, use the regression equations from this talk to find what an appropriate H/L value (or basal friction angle) is for these PDC volumes.
7. Use the LAHARZ H/L ratio tool to make energy cones for these values (but don't run LAHARZ again from these points!). Display these new energy cones on the map and label them something like 'High probability PDC zone' for the smallest volume/smallest H/L cone, etc.

Volume



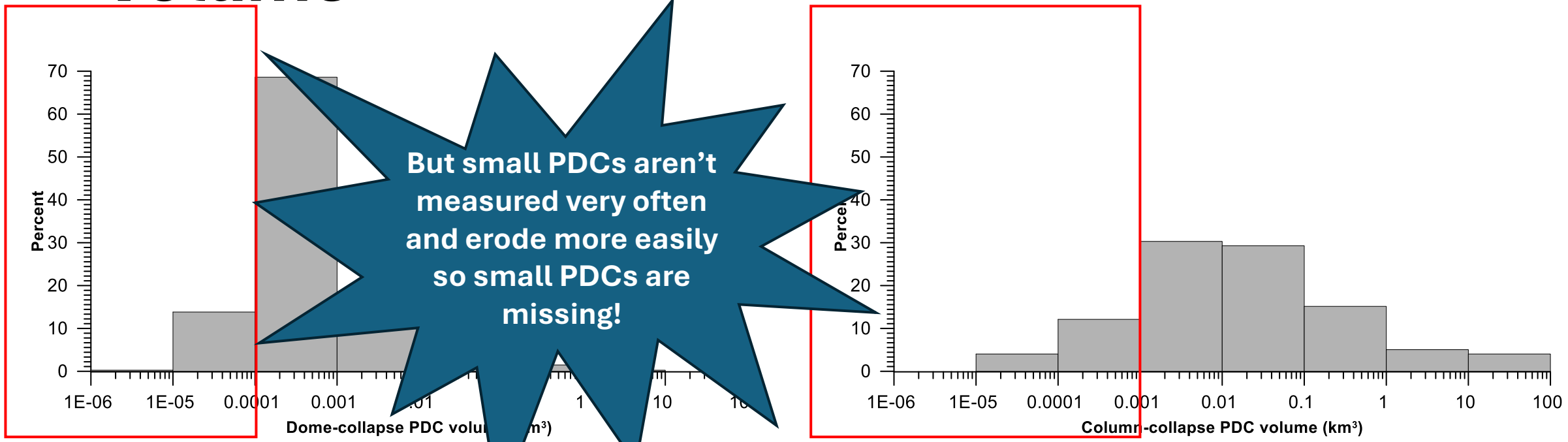


- Dome-collapse PDC volume is generally limited by the size of the dome
- Most dome-collapse PDCs are 0.0001-0.001 km^3 (0.1-1 M m^3)

- The column-collapse PDC volume distribution is more spread out
- Most column-collapse PDCs are 0.001-0.01 km^3 (1-10 M m^3)

Data from FlowDat (Ogburn 2025)

Volume



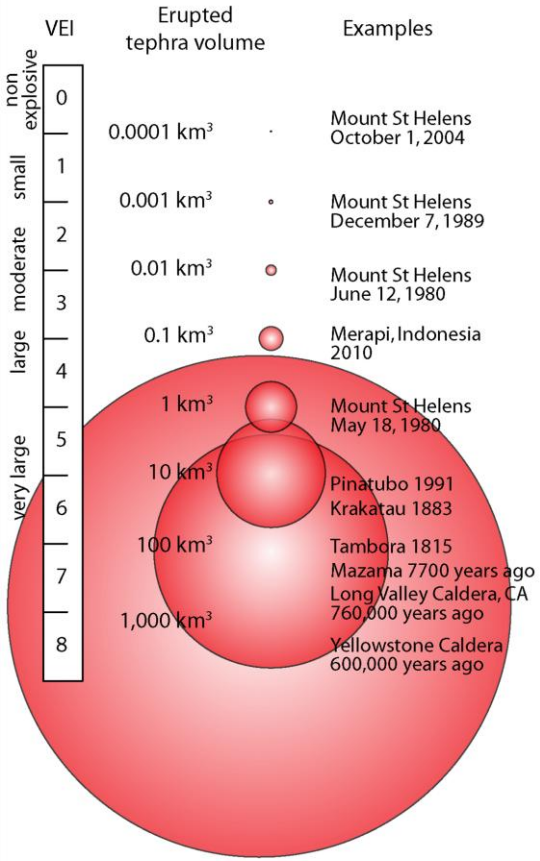
But small PDCs aren't measured very often and erode more easily so small PDCs are missing!

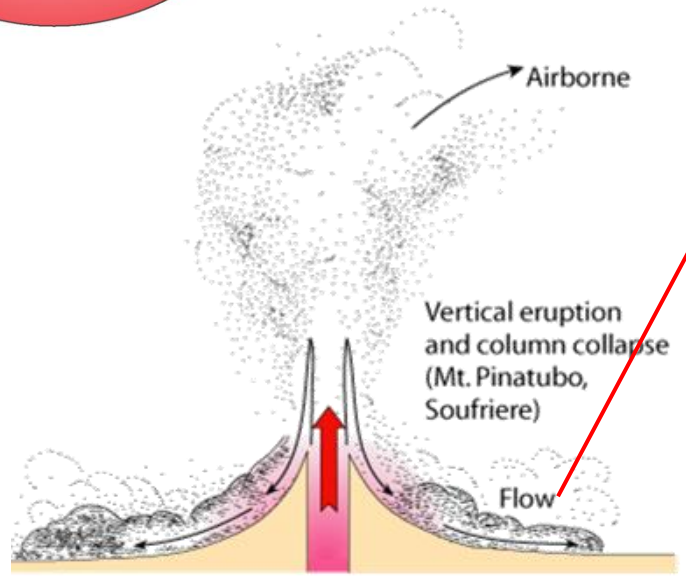
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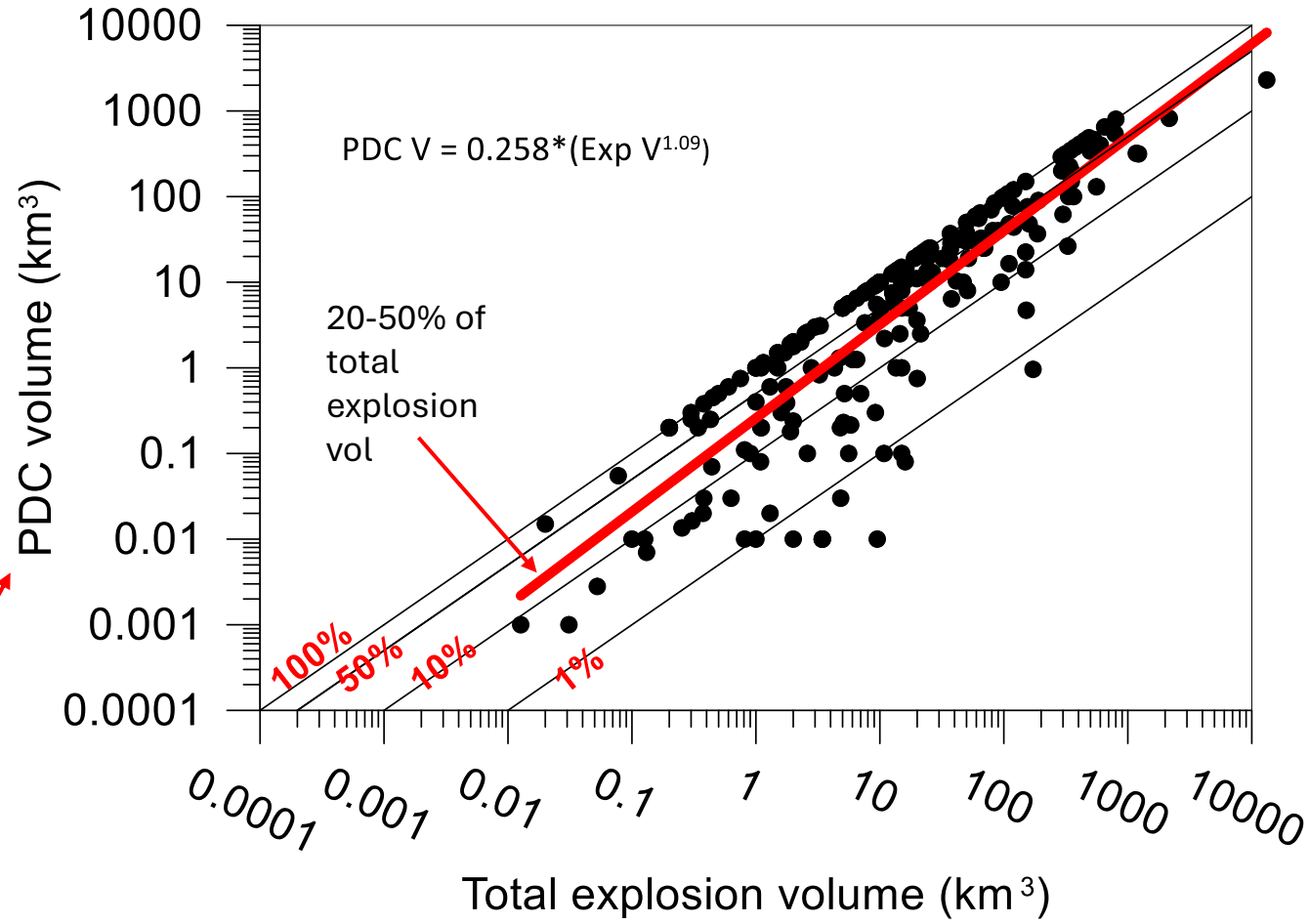
- The column-collapse PDC volume distribution is more spread out
- Most column-collapse PDCs are 0.001-0.01 km^3 (1-10 M m^3)

Data from FlowDat (Ogburn 2025)

How big are column-collapse PDCs for a given VEI?
How much erupted volume collapses to form PDCs?



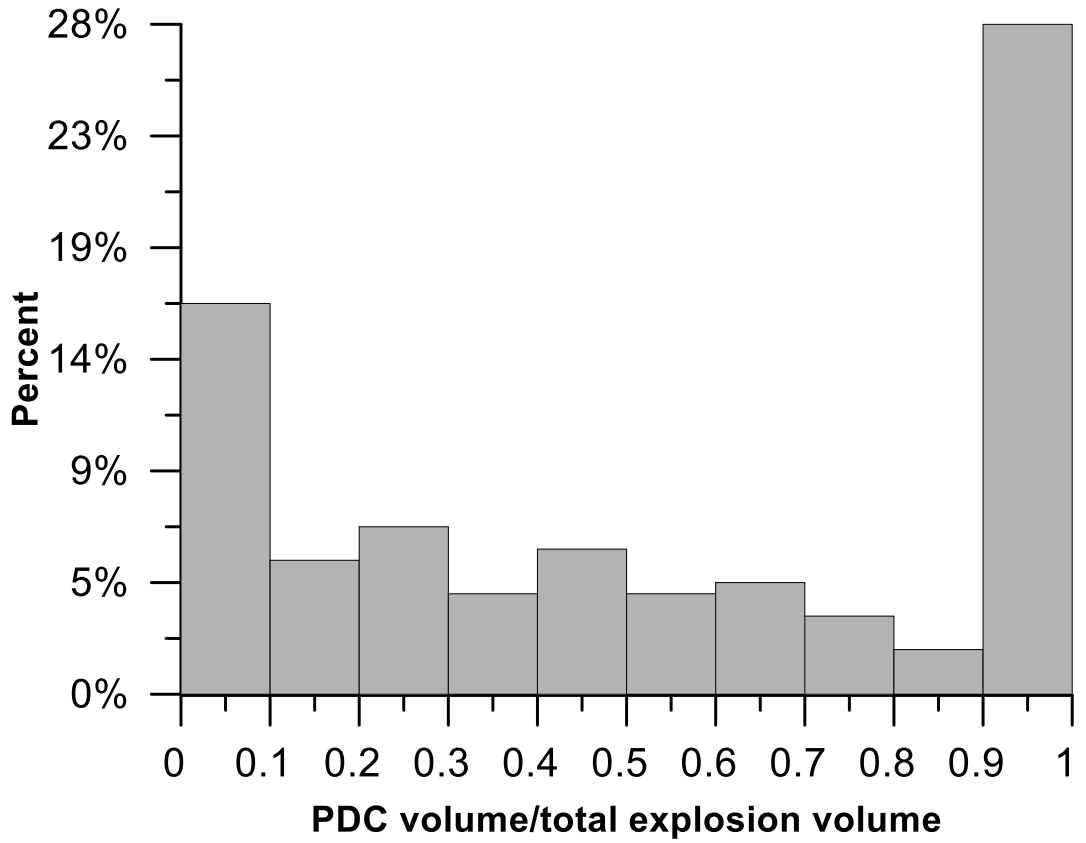




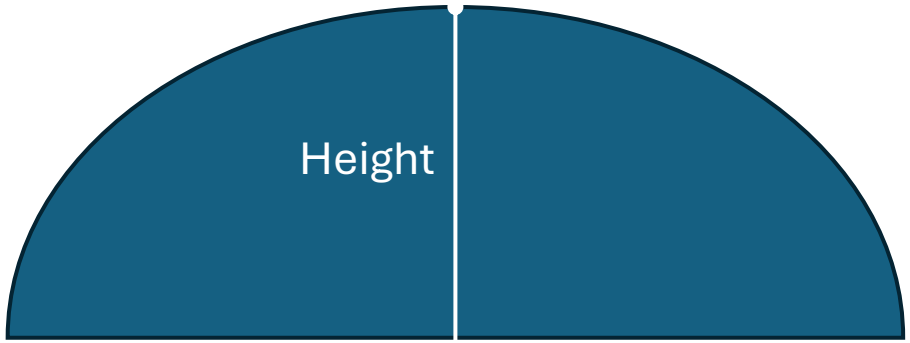
Data from LaMEVE (Crossweller et al. 2012) and FlowDat (Ogburn 2025)

If the most or all of the column collapsed (worst case scenario)

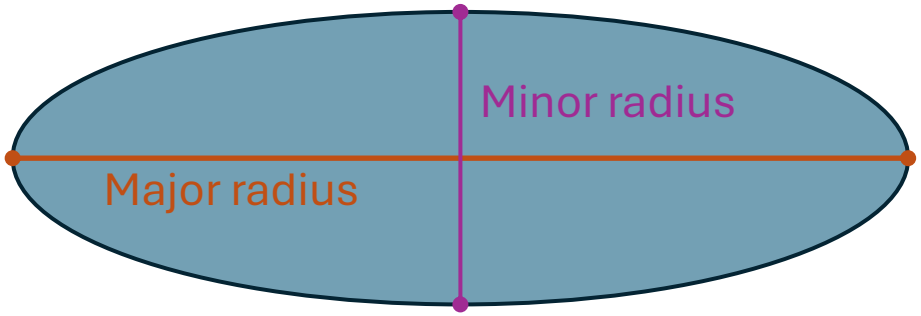
VEI	Max volume (km3)	Max volume (M m3)
1	0.001	1
2	0.01	10
3	0.1	100
4	1	1000
5	10	10,000
6	100	100,000
7	1000	1,000,00



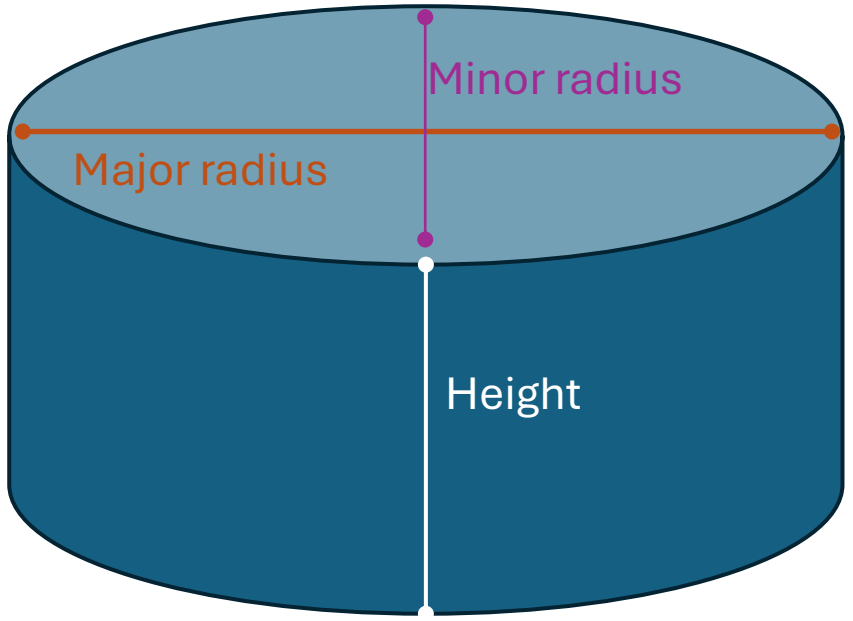
Source shape parameters: Pile size



Paraboloid (dome)



Paraboloid (dome)
Top view



Cylinder (column)

Source shape parameters: Pile orientation

